

Part I

Mathematical Foundations

Introduction and Motivation

Machine learning is about designing algorithms that automatically extract valuable information from data. The emphasis here is on “automatic”, i.e., machine learning is concerned about general-purpose methodologies that can be applied to many datasets, while producing something that is meaningful. There are three concepts that are at the core of machine learning: data, a model, and learning.

Since machine learning is inherently data driven, *data* is at the core of machine learning. The goal of machine learning is to design general-purpose methodologies to extract valuable patterns from data, ideally without much domain-specific expertise. For example, given a large corpus of documents (e.g., books in many libraries), machine learning methods can be used to automatically find relevant topics that are shared across documents (Hoffman et al., 2010). To achieve this goal, we design *models* that are typically related to the process that generates data, similar to the dataset we are given. For example, in a regression setting, the model would describe a function that maps inputs to real-valued outputs. To paraphrase Mitchell (1997): A model is said to learn from data if its performance on a given task improves after the data is taken into account. The goal is to find good models that generalize well to yet unseen data, which we may care about in the future. *Learning* can be understood as a way to automatically find patterns and structure in data by optimizing the parameters of the model.

While machine learning has seen many success stories, and software is readily available to design and train rich and flexible machine learning systems, we believe that the mathematical foundations of machine learning are important in order to understand fundamental principles upon which more complicated machine learning systems are built. Understanding these principles can facilitate creating new machine learning solutions, understanding and debugging existing approaches, and learning about the inherent assumptions and limitations of the methodologies we are working with.

1.1 Finding Words for Intuitions

A challenge we face regularly in machine learning is that concepts and words are slippery, and a particular component of the machine learning system can be abstracted to different mathematical concepts. For example, the word “algorithm” is used in at least two different senses in the context of machine learning. In the first sense, we use the phrase “machine learning algorithm” to mean a system that makes predictions based on input data. We refer to these algorithms as *predictors*. In the second sense, we use the exact same phrase “machine learning algorithm” to mean a system that adapts some internal parameters of the predictor so that it performs well on future unseen input data. Here we refer to this adaptation as *training* a system.

predictor

training

This book will not resolve the issue of ambiguity, but we want to highlight upfront that, depending on the context, the same expressions can mean different things. However, we attempt to make the context sufficiently clear to reduce the level of ambiguity.

The first part of this book introduces the mathematical concepts and foundations needed to talk about the three main components of a machine learning system: data, models, and learning. We will briefly outline these components here, and we will revisit them again in Chapter 8 once we have discussed the necessary mathematical concepts.

data as vectors

While not all data is numerical, it is often useful to consider data in a number format. In this book, we assume that *data* has already been appropriately converted into a numerical representation suitable for reading into a computer program. Therefore, we think of data as vectors. As another illustration of how subtle words are, there are (at least) three different ways to think about vectors: a vector as an array of numbers (a computer science view), a vector as an arrow with a direction and magnitude (a physics view), and a vector as an object that obeys addition and scaling (a mathematical view).

model

A *model* is typically used to describe a process for generating data, similar to the dataset at hand. Therefore, good models can also be thought of as simplified versions of the real (unknown) data-generating process, capturing aspects that are relevant for modeling the data and extracting hidden patterns from it. A good model can then be used to predict what would happen in the real world without performing real-world experiments.

learning

We now come to the crux of the matter, the *learning* component of machine learning. Assume we are given a dataset and a suitable model. *Training* the model means to use the data available to optimize some parameters of the model with respect to a utility function that evaluates how well the model predicts the training data. Most training methods can be thought of as an approach analogous to climbing a hill to reach its peak. In this analogy, the peak of the hill corresponds to a maximum of some

desired performance measure. However, in practice, we are interested in the model to perform well on unseen data. Performing well on data that we have already seen (training data) may only mean that we found a good way to memorize the data. However, this may not generalize well to unseen data, and, in practical applications, we often need to expose our machine learning system to situations that it has not encountered before.

Let us summarize the main concepts of machine learning that we cover in this book:

- We represent data as vectors.
- We choose an appropriate model, either using the probabilistic or optimization view.
- We learn from available data by using numerical optimization methods with the aim that the model performs well on data not used for training.

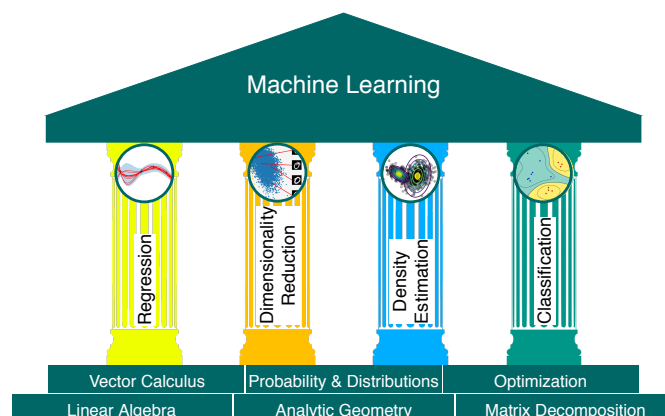
1.2 Two Ways to Read This Book

We can consider two strategies for understanding the mathematics for machine learning:

- **Bottom-up:** Building up the concepts from foundational to more advanced. This is often the preferred approach in more technical fields, such as mathematics. This strategy has the advantage that the reader at all times is able to rely on their previously learned concepts. Unfortunately, for a practitioner many of the foundational concepts are not particularly interesting by themselves, and the lack of motivation means that most foundational definitions are quickly forgotten.
- **Top-down:** Drilling down from practical needs to more basic requirements. This goal-driven approach has the advantage that the readers know at all times why they need to work on a particular concept, and there is a clear path of required knowledge. The downside of this strategy is that the knowledge is built on potentially shaky foundations, and the readers have to remember a set of words that they do not have any way of understanding.

We decided to write this book in a modular way to separate foundational (mathematical) concepts from applications so that this book can be read in both ways. The book is split into two parts, where Part I lays the mathematical foundations and Part II applies the concepts from Part I to a set of fundamental machine learning problems, which form four pillars of machine learning as illustrated in Figure 1.1: regression, dimensionality reduction, density estimation, and classification. Chapters in Part I mostly build upon the previous ones, but it is possible to skip a chapter and work backward if necessary. Chapters in Part II are only loosely coupled and can be read in any order. There are many pointers forward and backward

Figure 1.1 The foundations and four pillars of machine learning.



between the two parts of the book to link mathematical concepts with machine learning algorithms.

Of course there are more than two ways to read this book. Most readers learn using a combination of top-down and bottom-up approaches, sometimes building up basic mathematical skills before attempting more complex concepts, but also choosing topics based on applications of machine learning.

Part I Is about Mathematics

The four pillars of machine learning we cover in this book (see Figure 1.1) require a solid mathematical foundation, which is laid out in Part I.

linear algebra

We represent numerical data as vectors and represent a table of such data as a matrix. The study of vectors and matrices is called *linear algebra*, which we introduce in Chapter 2. The collection of vectors as a matrix is also described there.

analytic geometry

Given two vectors representing two objects in the real world, we want to make statements about their similarity. The idea is that vectors that are similar should be predicted to have similar outputs by our machine learning algorithm (our predictor). To formalize the idea of similarity between vectors, we need to introduce operations that take two vectors as input and return a numerical value representing their similarity. The construction of similarity and distances is central to *analytic geometry* and is discussed in Chapter 3.

matrix
decomposition

In Chapter 4, we introduce some fundamental concepts about matrices and *matrix decomposition*. Some operations on matrices are extremely useful in machine learning, and they allow for an intuitive interpretation of the data and more efficient learning.

We often consider data to be noisy observations of some true underlying signal. We hope that by applying machine learning we can identify the signal from the noise. This requires us to have a language for quantifying what “noise” means. We often would also like to have predictors that

allow us to express some sort of uncertainty, e.g., to quantify the confidence we have about the value of the prediction at a particular test data point. Quantification of uncertainty is the realm of *probability theory* and is covered in Chapter 6.

probability theory

To train machine learning models, we typically find parameters that maximize some performance measure. Many optimization techniques require the concept of a gradient, which tells us the direction in which to search for a solution. Chapter 5 is about *vector calculus* and details the concept of gradients, which we subsequently use in Chapter 7, where we talk about *optimization* to find maxima/minima of functions.

vector calculus

optimization

Part II Is about Machine Learning

The second part of the book introduces *four pillars of machine learning* as shown in Figure 1.1. We illustrate how the mathematical concepts introduced in the first part of the book are the foundation for each pillar. Broadly speaking, chapters are ordered by difficulty (in ascending order).

In Chapter 8, we restate the three components of machine learning (data, models, and parameter estimation) in a mathematical fashion. In addition, we provide some guidelines for building experimental set-ups that guard against overly optimistic evaluations of machine learning systems. Recall that the goal is to build a predictor that performs well on unseen data.

In Chapter 9, we will have a close look at *linear regression*, where our objective is to find functions that map inputs $\mathbf{x} \in \mathbb{R}^D$ to corresponding observed function values $y \in \mathbb{R}$, which we can interpret as the labels of their respective inputs. We will discuss classical model fitting (parameter estimation) via maximum likelihood and maximum a posteriori estimation, as well as Bayesian linear regression, where we integrate the parameters out instead of optimizing them.

linear regression

Chapter 10 focuses on *dimensionality reduction*, the second pillar in Figure 1.1, using principal component analysis. The key objective of dimensionality reduction is to find a compact, lower-dimensional representation of high-dimensional data $\mathbf{x} \in \mathbb{R}^D$, which is often easier to analyze than the original data. Unlike regression, dimensionality reduction is only concerned about modeling the data – there are no labels associated with a data point $\mathbf{x}$.

dimensionality
reduction

In Chapter 11, we will move to our third pillar: *density estimation*. The objective of density estimation is to find a probability distribution that describes a given dataset. We will focus on Gaussian mixture models for this purpose, and we will discuss an iterative scheme to find the parameters of this model. As in dimensionality reduction, there are no labels associated with the data points $\mathbf{x} \in \mathbb{R}^D$. However, we do not seek a low-dimensional representation of the data. Instead, we are interested in a density model that describes the data.

density estimation

Chapter 12 concludes the book with an in-depth discussion of the fourth

classification

pillar: *classification*. We will discuss classification in the context of support vector machines. Similar to regression (Chapter 9), we have inputs x and corresponding labels y . However, unlike regression, where the labels were real-valued, the labels in classification are integers, which requires special care.

1.3 Exercises and Feedback

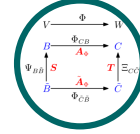
We provide some exercises in Part I, which can be done mostly by pen and paper. For Part II, we provide programming tutorials (jupyter notebooks) to explore some properties of the machine learning algorithms we discuss in this book.

We appreciate that Cambridge University Press strongly supports our aim to democratize education and learning by making this book freely available for download at

<https://mml-book.com>

where tutorials, errata, and additional materials can be found. Mistakes can be reported and feedback provided using the preceding URL.

Linear Algebra



algebra

When formalizing intuitive concepts, a common approach is to construct a set of objects (symbols) and a set of rules to manipulate these objects. This is known as an *algebra*. Linear algebra is the study of vectors and certain rules to manipulate vectors. The vectors many of us know from school are called “geometric vectors”, which are usually denoted by a small arrow above the letter, e.g., $\vec{x}$ and $\vec{y}$. In this book, we discuss more general concepts of vectors and use a bold letter to represent them, e.g., $\mathbf{x}$ and $\mathbf{y}$.

In general, vectors are special objects that can be added together and multiplied by scalars to produce another object of the same kind. From an abstract mathematical viewpoint, any object that satisfies these two properties can be considered a vector. Here are some examples of such vector objects:

1. Geometric vectors. This example of a vector may be familiar from high school mathematics and physics. Geometric vectors – see Figure 2.1(a) – are directed segments, which can be drawn (at least in two dimensions). Two geometric vectors $\vec{x}$, $\vec{y}$ can be added, such that $\vec{x} + \vec{y} = \vec{z}$ is another geometric vector. Furthermore, multiplication by a scalar $\lambda \vec{x}$, $\lambda \in \mathbb{R}$, is also a geometric vector. In fact, it is the original vector scaled by λ . Therefore, geometric vectors are instances of the vector concepts introduced previously. Interpreting vectors as geometric vectors enables us to use our intuitions about direction and magnitude to reason about mathematical operations.
2. Polynomials are also vectors; see Figure 2.1(b): Two polynomials can

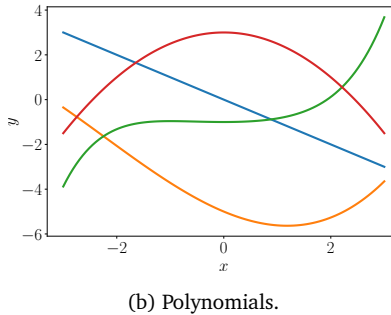
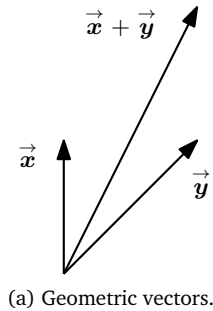


Figure 2.1
Different types of vectors. Vectors can be surprising objects, including (a) geometric vectors and (b) polynomials.

be added together, which results in another polynomial; and they can be multiplied by a scalar $\lambda \in \mathbb{R}$, and the result is a polynomial as well. Therefore, polynomials are (rather unusual) instances of vectors. Note that polynomials are very different from geometric vectors. While geometric vectors are concrete “drawings”, polynomials are abstract concepts. However, they are both vectors in the sense previously described.

3. Audio signals are vectors. Audio signals are represented as a series of numbers. We can add audio signals together, and their sum is a new audio signal. If we scale an audio signal, we also obtain an audio signal. Therefore, audio signals are a type of vector, too.
4. Elements of $\mathbb{R}^n$ (tuples of n real numbers) are vectors. $\mathbb{R}^n$ is more abstract than polynomials, and it is the concept we focus on in this book. For instance,

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3 \quad (2.1)$$

is an example of a triplet of numbers. Adding two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ component-wise results in another vector: $\mathbf{a} + \mathbf{b} = \mathbf{c} \in \mathbb{R}^n$. Moreover, multiplying $\mathbf{a} \in \mathbb{R}^n$ by $\lambda \in \mathbb{R}$ results in a scaled vector $\lambda \mathbf{a} \in \mathbb{R}^n$. Considering vectors as elements of $\mathbb{R}^n$ has an additional benefit that it loosely corresponds to arrays of real numbers on a computer. Many programming languages support array operations, which allow for convenient implementation of algorithms that involve vector operations.

Be careful to check whether array operations actually perform vector operations when implementing on a computer.

Pavel Grinfeld’s series on linear algebra:
<http://tinyurl.com/naclwm>
 Gilbert Strang’s course on linear algebra:
<http://tinyurl.com/29p5q8j>
 3Blue1Brown series on linear algebra:
<https://tinyurl.com/h5g4kps>

Linear algebra focuses on the similarities between these vector concepts. We can add them together and multiply them by scalars. We will largely focus on vectors in $\mathbb{R}^n$ since most algorithms in linear algebra are formulated in $\mathbb{R}^n$. We will see in Chapter 8 that we often consider data to be represented as vectors in $\mathbb{R}^n$. In this book, we will focus on finite-dimensional vector spaces, in which case there is a 1:1 correspondence between any kind of vector and $\mathbb{R}^n$. When it is convenient, we will use intuitions about geometric vectors and consider array-based algorithms.

One major idea in mathematics is the idea of “closure”. This is the question: What is the set of all things that can result from my proposed operations? In the case of vectors: What is the set of vectors that can result by starting with a small set of vectors, and adding them to each other and scaling them? This results in a vector space (Section 2.4). The concept of a vector space and its properties underlie much of machine learning. The concepts introduced in this chapter are summarized in Figure 2.2.

This chapter is mostly based on the lecture notes and books by Drumm and Weil (2001), Strang (2003), Hogben (2013), Liesen and Mehrmann (2015), as well as Pavel Grinfeld’s Linear Algebra series. Other excellent

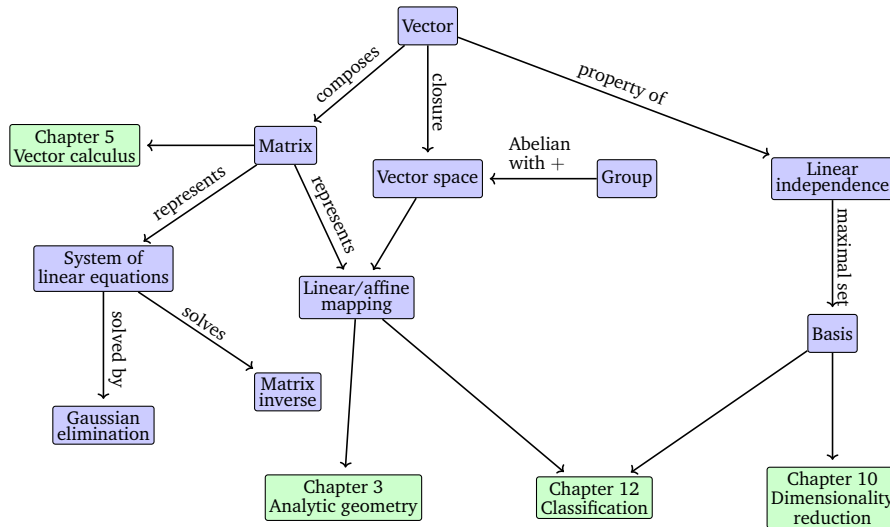


Figure 2.2 A mind map of the concepts introduced in this chapter, along with where they are used in other parts of the book.

resources are Gilbert Strang's Linear Algebra course at MIT and the Linear Algebra Series by 3Blue1Brown.

Linear algebra plays an important role in machine learning and general mathematics. The concepts introduced in this chapter are further expanded to include the idea of geometry in Chapter 3. In Chapter 5, we will discuss vector calculus, where a principled knowledge of matrix operations is essential. In Chapter 10, we will use projections (to be introduced in Section 3.8) for dimensionality reduction with principal component analysis (PCA). In Chapter 9, we will discuss linear regression, where linear algebra plays a central role for solving least-squares problems.

2.1 Systems of Linear Equations

Systems of linear equations play a central part of linear algebra. Many problems can be formulated as systems of linear equations, and linear algebra gives us the tools for solving them.

Example 2.1

A company produces products $N_1, \dots, N_n$ for which resources $R_1, \dots, R_m$ are required. To produce a unit of product N_j , a_{ij} units of resource R_i are needed, where $i = 1, \dots, m$ and $j = 1, \dots, n$.

The objective is to find an optimal production plan, i.e., a plan of how many units x_j of product N_j should be produced if a total of b_i units of resource R_i are available and (ideally) no resources are left over.

If we produce $x_1, \dots, x_n$ units of the corresponding products, we need

a total of

$$a_{i1}x_1 + \cdots + a_{in}x_n \quad (2.2)$$

many units of resource R_i . An optimal production plan $(x_1, \dots, x_n) \in \mathbb{R}^n$, therefore, has to satisfy the following system of equations:

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned} \quad (2.3)$$

where $a_{ij} \in \mathbb{R}$ and $b_i \in \mathbb{R}$.

system of linear
equations
solution

Equation (2.3) is the general form of a *system of linear equations*, and $x_1, \dots, x_n$ are the *unknowns* of this system. Every n -tuple $(x_1, \dots, x_n) \in \mathbb{R}^n$ that satisfies (2.3) is a *solution* of the linear equation system.

Example 2.2

The system of linear equations

$$\begin{aligned} x_1 + x_2 + x_3 &= 3 & (1) \\ x_1 - x_2 + 2x_3 &= 2 & (2) \\ 2x_1 + 3x_3 &= 1 & (3) \end{aligned} \quad (2.4)$$

has *no solution*: Adding the first two equations yields $2x_1 + 3x_3 = 5$, which contradicts the third equation (3).

Let us have a look at the system of linear equations

$$\begin{aligned} x_1 + x_2 + x_3 &= 3 & (1) \\ x_1 - x_2 + 2x_3 &= 2 & (2) \\ x_2 + x_3 &= 2 & (3) \end{aligned} \quad (2.5)$$

From the first and third equation, it follows that $x_1 = 1$. From (1)+(2), we get $2x_1 + 3x_3 = 5$, i.e., $x_3 = 1$. From (3), we then get that $x_2 = 1$. Therefore, $(1, 1, 1)$ is the only possible and *unique solution* (verify that $(1, 1, 1)$ is a solution by plugging in).

As a third example, we consider

$$\begin{aligned} x_1 + x_2 + x_3 &= 3 & (1) \\ x_1 - x_2 + 2x_3 &= 2 & (2) \\ 2x_1 + 3x_3 &= 5 & (3) \end{aligned} \quad (2.6)$$

Since (1)+(2)=(3), we can omit the third equation (redundancy). From (1) and (2), we get $2x_1 = 5 - 3x_3$ and $2x_2 = 1 + x_3$. We define $x_3 = a \in \mathbb{R}$ as a free variable, such that any triplet

$$\left(\frac{5}{2} - \frac{3}{2}a, \frac{1}{2} + \frac{1}{2}a, a \right), \quad a \in \mathbb{R} \quad (2.7)$$

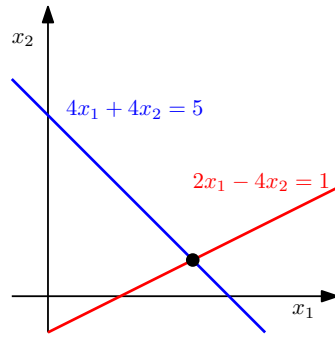


Figure 2.3 The solution space of a system of two linear equations with two variables can be geometrically interpreted as the intersection of two lines. Every linear equation represents a line.

is a solution of the system of linear equations, i.e., we obtain a solution set that contains *infinitely many* solutions.

In general, for a real-valued system of linear equations we obtain either no, exactly one, or infinitely many solutions. Linear regression (Chapter 9) solves a version of Example 2.1 when we cannot solve the system of linear equations.

Remark (Geometric Interpretation of Systems of Linear Equations). In a system of linear equations with two variables x_1, x_2 , each linear equation defines a line on the x_1x_2 -plane. Since a solution to a system of linear equations must satisfy all equations simultaneously, the solution set is the intersection of these lines. This intersection set can be a line (if the linear equations describe the same line), a point, or empty (when the lines are parallel). An illustration is given in Figure 2.3 for the system

$$\begin{aligned} 4x_1 + 4x_2 &= 5 \\ 2x_1 - 4x_2 &= 1 \end{aligned} \quad (2.8)$$

where the solution space is the point $(x_1, x_2) = (1, \frac{1}{4})$. Similarly, for three variables, each linear equation determines a plane in three-dimensional space. When we intersect these planes, i.e., satisfy all linear equations at the same time, we can obtain a solution set that is a plane, a line, a point or empty (when the planes have no common intersection). $\diamond$

For a systematic approach to solving systems of linear equations, we will introduce a useful compact notation. We collect the coefficients a_{ij} into vectors and collect the vectors into matrices. In other words, we write the system from (2.3) in the following form:

$$\begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} x_1 + \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} x_2 + \cdots + \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} x_n = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \quad (2.9)$$

$$\Leftrightarrow \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}. \quad (2.10)$$

In the following, we will have a close look at these *matrices* and define computation rules. We will return to solving linear equations in Section 2.3.

2.2 Matrices

Matrices play a central role in linear algebra. They can be used to compactly represent systems of linear equations, but they also represent linear functions (linear mappings) as we will see later in Section 2.7. Before we discuss some of these interesting topics, let us first define what a matrix is and what kind of operations we can do with matrices. We will see more properties of matrices in Chapter 4.

matrix

Definition 2.1 (Matrix). With $m, n \in \mathbb{N}$ a real-valued (m, n) *matrix* $\mathbf{A}$ is an $m \cdot n$ -tuple of elements a_{ij} , $i = 1, \dots, m$, $j = 1, \dots, n$, which is ordered according to a rectangular scheme consisting of m rows and n columns:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad a_{ij} \in \mathbb{R}. \quad (2.11)$$

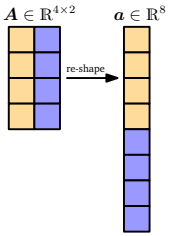
row

column

row vector

column vector

Figure 2.4 By stacking its columns, a matrix $\mathbf{A}$ can be represented as a long vector $\mathbf{a}$.



Note the size of the matrices.

$\mathbf{C} =$
`np.einsum('il, lj', A, B)`

By convention $(1, n)$ -matrices are called *rows* and $(m, 1)$ -matrices are called *columns*. These special matrices are also called *row/column vectors*.

$\mathbb{R}^{m \times n}$ is the set of all real-valued (m, n) -matrices. $\mathbf{A} \in \mathbb{R}^{m \times n}$ can be equivalently represented as $\mathbf{a} \in \mathbb{R}^{mn}$ by stacking all n columns of the matrix into a long vector; see Figure 2.4.

2.2.1 Matrix Addition and Multiplication

The sum of two matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{m \times n}$ is defined as the element-wise sum, i.e.,

$$\mathbf{A} + \mathbf{B} := \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1n} + b_{1n} \\ \vdots & & \vdots \\ a_{m1} + b_{m1} & \cdots & a_{mn} + b_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}. \quad (2.12)$$

For matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times k}$, the elements c_{ij} of the product $\mathbf{C} = \mathbf{AB} \in \mathbb{R}^{m \times k}$ are computed as

$$c_{ij} = \sum_{l=1}^n a_{il} b_{lj}, \quad i = 1, \dots, m, \quad j = 1, \dots, k. \quad (2.13)$$

This means, to compute element c_{ij} we multiply the elements of the i th row of $\mathbf{A}$ with the j th column of $\mathbf{B}$ and sum them up. Later in Section 3.2, we will call this the *dot product* of the corresponding row and column. In cases, where we need to be explicit that we are performing multiplication, we use the notation $\mathbf{A} \cdot \mathbf{B}$ to denote multiplication (explicitly showing “.”).

Remark. Matrices can only be multiplied if their “neighboring” dimensions match. For instance, an $n \times k$ -matrix $\mathbf{A}$ can be multiplied with a $k \times m$ -matrix $\mathbf{B}$, but only from the left side:

$$\underbrace{\mathbf{A}}_{n \times k} \underbrace{\mathbf{B}}_{k \times m} = \underbrace{\mathbf{C}}_{n \times m} \quad (2.14)$$

The product $\mathbf{BA}$ is not defined if $m \neq n$ since the neighboring dimensions do not match. $\diamond$

Remark. Matrix multiplication is *not* defined as an element-wise operation on matrix elements, i.e., $c_{ij} \neq a_{ij}b_{ij}$ (even if the size of $\mathbf{A}, \mathbf{B}$ was chosen appropriately). This kind of element-wise multiplication often appears in programming languages when we multiply (multi-dimensional) arrays with each other, and is called a *Hadamard product*. $\diamond$

There are n columns in $\mathbf{A}$ and n rows in $\mathbf{B}$ so that we can compute $a_{il}b_{lj}$ for $l = 1, \dots, n$.

Commonly, the dot product between two vectors $\mathbf{a}, \mathbf{b}$ is denoted by $\mathbf{a}^\top \mathbf{b}$ or $\langle \mathbf{a}, \mathbf{b} \rangle$.

Hadamard product

Example 2.3

For $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 3}$, $\mathbf{B} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}$, we obtain

$$\mathbf{AB} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad (2.15)$$

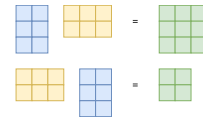
$$\mathbf{BA} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 4 & 2 \\ -2 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 3}. \quad (2.16)$$

From this example, we can already see that matrix multiplication is not commutative, i.e., $\mathbf{AB} \neq \mathbf{BA}$; see also Figure 2.5 for an illustration.

Definition 2.2 (Identity Matrix). In $\mathbb{R}^{n \times n}$, we define the *identity matrix*

$$\mathbf{I}_n := \begin{bmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 1 \end{bmatrix} \in \mathbb{R}^{n \times n} \quad (2.17)$$

Figure 2.5 Even if both matrix multiplications $\mathbf{AB}$ and $\mathbf{BA}$ are defined, the dimensions of the results can be different.



identity matrix

as the $n \times n$ -matrix containing 1 on the diagonal and 0 everywhere else.

Now that we defined matrix multiplication, matrix addition and the identity matrix, let us have a look at some properties of matrices:

associativity

▪ *Associativity:*

$$\forall \mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p}, \mathbf{C} \in \mathbb{R}^{p \times q} : (\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}) \quad (2.18)$$

distributivity

▪ *Distributivity:*

$$\forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}, \mathbf{C}, \mathbf{D} \in \mathbb{R}^{n \times p} : (\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC} \quad (2.19a)$$

$$\mathbf{A}(\mathbf{C} + \mathbf{D}) = \mathbf{AC} + \mathbf{AD} \quad (2.19b)$$

▪ *Multiplication with the identity matrix:*

$$\forall \mathbf{A} \in \mathbb{R}^{m \times n} : \mathbf{I}_m \mathbf{A} = \mathbf{A} \mathbf{I}_n = \mathbf{A} \quad (2.20)$$

Note that $\mathbf{I}_m \neq \mathbf{I}_n$ for $m \neq n$.

2.2.2 Inverse and Transpose

A square matrix
possesses the same
number of columns
and rows.
inverse

Definition 2.3 (Inverse). Consider a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$. Let matrix $\mathbf{B} \in \mathbb{R}^{n \times n}$ have the property that $\mathbf{AB} = \mathbf{I}_n = \mathbf{BA}$. $\mathbf{B}$ is called the *inverse* of $\mathbf{A}$ and denoted by $\mathbf{A}^{-1}$.

regular
invertible
nonsingular
singular
noninvertible

Unfortunately, not every matrix $\mathbf{A}$ possesses an inverse $\mathbf{A}^{-1}$. If this inverse does exist, $\mathbf{A}$ is called *regular/invertible/nonsingular*, otherwise *singular/noninvertible*. When the matrix inverse exists, it is unique. In Section 2.3, we will discuss a general way to compute the inverse of a matrix by solving a system of linear equations.

Remark (Existence of the Inverse of a 2×2 -matrix). Consider a matrix

$$\mathbf{A} := \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}. \quad (2.21)$$

If we multiply $\mathbf{A}$ with

$$\mathbf{A}' := \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \quad (2.22)$$

we obtain

$$\mathbf{AA}' = \begin{bmatrix} a_{11}a_{22} - a_{12}a_{21} & 0 \\ 0 & a_{11}a_{22} - a_{12}a_{21} \end{bmatrix} = (a_{11}a_{22} - a_{12}a_{21})\mathbf{I}. \quad (2.23)$$

Therefore,

$$\mathbf{A}^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \quad (2.24)$$

if and only if $a_{11}a_{22} - a_{12}a_{21} \neq 0$. In Section 4.1, we will see that $a_{11}a_{22} -$

$a_{12}a_{21}$ is the determinant of a 2×2 -matrix. Furthermore, we can generally use the determinant to check whether a matrix is invertible. $\diamond$

Example 2.4 (Inverse Matrix)

The matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 4 & 5 \\ 6 & 7 & 7 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -7 & -7 & 6 \\ 2 & 1 & -1 \\ 4 & 5 & -4 \end{bmatrix} \quad (2.25)$$

are inverse to each other since $\mathbf{AB} = \mathbf{I} = \mathbf{BA}$.

Definition 2.4 (Transpose). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ the matrix $\mathbf{B} \in \mathbb{R}^{n \times m}$ with $b_{ij} = a_{ji}$ is called the *transpose* of $\mathbf{A}$. We write $\mathbf{B} = \mathbf{A}^\top$.

In general, $\mathbf{A}^\top$ can be obtained by writing the columns of $\mathbf{A}$ as the rows of $\mathbf{A}^\top$. The following are important properties of inverses and transposes:

$$\mathbf{AA}^{-1} = \mathbf{I} = \mathbf{A}^{-1}\mathbf{A} \quad (2.26)$$

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} \quad (2.27)$$

$$(\mathbf{A} + \mathbf{B})^{-1} \neq \mathbf{A}^{-1} + \mathbf{B}^{-1} \quad (2.28)$$

$$(\mathbf{A}^\top)^\top = \mathbf{A} \quad (2.29)$$

$$(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top \quad (2.30)$$

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top \quad (2.31)$$

Definition 2.5 (Symmetric Matrix). A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is *symmetric* if $\mathbf{A} = \mathbf{A}^\top$.

Note that only (n, n) -matrices can be symmetric. Generally, we call (n, n) -matrices also *square matrices* because they possess the same number of rows and columns. Moreover, if $\mathbf{A}$ is invertible, then so is $\mathbf{A}^\top$, and $(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1} =: \mathbf{A}^{-\top}$.

Remark (Sum and Product of Symmetric Matrices). The sum of symmetric matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ is always symmetric. However, although their product is always defined, it is generally not symmetric:

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}. \quad (2.32)$$

$\diamond$

2.2.3 Multiplication by a Scalar

Let us look at what happens to matrices when they are multiplied by a scalar $\lambda \in \mathbb{R}$. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\lambda \in \mathbb{R}$. Then $\lambda \mathbf{A} = \mathbf{K}$, $K_{ij} = \lambda a_{ij}$. Practically, λ scales each element of $\mathbf{A}$. For $\lambda, \psi \in \mathbb{R}$, the following holds:

transpose

The main diagonal (sometimes called “principal diagonal”, “primary diagonal”, “leading diagonal”, or “major diagonal”) of a matrix $\mathbf{A}$ is the collection of entries A_{ij} where $i = j$.

The scalar case of (2.28) is

$$\frac{1}{2+4} = \frac{1}{6} \neq \frac{1}{2} + \frac{1}{4}.$$

symmetric matrix

square matrix

associativity

▪ **Associativity:**

$$(\lambda\psi)\mathbf{C} = \lambda(\psi\mathbf{C}), \quad \mathbf{C} \in \mathbb{R}^{m \times n}$$

$$\lambda(\mathbf{BC}) = (\lambda\mathbf{B})\mathbf{C} = \mathbf{B}(\lambda\mathbf{C}) = (\mathbf{BC})\lambda, \quad \mathbf{B} \in \mathbb{R}^{m \times n}, \mathbf{C} \in \mathbb{R}^{n \times k}.$$

Note that this allows us to move scalar values around.

distributivity

$$(\lambda\mathbf{C})^\top = \mathbf{C}^\top \lambda^\top = \mathbf{C}^\top \lambda = \lambda\mathbf{C}^\top \text{ since } \lambda = \lambda^\top \text{ for all } \lambda \in \mathbb{R}.$$

▪ **Distributivity:**

$$(\lambda + \psi)\mathbf{C} = \lambda\mathbf{C} + \psi\mathbf{C}, \quad \mathbf{C} \in \mathbb{R}^{m \times n}$$

$$\lambda(\mathbf{B} + \mathbf{C}) = \lambda\mathbf{B} + \lambda\mathbf{C}, \quad \mathbf{B}, \mathbf{C} \in \mathbb{R}^{m \times n}$$

Example 2.5 (Distributivity)

If we define

$$\mathbf{C} := \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad (2.33)$$

then for any $\lambda, \psi \in \mathbb{R}$ we obtain

$$(\lambda + \psi)\mathbf{C} = \begin{bmatrix} (\lambda + \psi)1 & (\lambda + \psi)2 \\ (\lambda + \psi)3 & (\lambda + \psi)4 \end{bmatrix} = \begin{bmatrix} \lambda + \psi & 2\lambda + 2\psi \\ 3\lambda + 3\psi & 4\lambda + 4\psi \end{bmatrix} \quad (2.34a)$$

$$= \begin{bmatrix} \lambda & 2\lambda \\ 3\lambda & 4\lambda \end{bmatrix} + \begin{bmatrix} \psi & 2\psi \\ 3\psi & 4\psi \end{bmatrix} = \lambda\mathbf{C} + \psi\mathbf{C}. \quad (2.34b)$$

2.2.4 Compact Representations of Systems of Linear Equations

If we consider the system of linear equations

$$\begin{aligned} 2x_1 + 3x_2 + 5x_3 &= 1 \\ 4x_1 - 2x_2 - 7x_3 &= 8 \\ 9x_1 + 5x_2 - 3x_3 &= 2 \end{aligned} \quad (2.35)$$

and use the rules for matrix multiplication, we can write this equation system in a more compact form as

$$\begin{bmatrix} 2 & 3 & 5 \\ 4 & -2 & -7 \\ 9 & 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ 2 \end{bmatrix}. \quad (2.36)$$

Note that x_1 scales the first column, x_2 the second one, and x_3 the third one.

Generally, a system of linear equations can be compactly represented in their matrix form as $\mathbf{Ax} = \mathbf{b}$; see (2.3), and the product $\mathbf{Ax}$ is a (linear) combination of the columns of $\mathbf{A}$. We will discuss linear combinations in more detail in Section 2.5.

2.3 Solving Systems of Linear Equations

In (2.3), we introduced the general form of an equation system, i.e.,

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m, \end{aligned} \quad (2.37)$$

where $a_{ij} \in \mathbb{R}$ and $b_i \in \mathbb{R}$ are known constants and x_j are unknowns, $i = 1, \dots, m$, $j = 1, \dots, n$. Thus far, we saw that matrices can be used as a compact way of formulating systems of linear equations so that we can write $\mathbf{Ax} = \mathbf{b}$, see (2.10). Moreover, we defined basic matrix operations, such as addition and multiplication of matrices. In the following, we will focus on solving systems of linear equations and provide an algorithm for finding the inverse of a matrix.

2.3.1 Particular and General Solution

Before discussing how to generally solve systems of linear equations, let us have a look at an example. Consider the system of equations

$$\begin{bmatrix} 1 & 0 & 8 & -4 \\ 0 & 1 & 2 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 42 \\ 8 \end{bmatrix}. \quad (2.38)$$

The system has two equations and four unknowns. Therefore, in general we would expect infinitely many solutions. This system of equations is in a particularly easy form, where the first two columns consist of a 1 and a 0. Remember that we want to find scalars $x_1, \dots, x_4$, such that $\sum_{i=1}^4 x_i \mathbf{c}_i = \mathbf{b}$, where we define $\mathbf{c}_i$ to be the i th column of the matrix and $\mathbf{b}$ the right-hand-side of (2.38). A solution to the problem in (2.38) can be found immediately by taking 42 times the first column and 8 times the second column so that

$$\mathbf{b} = \begin{bmatrix} 42 \\ 8 \end{bmatrix} = 42 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 8 \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (2.39)$$

Therefore, a solution is $[42, 8, 0, 0]^\top$. This solution is called a *particular solution* or *special solution*. However, this is not the only solution of this system of linear equations. To capture all the other solutions, we need to be creative in generating $\mathbf{0}$ in a non-trivial way using the columns of the matrix: Adding $\mathbf{0}$ to our special solution does not change the special solution. To do so, we express the third column using the first two columns (which are of this very simple form)

$$\begin{bmatrix} 8 \\ 2 \end{bmatrix} = 8 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (2.40)$$

particular solution
special solution

so that $\mathbf{0} = 8\mathbf{c}_1 + 2\mathbf{c}_2 - 1\mathbf{c}_3 + 0\mathbf{c}_4$ and $(x_1, x_2, x_3, x_4) = (8, 2, -1, 0)$. In fact, any scaling of this solution by $\lambda_1 \in \mathbb{R}$ produces the $\mathbf{0}$ vector, i.e.,

$$\begin{bmatrix} 1 & 0 & 8 & -4 \\ 0 & 1 & 2 & 12 \end{bmatrix} \left(\lambda_1 \begin{bmatrix} 8 \\ 2 \\ -1 \\ 0 \end{bmatrix} \right) = \lambda_1 (8\mathbf{c}_1 + 2\mathbf{c}_2 - \mathbf{c}_3) = \mathbf{0}. \quad (2.41)$$

Following the same line of reasoning, we express the fourth column of the matrix in (2.38) using the first two columns and generate another set of non-trivial versions of $\mathbf{0}$ as

$$\begin{bmatrix} 1 & 0 & 8 & -4 \\ 0 & 1 & 2 & 12 \end{bmatrix} \left(\lambda_2 \begin{bmatrix} -4 \\ 12 \\ 0 \\ -1 \end{bmatrix} \right) = \lambda_2 (-4\mathbf{c}_1 + 12\mathbf{c}_2 - \mathbf{c}_4) = \mathbf{0} \quad (2.42)$$

for any $\lambda_2 \in \mathbb{R}$. Putting everything together, we obtain all solutions of the equation system in (2.38), which is called the *general solution*, as the set

$$\left\{ \mathbf{x} \in \mathbb{R}^4 : \mathbf{x} = \begin{bmatrix} 42 \\ 8 \\ 0 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 8 \\ 2 \\ -1 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} -4 \\ 12 \\ 0 \\ -1 \end{bmatrix}, \lambda_1, \lambda_2 \in \mathbb{R} \right\}. \quad (2.43)$$

Remark. The general approach we followed consisted of the following three steps:

1. Find a particular solution to $\mathbf{Ax} = \mathbf{b}$.
2. Find all solutions to $\mathbf{Ax} = \mathbf{0}$.
3. Combine the solutions from steps 1. and 2. to the general solution.

Neither the general nor the particular solution is unique. $\diamond$

The system of linear equations in the preceding example was easy to solve because the matrix in (2.38) has this particularly convenient form, which allowed us to find the particular and the general solution by inspection. However, general equation systems are not of this simple form. Fortunately, there exists a constructive algorithmic way of transforming any system of linear equations into this particularly simple form: Gaussian elimination. Key to Gaussian elimination are elementary transformations of systems of linear equations, which transform the equation system into a simple form. Then, we can apply the three steps to the simple form that we just discussed in the context of the example in (2.38).

2.3.2 Elementary Transformations

elementary
transformations

Key to solving a system of linear equations are *elementary transformations* that keep the solution set the same, but that transform the equation system into a simpler form:

- Exchange of two equations (rows in the matrix representing the system of equations)
- Multiplication of an equation (row) with a constant $\lambda \in \mathbb{R} \setminus \{0\}$
- Addition of two equations (rows)

Example 2.6

For $a \in \mathbb{R}$, we seek all solutions of the following system of equations:

$$\begin{array}{rrrrrrrrcl} -2x_1 & + & 4x_2 & - & 2x_3 & - & x_4 & + & 4x_5 & = & -3 \\ 4x_1 & - & 8x_2 & + & 3x_3 & - & 3x_4 & + & x_5 & = & 2 \\ x_1 & - & 2x_2 & + & x_3 & - & x_4 & + & x_5 & = & 0 \\ x_1 & - & 2x_2 & & & - & 3x_4 & + & 4x_5 & = & a \end{array} \quad (2.44)$$

We start by converting this system of equations into the compact matrix notation $\mathbf{Ax} = \mathbf{b}$. We no longer mention the variables $\mathbf{x}$ explicitly and build the *augmented matrix* (in the form $[\mathbf{A} | \mathbf{b}]$)

augmented matrix

$$\left[\begin{array}{ccccc|c} -2 & 4 & -2 & -1 & 4 & -3 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ 1 & -2 & 1 & -1 & 1 & 0 \\ 1 & -2 & 0 & -3 & 4 & a \end{array} \right] \begin{array}{l} \text{Swap with } R_3 \\ \\ \text{Swap with } R_1 \end{array}$$

where we used the vertical line to separate the left-hand side from the right-hand side in (2.44). We use $\rightsquigarrow$ to indicate a transformation of the augmented matrix using elementary transformations.

Swapping Rows 1 and 3 leads to

$$\left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ -2 & 4 & -2 & -1 & 4 & -3 \\ 1 & -2 & 0 & -3 & 4 & a \end{array} \right] \begin{array}{l} \\ -4R_1 \\ +2R_1 \\ -R_1 \end{array}$$

The augmented matrix $[\mathbf{A} | \mathbf{b}]$ compactly represents the system of linear equations $\mathbf{Ax} = \mathbf{b}$.

When we now apply the indicated transformations (e.g., subtract Row 1 four times from Row 2), we obtain

$$\begin{array}{l} \left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & -3 & 2 \\ 0 & 0 & 0 & -3 & 6 & -3 \\ 0 & 0 & -1 & -2 & 3 & a \end{array} \right] \begin{array}{l} \\ \\ -R_2 - R_3 \end{array} \\ \rightsquigarrow \left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & -3 & 2 \\ 0 & 0 & 0 & -3 & 6 & -3 \\ 0 & 0 & 0 & 0 & 0 & a+1 \end{array} \right] \begin{array}{l} \\ \cdot(-1) \\ \cdot(-\frac{1}{3}) \end{array} \\ \rightsquigarrow \left[\begin{array}{ccccc|c} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 3 & -2 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 & a+1 \end{array} \right] \end{array}$$

row-echelon form

This (augmented) matrix is in a convenient form, the *row-echelon form* (REF). Reverting this compact notation back into the explicit notation with the variables we seek, we obtain

$$\begin{array}{ccccccccc} x_1 & - & 2x_2 & + & x_3 & - & x_4 & + & x_5 & = & 0 \\ & & & & x_3 & - & x_4 & + & 3x_5 & = & -2 \\ & & & & & & x_4 & - & 2x_5 & = & 1 \\ & & & & & & & & 0 & = & a + 1 \end{array} \quad (2.45)$$

particular solution

Only for $a = -1$ this system can be solved. A *particular solution* is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}. \quad (2.46)$$

general solution

The *general solution*, which captures the set of all possible solutions, is

$$\left\{ \mathbf{x} \in \mathbb{R}^5 : \mathbf{x} = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 0 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R} \right\}. \quad (2.47)$$

In the following, we will detail a constructive way to obtain a particular and general solution of a system of linear equations.

pivot

Remark (Pivots and Staircase Structure). The leading coefficient of a row (first nonzero number from the left) is called the *pivot* and is always strictly to the right of the pivot of the row above it. Therefore, any equation system in row-echelon form always has a “staircase” structure. $\diamond$

row-echelon form

Definition 2.6 (Row-Echelon Form). A matrix is in *row-echelon form* if

- All rows that contain only zeros are at the bottom of the matrix; correspondingly, all rows that contain at least one nonzero element are on top of rows that contain only zeros.
- Looking at nonzero rows only, the first nonzero number from the left (also called the *pivot* or the *leading coefficient*) is always strictly to the right of the pivot of the row above it.

pivot

leading coefficient

In other texts, it is sometimes required that the pivot is 1.

basic variable

free variable

Remark (Basic and Free Variables). The variables corresponding to the pivots in the row-echelon form are called *basic variables* and the other variables are *free variables*. For example, in (2.45), x_1, x_3, x_4 are basic variables, whereas x_2, x_5 are free variables. $\diamond$

Remark (Obtaining a Particular Solution). The row-echelon form makes

our lives easier when we need to determine a particular solution. To do this, we express the right-hand side of the equation system using the pivot columns, such that $\mathbf{b} = \sum_{i=1}^P \lambda_i \mathbf{p}_i$, where $\mathbf{p}_i$, $i = 1, \dots, P$, are the pivot columns. The λ_i are determined easiest if we start with the rightmost pivot column and work our way to the left.

In the previous example, we would try to find $\lambda_1, \lambda_2, \lambda_3$ so that

$$\lambda_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \lambda_3 \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}. \quad (2.48)$$

From here, we find relatively directly that $\lambda_3 = 1, \lambda_2 = -1, \lambda_1 = 2$. When we put everything together, we must not forget the non-pivot columns for which we set the coefficients implicitly to 0. Therefore, we get the particular solution $\mathbf{x} = [2, 0, -1, 1, 0]^\top$. $\diamond$

Remark (Reduced Row Echelon Form). An equation system is in *reduced row-echelon form* (also: *row-reduced echelon form* or *row canonical form*) if

reduced
row-echelon form

- It is in row-echelon form.
- Every pivot is 1.
- The pivot is the only nonzero entry in its column.

$\diamond$

The reduced row-echelon form will play an important role later in Section 2.3.3 because it allows us to determine the general solution of a system of linear equations in a straightforward way.

Remark (Gaussian Elimination). *Gaussian elimination* is an algorithm that performs elementary transformations to bring a system of linear equations into reduced row-echelon form. $\diamond$

Gaussian
elimination

Example 2.7 (Reduced Row Echelon Form)

Verify that the following matrix is in reduced row-echelon form (the pivots are in **bold**):

$$\mathbf{A} = \begin{bmatrix} \mathbf{1} & 3 & 0 & 0 & 3 \\ 0 & 0 & \mathbf{1} & 0 & 9 \\ 0 & 0 & 0 & \mathbf{1} & -4 \end{bmatrix}. \quad (2.49)$$

The key idea for finding the solutions of $\mathbf{Ax} = \mathbf{0}$ is to look at the *non-pivot columns*, which we will need to express as a (linear) combination of the pivot columns. The reduced row echelon form makes this relatively straightforward, and we express the non-pivot columns in terms of sums and multiples of the pivot columns that are on their left: The second column is 3 times the first column (we can ignore the pivot columns on the right of the second column). Therefore, to obtain $\mathbf{0}$, we need to subtract

the second column from three times the first column. Now, we look at the fifth column, which is our second non-pivot column. The fifth column can be expressed as 3 times the first pivot column, 9 times the second pivot column, and -4 times the third pivot column. We need to keep track of the indices of the pivot columns and translate this into 3 times the first column, 0 times the second column (which is a non-pivot column), 9 times the third column (which is our second pivot column), and -4 times the fourth column (which is the third pivot column). Then we need to subtract the fifth column to obtain $\mathbf{0}$. In the end, we are still solving a homogeneous equation system.

To summarize, all solutions of $\mathbf{Ax} = \mathbf{0}$, $\mathbf{x} \in \mathbb{R}^5$ are given by

$$\left\{ \mathbf{x} \in \mathbb{R}^5 : \mathbf{x} = \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 3 \\ 0 \\ 9 \\ -4 \\ -1 \end{bmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R} \right\}. \quad (2.50)$$

2.3.3 The Minus-1 Trick

In the following, we introduce a practical trick for reading out the solutions $\mathbf{x}$ of a homogeneous system of linear equations $\mathbf{Ax} = \mathbf{0}$, where $\mathbf{A} \in \mathbb{R}^{k \times n}$, $\mathbf{x} \in \mathbb{R}^n$.

To start, we assume that $\mathbf{A}$ is in reduced row-echelon form without any rows that just contain zeros, i.e.,

$$\mathbf{A} = \begin{bmatrix} 0 & \cdots & 0 & \mathbf{1} & * & \cdots & * & 0 & * & \cdots & * & 0 & * & \cdots & * \\ \vdots & & \vdots & 0 & 0 & \cdots & 0 & \mathbf{1} & * & \cdots & * & \vdots & \vdots & & \vdots \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & 0 & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & 0 & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \mathbf{1} & * & \cdots & * \end{bmatrix}, \quad (2.51)$$

where $*$ can be an arbitrary real number, with the constraints that the first nonzero entry per row must be 1 and all other entries in the corresponding column must be 0. The columns $j_1, \dots, j_k$ with the pivots (marked in **bold**) are the standard unit vectors $\mathbf{e}_1, \dots, \mathbf{e}_k \in \mathbb{R}^k$. We extend this matrix to an $n \times n$ -matrix $\tilde{\mathbf{A}}$ by adding $n - k$ rows of the form

$$[0 \quad \cdots \quad 0 \quad -1 \quad 0 \quad \cdots \quad 0] \quad (2.52)$$

so that the diagonal of the augmented matrix $\tilde{\mathbf{A}}$ contains either 1 or -1 . Then, the columns of $\tilde{\mathbf{A}}$ that contain the -1 as pivots are solutions of

the homogeneous equation system $\mathbf{A}\mathbf{x} = \mathbf{0}$. To be more precise, these columns form a basis (Section 2.6.1) of the solution space of $\mathbf{A}\mathbf{x} = \mathbf{0}$, which we will later call the *kernel* or *null space* (see Section 2.7.3).

kernel
null space

Example 2.8 (Minus-1 Trick)

Let us revisit the matrix in (2.49), which is already in REF:

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix}. \quad (2.53)$$

We now augment this matrix to a 5×5 matrix by adding rows of the form (2.52) at the places where the pivots on the diagonal are missing and obtain

$$\tilde{\mathbf{A}} = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}. \quad (2.54)$$

From this form, we can immediately read out the solutions of $\mathbf{A}\mathbf{x} = \mathbf{0}$ by taking the columns of $\tilde{\mathbf{A}}$, which contain -1 on the diagonal:

$$\left\{ \mathbf{x} \in \mathbb{R}^5 : \mathbf{x} = \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 3 \\ 0 \\ 9 \\ -4 \\ -1 \end{bmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R} \right\}, \quad (2.55)$$

which is identical to the solution in (2.50) that we obtained by “insight”.

Calculating the Inverse

To compute the inverse $\mathbf{A}^{-1}$ of $\mathbf{A} \in \mathbb{R}^{n \times n}$, we need to find a matrix $\mathbf{X}$ that satisfies $\mathbf{A}\mathbf{X} = \mathbf{I}_n$. Then, $\mathbf{X} = \mathbf{A}^{-1}$. We can write this down as a set of simultaneous linear equations $\mathbf{A}\mathbf{X} = \mathbf{I}_n$, where we solve for $\mathbf{X} = [\mathbf{x}_1 | \cdots | \mathbf{x}_n]$. We use the augmented matrix notation for a compact representation of this set of systems of linear equations and obtain

$$[\mathbf{A} | \mathbf{I}_n] \rightsquigarrow \cdots \rightsquigarrow [\mathbf{I}_n | \mathbf{A}^{-1}]. \quad (2.56)$$

This means that if we bring the augmented equation system into reduced row-echelon form, we can read out the inverse on the right-hand side of the equation system. Hence, determining the inverse of a matrix is equivalent to solving systems of linear equations.

Example 2.9 (Calculating an Inverse Matrix by Gaussian Elimination)

To determine the inverse of

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad (2.57)$$

we write down the augmented matrix

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right]$$

and use Gaussian elimination to bring it into reduced row-echelon form

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -1 & 2 & -2 & 2 \\ 0 & 1 & 0 & 0 & 1 & -1 & 2 & -2 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & -1 & 2 \end{array} \right],$$

such that the desired inverse is given as its right-hand side:

$$\mathbf{A}^{-1} = \begin{bmatrix} -1 & 2 & -2 & 2 \\ 1 & -1 & 2 & -2 \\ 1 & -1 & 1 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}. \quad (2.58)$$

We can verify that (2.58) is indeed the inverse by performing the multiplication $\mathbf{A}\mathbf{A}^{-1}$ and observing that we recover $\mathbf{I}_4$.

2.3.4 Algorithms for Solving a System of Linear Equations

In the following, we briefly discuss approaches to solving a system of linear equations of the form $\mathbf{A}\mathbf{x} = \mathbf{b}$. We make the assumption that a solution exists. Should there be no solution, we need to resort to approximate solutions, which we do not cover in this chapter. One way to solve the approximate problem is using the approach of linear regression, which we discuss in detail in Chapter 9.

In special cases, we may be able to determine the inverse $\mathbf{A}^{-1}$, such that the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is given as $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. However, this is only possible if $\mathbf{A}$ is a square matrix and invertible, which is often not the case. Otherwise, under mild assumptions (i.e., $\mathbf{A}$ needs to have linearly independent columns) we can use the transformation

$$\mathbf{A}\mathbf{x} = \mathbf{b} \iff \mathbf{A}^\top \mathbf{A}\mathbf{x} = \mathbf{A}^\top \mathbf{b} \iff \mathbf{x} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} \quad (2.59)$$

and use the *Moore-Penrose pseudo-inverse* $(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$ to determine the solution (2.59) that solves $\mathbf{A}\mathbf{x} = \mathbf{b}$, which also corresponds to the minimum norm least-squares solution. A disadvantage of this approach is that it requires many computations for the matrix-matrix product and computing the inverse of $\mathbf{A}^\top \mathbf{A}$. Moreover, for reasons of numerical precision it is generally not recommended to compute the inverse or pseudo-inverse. In the following, we therefore briefly discuss alternative approaches to solving systems of linear equations.

Moore-Penrose
pseudo-inverse

Gaussian elimination plays an important role when computing determinants (Section 4.1), checking whether a set of vectors is linearly independent (Section 2.5), computing the inverse of a matrix (Section 2.2.2), computing the rank of a matrix (Section 2.6.2), and determining a basis of a vector space (Section 2.6.1). Gaussian elimination is an intuitive and constructive way to solve a system of linear equations with thousands of variables. However, for systems with millions of variables, it is impractical as the required number of arithmetic operations scales cubically in the number of simultaneous equations.

In practice, systems of many linear equations are solved indirectly, by either stationary iterative methods, such as the Richardson method, the Jacobi method, the Gauß-Seidel method, and the successive over-relaxation method, or Krylov subspace methods, such as conjugate gradients, generalized minimal residual, or biconjugate gradients. We refer to the books by Stoer and Burlirsch (2002), Strang (2003), and Liesen and Mehrmann (2015) for further details.

Let $\mathbf{x}_*$ be a solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$. The key idea of these iterative methods is to set up an iteration of the form

$$\mathbf{x}^{(k+1)} = \mathbf{C}\mathbf{x}^{(k)} + \mathbf{d} \quad (2.60)$$

for suitable $\mathbf{C}$ and $\mathbf{d}$ that reduces the residual error $\|\mathbf{x}^{(k+1)} - \mathbf{x}_*\|$ in every iteration and converges to $\mathbf{x}_*$. We will introduce norms $\|\cdot\|$, which allow us to compute similarities between vectors, in Section 3.1.

2.4 Vector Spaces

Thus far, we have looked at systems of linear equations and how to solve them (Section 2.3). We saw that systems of linear equations can be compactly represented using matrix-vector notation (2.10). In the following, we will have a closer look at vector spaces, i.e., a structured space in which vectors live.

In the beginning of this chapter, we informally characterized vectors as objects that can be added together and multiplied by a scalar, and they remain objects of the same type. Now, we are ready to formalize this, and we will start by introducing the concept of a group, which is a set of elements and an operation defined on these elements that keeps some structure of the set intact.

2.4.1 Groups

Groups play an important role in computer science. Besides providing a fundamental framework for operations on sets, they are heavily used in cryptography, coding theory, and graphics.

Definition 2.7 (Group). Consider a set $\mathcal{G}$ and an operation $\otimes : \mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G}$ defined on $\mathcal{G}$. Then $G := (\mathcal{G}, \otimes)$ is called a *group* if the following hold:

group

closure

associativity

neutral element

inverse element

1. *Closure of $\mathcal{G}$ under $\otimes$:* $\forall x, y \in \mathcal{G} : x \otimes y \in \mathcal{G}$
2. *Associativity:* $\forall x, y, z \in \mathcal{G} : (x \otimes y) \otimes z = x \otimes (y \otimes z)$
3. *Neutral element:* $\exists e \in \mathcal{G} \forall x \in \mathcal{G} : x \otimes e = x$ and $e \otimes x = x$
4. *Inverse element:* $\forall x \in \mathcal{G} \exists y \in \mathcal{G} : x \otimes y = e$ and $y \otimes x = e$, where e is the neutral element. We often write x^{-1} to denote the inverse element of x .

Remark. The inverse element is defined with respect to the operation $\otimes$ and does not necessarily mean $\frac{1}{x}$. $\diamond$

Abelian group

If additionally $\forall x, y \in \mathcal{G} : x \otimes y = y \otimes x$, then $G = (\mathcal{G}, \otimes)$ is an *Abelian group* (commutative).

Example 2.10 (Groups)

Let us have a look at some examples of sets with associated operations and see whether they are groups:

 $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$

- $(\mathbb{Z}, +)$ is an Abelian group.
- $(\mathbb{N}_0, +)$ is not a group: Although $(\mathbb{N}_0, +)$ possesses a neutral element (0), the inverse elements are missing.
- $(\mathbb{Z}, \cdot)$ is not a group: Although $(\mathbb{Z}, \cdot)$ contains a neutral element (1), the inverse elements for any $z \in \mathbb{Z}, z \neq \pm 1$, are missing.
- $(\mathbb{R}, \cdot)$ is not a group since 0 does not possess an inverse element.
- $(\mathbb{R} \setminus \{0\}, \cdot)$ is Abelian.
- $(\mathbb{R}^n, +), (\mathbb{Z}^n, +), n \in \mathbb{N}$ are Abelian if $+$ is defined componentwise, i.e.,

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n). \quad (2.61)$$

Then, $(x_1, \dots, x_n)^{-1} := (-x_1, \dots, -x_n)$ is the inverse element and $e = (0, \dots, 0)$ is the neutral element.

- $(\mathbb{R}^{m \times n}, +)$, the set of $m \times n$ -matrices is Abelian (with componentwise addition as defined in (2.61)).
- Let us have a closer look at $(\mathbb{R}^{n \times n}, \cdot)$, i.e., the set of $n \times n$ -matrices with matrix multiplication as defined in (2.13).
 - Closure and associativity follow directly from the definition of matrix multiplication.
 - Neutral element: The identity matrix $\mathbf{I}_n$ is the neutral element with respect to matrix multiplication “ $\cdot$ ” in $(\mathbb{R}^{n \times n}, \cdot)$.

- Inverse element: If the inverse exists ($\mathbf{A}$ is regular), then $\mathbf{A}^{-1}$ is the inverse element of $\mathbf{A} \in \mathbb{R}^{n \times n}$, and in exactly this case $(\mathbb{R}^{n \times n}, \cdot)$ is a group, called the *general linear group*.

Definition 2.8 (General Linear Group). The set of regular (invertible) matrices $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a group with respect to matrix multiplication as defined in (2.13) and is called *general linear group* $GL(n, \mathbb{R})$. However, since matrix multiplication is not commutative, the group is not Abelian.

general linear group

2.4.2 Vector Spaces

When we discussed groups, we looked at sets $\mathcal{G}$ and inner operations on $\mathcal{G}$, i.e., mappings $\mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G}$ that only operate on elements in $\mathcal{G}$. In the following, we will consider sets that in addition to an inner operation $+$ also contain an outer operation $\cdot$, the multiplication of a vector $\mathbf{x} \in \mathcal{G}$ by a scalar $\lambda \in \mathbb{R}$. We can think of the inner operation as a form of addition, and the outer operation as a form of scaling. Note that the inner/outer operations have nothing to do with inner/outer products.

Definition 2.9 (Vector Space). A real-valued *vector space* $V = (\mathcal{V}, +, \cdot)$ is a set $\mathcal{V}$ with two operations

vector space

$$+ : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V} \quad (2.62)$$

$$\cdot : \mathbb{R} \times \mathcal{V} \rightarrow \mathcal{V} \quad (2.63)$$

where

1. $(\mathcal{V}, +)$ is an Abelian group
2. Distributivity:
 1. $\forall \lambda \in \mathbb{R}, \mathbf{x}, \mathbf{y} \in \mathcal{V} : \lambda \cdot (\mathbf{x} + \mathbf{y}) = \lambda \cdot \mathbf{x} + \lambda \cdot \mathbf{y}$
 2. $\forall \lambda, \psi \in \mathbb{R}, \mathbf{x} \in \mathcal{V} : (\lambda + \psi) \cdot \mathbf{x} = \lambda \cdot \mathbf{x} + \psi \cdot \mathbf{x}$
3. Associativity (outer operation): $\forall \lambda, \psi \in \mathbb{R}, \mathbf{x} \in \mathcal{V} : \lambda \cdot (\psi \cdot \mathbf{x}) = (\lambda\psi) \cdot \mathbf{x}$
4. Neutral element with respect to the outer operation: $\forall \mathbf{x} \in \mathcal{V} : 1 \cdot \mathbf{x} = \mathbf{x}$

The elements $\mathbf{x} \in V$ are called *vectors*. The neutral element of $(\mathcal{V}, +)$ is the zero vector $\mathbf{0} = [0, \dots, 0]^\top$, and the inner operation $+$ is called *vector addition*. The elements $\lambda \in \mathbb{R}$ are called *scalars* and the outer operation $\cdot$ is a *multiplication by scalars*. Note that a scalar product is something different, and we will get to this in Section 3.2.

vector

vector addition

scalar

multiplication by
scalars

Remark. A “vector multiplication” $\mathbf{a}\mathbf{b}$, $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, is not defined. Theoretically, we could define an element-wise multiplication, such that $\mathbf{c} = \mathbf{a}\mathbf{b}$ with $c_j = a_j b_j$. This “array multiplication” is common to many programming languages but makes mathematically limited sense using the standard rules for matrix multiplication: By treating vectors as $n \times 1$ matrices

outer product

(which we usually do), we can use the matrix multiplication as defined in (2.13). However, then the dimensions of the vectors do not match. Only the following multiplications for vectors are defined: $\mathbf{a}\mathbf{b}^\top \in \mathbb{R}^{n \times n}$ (*outer product*), $\mathbf{a}^\top \mathbf{b} \in \mathbb{R}$ (inner/scalar/dot product). $\diamond$

Example 2.11 (Vector Spaces)

Let us have a look at some important examples:

- $\mathcal{V} = \mathbb{R}^n, n \in \mathbb{N}$ is a vector space with operations defined as follows:
 - Addition: $\mathbf{x} + \mathbf{y} = (x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$
 - Multiplication by scalars: $\lambda \mathbf{x} = \lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n)$ for all $\lambda \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^n$
- $\mathcal{V} = \mathbb{R}^{m \times n}, m, n \in \mathbb{N}$ is a vector space with
 - Addition: $\mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1n} + b_{1n} \\ \vdots & & \vdots \\ a_{m1} + b_{m1} & \cdots & a_{mn} + b_{mn} \end{bmatrix}$ is defined elementwise for all $\mathbf{A}, \mathbf{B} \in \mathcal{V}$
 - Multiplication by scalars: $\lambda \mathbf{A} = \begin{bmatrix} \lambda a_{11} & \cdots & \lambda a_{1n} \\ \vdots & & \vdots \\ \lambda a_{m1} & \cdots & \lambda a_{mn} \end{bmatrix}$ as defined in Section 2.2. Remember that $\mathbb{R}^{m \times n}$ is equivalent to $\mathbb{R}^{mn}$.
- $\mathcal{V} = \mathbb{C}$, with the standard definition of addition of complex numbers.

Remark. In the following, we will denote a vector space $(\mathcal{V}, +, \cdot)$ by V when $+$ and $\cdot$ are the standard vector addition and scalar multiplication. Moreover, we will use the notation $\mathbf{x} \in V$ for vectors in $\mathcal{V}$ to simplify notation. $\diamond$

column vector

Remark. The vector spaces $\mathbb{R}^n, \mathbb{R}^{n \times 1}, \mathbb{R}^{1 \times n}$ are only different in the way we write vectors. In the following, we will not make a distinction between $\mathbb{R}^n$ and $\mathbb{R}^{n \times 1}$, which allows us to write n -tuples as *column vectors*

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}. \quad (2.64)$$

row vector

transpose

This simplifies the notation regarding vector space operations. However, we do distinguish between $\mathbb{R}^{n \times 1}$ and $\mathbb{R}^{1 \times n}$ (the *row vectors*) to avoid confusion with matrix multiplication. By default, we write $\mathbf{x}$ to denote a column vector, and a row vector is denoted by $\mathbf{x}^\top$, the *transpose* of $\mathbf{x}$. $\diamond$

2.4.3 Vector Subspaces

In the following, we will introduce vector subspaces. Intuitively, they are sets contained in the original vector space with the property that when we perform vector space operations on elements within this subspace, we will never leave it. In this sense, they are “closed”. Vector subspaces are a key idea in machine learning. For example, Chapter 10 demonstrates how to use vector subspaces for dimensionality reduction.

Definition 2.10 (Vector Subspace). Let $V = (\mathcal{V}, +, \cdot)$ be a vector space and $\mathcal{U} \subseteq \mathcal{V}$, $\mathcal{U} \neq \emptyset$. Then $U = (\mathcal{U}, +, \cdot)$ is called *vector subspace* of V (or *linear subspace*) if U is a vector space with the vector space operations $+$ and $\cdot$ restricted to $\mathcal{U} \times \mathcal{U}$ and $\mathbb{R} \times \mathcal{U}$. We write $U \subseteq V$ to denote a subspace U of V .

vector subspace
linear subspace

If $\mathcal{U} \subseteq \mathcal{V}$ and V is a vector space, then U naturally inherits many properties directly from V because they hold for all $\mathbf{x} \in \mathcal{V}$, and in particular for all $\mathbf{x} \in \mathcal{U} \subseteq \mathcal{V}$. This includes the Abelian group properties, the distributivity, the associativity and the neutral element. To determine whether $(\mathcal{U}, +, \cdot)$ is a subspace of V we still do need to show

1. $\mathcal{U} \neq \emptyset$, in particular: $\mathbf{0} \in \mathcal{U}$
2. Closure of U :
 - a. With respect to the outer operation: $\forall \lambda \in \mathbb{R} \forall \mathbf{x} \in \mathcal{U} : \lambda \mathbf{x} \in \mathcal{U}$.
 - b. With respect to the inner operation: $\forall \mathbf{x}, \mathbf{y} \in \mathcal{U} : \mathbf{x} + \mathbf{y} \in \mathcal{U}$.

Example 2.12 (Vector Subspaces)

Let us have a look at some examples:

- For every vector space V , the trivial subspaces are V itself and $\{\mathbf{0}\}$.
- Only example D in Figure 2.6 is a subspace of $\mathbb{R}^2$ (with the usual inner/outer operations). In A and C , the closure property is violated; B does not contain $\mathbf{0}$.
- The solution set of a homogeneous system of linear equations $A\mathbf{x} = \mathbf{0}$ with n unknowns $\mathbf{x} = [x_1, \dots, x_n]^\top$ is a subspace of $\mathbb{R}^n$.
- The solution of an inhomogeneous system of linear equations $A\mathbf{x} = \mathbf{b}$, $\mathbf{b} \neq \mathbf{0}$ is not a subspace of $\mathbb{R}^n$.
- The intersection of arbitrarily many subspaces is a subspace itself.

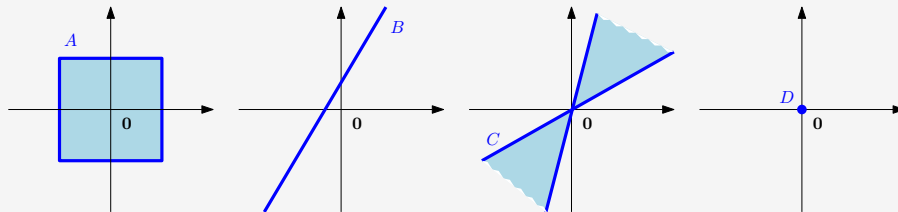


Figure 2.6 Not all subsets of $\mathbb{R}^2$ are subspaces. In A and C , the closure property is violated; B does not contain $\mathbf{0}$. Only D is a subspace.

Remark. Every subspace $U \subseteq (\mathbb{R}^n, +, \cdot)$ is the solution space of a homogeneous system of linear equations $A\mathbf{x} = \mathbf{0}$ for $\mathbf{x} \in \mathbb{R}^n$. $\diamond$

2.5 Linear Independence

In the following, we will have a close look at what we can do with vectors (elements of the vector space). In particular, we can add vectors together and multiply them with scalars. The closure property guarantees that we end up with another vector in the same vector space. It is possible to find a set of vectors with which we can represent every vector in the vector space by adding them together and scaling them. This set of vectors is a *basis*, and we will discuss them in Section 2.6.1. Before we get there, we will need to introduce the concepts of linear combinations and linear independence.

Definition 2.11 (Linear Combination). Consider a vector space V and a finite number of vectors $\mathbf{x}_1, \dots, \mathbf{x}_k \in V$. Then, every $\mathbf{v} \in V$ of the form

$$\mathbf{v} = \lambda_1 \mathbf{x}_1 + \dots + \lambda_k \mathbf{x}_k = \sum_{i=1}^k \lambda_i \mathbf{x}_i \in V \quad (2.65)$$

linear combination

with $\lambda_1, \dots, \lambda_k \in \mathbb{R}$ is a *linear combination* of the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$.

The $\mathbf{0}$ -vector can always be written as the linear combination of k vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ because $\mathbf{0} = \sum_{i=1}^k 0\mathbf{x}_i$ is always true. In the following, we are interested in non-trivial linear combinations of a set of vectors to represent $\mathbf{0}$, i.e., linear combinations of vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$, where not all coefficients λ_i in (2.65) are 0.

Definition 2.12 (Linear (In)dependence). Let us consider a vector space V with $k \in \mathbb{N}$ and $\mathbf{x}_1, \dots, \mathbf{x}_k \in V$. If there is a non-trivial linear combination, such that $\mathbf{0} = \sum_{i=1}^k \lambda_i \mathbf{x}_i$ with at least one $\lambda_i \neq 0$, the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are *linearly dependent*. If only the trivial solution exists, i.e., $\lambda_1 = \dots = \lambda_k = 0$ the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are *linearly independent*.

linearly dependent
linearly
independent

Linear independence is one of the most important concepts in linear algebra. Intuitively, a set of linearly independent vectors consists of vectors that have no redundancy, i.e., if we remove any of those vectors from the set, we will lose something. Throughout the next sections, we will formalize this intuition more.

Example 2.13 (Linearly Dependent Vectors)

A geographic example may help to clarify the concept of linear independence. A person in Nairobi (Kenya) describing where Kigali (Rwanda) is might say, “You can get to Kigali by first going 506 km Northwest to Kampala (Uganda) and then 374 km Southwest.” This is sufficient information

to describe the location of Kigali because the geographic coordinate system may be considered a two-dimensional vector space (ignoring altitude and the Earth's curved surface). The person may add, “It is about 751 km West of here.” Although this last statement is true, it is not necessary to find Kigali given the previous information (see Figure 2.7 for an illustration). In this example, the “506 km Northwest” vector (blue) and the “374 km Southwest” vector (purple) are linearly independent. This means the Southwest vector cannot be described in terms of the Northwest vector, and vice versa. However, the third “751 km West” vector (black) is a linear combination of the other two vectors, and it makes the set of vectors linearly dependent. Equivalently, given “751 km West” and “374 km Southwest” can be linearly combined to obtain “506 km Northwest”.

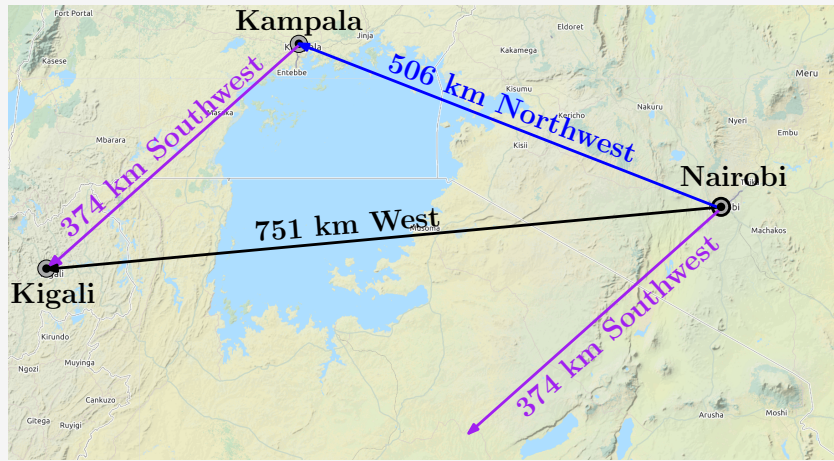


Figure 2.7
Geographic example
(with crude
approximations to
cardinal directions)
of linearly
dependent vectors
in a
two-dimensional
space (plane).

Remark. The following properties are useful to find out whether vectors are linearly independent:

- k vectors are either linearly dependent or linearly independent. There is no third option.
- If at least one of the vectors $x_1, \dots, x_k$ is $\mathbf{0}$ then they are linearly dependent. The same holds if two vectors are identical.
- The vectors $\{x_1, \dots, x_k : x_i \neq \mathbf{0}, i = 1, \dots, k\}$, $k \geq 2$, are linearly dependent if and only if (at least) one of them is a linear combination of the others. In particular, if one vector is a multiple of another vector, i.e., $x_i = \lambda x_j$, $\lambda \in \mathbb{R}$ then the set $\{x_1, \dots, x_k : x_i \neq \mathbf{0}, i = 1, \dots, k\}$ is linearly dependent.
- A practical way of checking whether vectors $x_1, \dots, x_k \in V$ are linearly independent is to use Gaussian elimination: Write all vectors as columns of a matrix A and perform Gaussian elimination until the matrix is in row echelon form (the reduced row-echelon form is unnecessary here):

- The pivot columns indicate the vectors, which are linearly independent of the vectors on the left. Note that there is an ordering of vectors when the matrix is built.
- The non-pivot columns can be expressed as linear combinations of the pivot columns on their left. For instance, the row-echelon form

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad (2.66)$$

tells us that the first and third columns are pivot columns. The second column is a non-pivot column because it is three times the first column.

All column vectors are linearly independent if and only if all columns are pivot columns. If there is at least one non-pivot column, the columns (and, therefore, the corresponding vectors) are linearly dependent.

◇

Example 2.14

Consider $\mathbb{R}^4$ with

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} -1 \\ -2 \\ 1 \\ 1 \end{bmatrix}. \quad (2.67)$$

To check whether they are linearly dependent, we follow the general approach and solve

$$\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \lambda_3 \mathbf{x}_3 = \lambda_1 \begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix} + \lambda_3 \begin{bmatrix} -1 \\ -2 \\ 1 \\ 1 \end{bmatrix} = \mathbf{0} \quad (2.68)$$

for $\lambda_1, \dots, \lambda_3$. We write the vectors $\mathbf{x}_i$, $i = 1, 2, 3$, as the columns of a matrix and apply elementary row operations until we identify the pivot columns:

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & -2 \\ -3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix} \rightsquigarrow \dots \rightsquigarrow \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}. \quad (2.69)$$

Here, every column of the matrix is a pivot column. Therefore, there is no non-trivial solution, and we require $\lambda_1 = 0, \lambda_2 = 0, \lambda_3 = 0$ to solve the equation system. Hence, the vectors $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly independent.

Remark. Consider a vector space V with k linearly independent vectors $\mathbf{b}_1, \dots, \mathbf{b}_k$ and m linear combinations

$$\begin{aligned} \mathbf{x}_1 &= \sum_{i=1}^k \lambda_{i1} \mathbf{b}_i, \\ &\vdots \\ \mathbf{x}_m &= \sum_{i=1}^k \lambda_{im} \mathbf{b}_i. \end{aligned} \quad (2.70)$$

Defining $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_k]$ as the matrix whose columns are the linearly independent vectors $\mathbf{b}_1, \dots, \mathbf{b}_k$, we can write

$$\mathbf{x}_j = \mathbf{B} \boldsymbol{\lambda}_j, \quad \boldsymbol{\lambda}_j = \begin{bmatrix} \lambda_{1j} \\ \vdots \\ \lambda_{kj} \end{bmatrix}, \quad j = 1, \dots, m, \quad (2.71)$$

in a more compact form.

We want to test whether $\mathbf{x}_1, \dots, \mathbf{x}_m$ are linearly independent. For this purpose, we follow the general approach of testing when $\sum_{j=1}^m \psi_j \mathbf{x}_j = \mathbf{0}$. With (2.71), we obtain

$$\sum_{j=1}^m \psi_j \mathbf{x}_j = \sum_{j=1}^m \psi_j \mathbf{B} \boldsymbol{\lambda}_j = \mathbf{B} \sum_{j=1}^m \psi_j \boldsymbol{\lambda}_j. \quad (2.72)$$

This means that $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ are linearly independent if and only if the column vectors $\{\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_m\}$ are linearly independent. $\diamond$

Remark. In a vector space V , m linear combinations of k vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linearly dependent if $m > k$. $\diamond$

Example 2.15

Consider a set of linearly independent vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \mathbf{b}_4 \in \mathbb{R}^n$ and

$$\begin{aligned} \mathbf{x}_1 &= \mathbf{b}_1 - 2\mathbf{b}_2 + \mathbf{b}_3 - \mathbf{b}_4 \\ \mathbf{x}_2 &= -4\mathbf{b}_1 - 2\mathbf{b}_2 + 4\mathbf{b}_4 \\ \mathbf{x}_3 &= 2\mathbf{b}_1 + 3\mathbf{b}_2 - \mathbf{b}_3 - 3\mathbf{b}_4 \\ \mathbf{x}_4 &= 17\mathbf{b}_1 - 10\mathbf{b}_2 + 11\mathbf{b}_3 + \mathbf{b}_4 \end{aligned} \quad (2.73)$$

Are the vectors $\mathbf{x}_1, \dots, \mathbf{x}_4 \in \mathbb{R}^n$ linearly independent? To answer this question, we investigate whether the column vectors

$$\left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -4 \\ -2 \\ 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ -1 \\ -3 \end{bmatrix}, \begin{bmatrix} 17 \\ -10 \\ 11 \\ 1 \end{bmatrix} \right\} \quad (2.74)$$

are linearly independent. The reduced row-echelon form of the corresponding linear equation system with coefficient matrix

$$\mathbf{A} = \begin{bmatrix} 1 & -4 & 2 & 17 \\ -2 & -2 & 3 & -10 \\ 1 & 0 & -1 & 11 \\ -1 & 4 & -3 & 1 \end{bmatrix} \quad (2.75)$$

is given as

$$\begin{bmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -15 \\ 0 & 0 & 1 & -18 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (2.76)$$

We see that the corresponding linear equation system is non-trivially solvable: The last column is not a pivot column, and $x_4 = -7x_1 - 15x_2 - 18x_3$. Therefore, $x_1, \dots, x_4$ are linearly dependent as x_4 can be expressed as a linear combination of $x_1, \dots, x_3$.

2.6 Basis and Rank

In a vector space V , we are particularly interested in sets of vectors $\mathcal{A}$ that possess the property that any vector $v \in V$ can be obtained by a linear combination of vectors in $\mathcal{A}$. These vectors are special vectors, and in the following, we will characterize them.

2.6.1 Generating Set and Basis

generating set
span

Definition 2.13 (Generating Set and Span). Consider a vector space $V = (\mathcal{V}, +, \cdot)$ and set of vectors $\mathcal{A} = \{x_1, \dots, x_k\} \subseteq \mathcal{V}$. If every vector $v \in \mathcal{V}$ can be expressed as a linear combination of $x_1, \dots, x_k$, $\mathcal{A}$ is called a *generating set* of V . The set of all linear combinations of vectors in $\mathcal{A}$ is called the *span* of $\mathcal{A}$. If $\mathcal{A}$ spans the vector space V , we write $V = \text{span}[\mathcal{A}]$ or $V = \text{span}[x_1, \dots, x_k]$.

Generating sets are sets of vectors that span vector (sub)spaces, i.e., every vector can be represented as a linear combination of the vectors in the generating set. Now, we will be more specific and characterize the smallest generating set that spans a vector (sub)space.

minimal

basis

Definition 2.14 (Basis). Consider a vector space $V = (\mathcal{V}, +, \cdot)$ and $\mathcal{A} \subseteq \mathcal{V}$. A generating set $\mathcal{A}$ of V is called *minimal* if there exists no smaller set $\tilde{\mathcal{A}} \subsetneq \mathcal{A} \subseteq \mathcal{V}$ that spans V . Every linearly independent generating set of V is minimal and is called a *basis* of V .

Let $V = (\mathcal{V}, +, \cdot)$ be a vector space and $\mathcal{B} \subseteq \mathcal{V}, \mathcal{B} \neq \emptyset$. Then, the following statements are equivalent:

- $\mathcal{B}$ is a basis of V .
- $\mathcal{B}$ is a minimal generating set.
- $\mathcal{B}$ is a maximal linearly independent set of vectors in V , i.e., adding any other vector to this set will make it linearly dependent.
- Every vector $x \in V$ is a linear combination of vectors from $\mathcal{B}$, and every linear combination is unique, i.e., with

$$x = \sum_{i=1}^k \lambda_i b_i = \sum_{i=1}^k \psi_i b_i \quad (2.77)$$

and $\lambda_i, \psi_i \in \mathbb{R}, b_i \in \mathcal{B}$ it follows that $\lambda_i = \psi_i, i = 1, \dots, k$.

A basis is a minimal generating set and a maximal linearly independent set of vectors.

Example 2.16

- In $\mathbb{R}^3$, the *canonical/standard basis* is

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}. \quad (2.78)$$

- Different bases in $\mathbb{R}^3$ are

$$\mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}, \mathcal{B}_2 = \left\{ \begin{bmatrix} 0.5 \\ 0.8 \\ 0.4 \end{bmatrix}, \begin{bmatrix} 1.8 \\ 0.3 \\ 0.3 \end{bmatrix}, \begin{bmatrix} -2.2 \\ -1.3 \\ 3.5 \end{bmatrix} \right\}. \quad (2.79)$$

- The set

$$\mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ -4 \end{bmatrix} \right\} \quad (2.80)$$

is linearly independent, but not a generating set (and no basis) of $\mathbb{R}^4$: For instance, the vector $[1, 0, 0, 0]^\top$ cannot be obtained by a linear combination of elements in $\mathcal{A}$.

canonical basis

Remark. Every vector space V possesses a basis $\mathcal{B}$. The preceding examples show that there can be many bases of a vector space V , i.e., there is no unique basis. However, all bases possess the same number of elements, the *basis vectors*.



basis vector

We only consider finite-dimensional vector spaces V . In this case, the *dimension* of V is the number of basis vectors of V , and we write $\dim(V)$. If $U \subseteq V$ is a subspace of V , then $\dim(U) \leq \dim(V)$ and $\dim(U) =$

dimension

The dimension of a vector space corresponds to the number of its basis vectors.

$\dim(V)$ if and only if $U = V$. Intuitively, the dimension of a vector space can be thought of as the number of independent directions in this vector space.

Remark. The dimension of a vector space is not necessarily the number of elements in a vector. For instance, the vector space $V = \text{span}\left[\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right]$ is one-dimensional, although the basis vector possesses two elements. $\diamond$

Remark. A basis of a subspace $U = \text{span}[\mathbf{x}_1, \dots, \mathbf{x}_m] \subseteq \mathbb{R}^n$ can be found by executing the following steps:

1. Write the spanning vectors as columns of a matrix $\mathbf{A}$
2. Determine the row-echelon form of $\mathbf{A}$.
3. The spanning vectors associated with the pivot columns are a basis of U .

$\diamond$

Example 2.17 (Determining a Basis)

For a vector subspace $U \subseteq \mathbb{R}^5$, spanned by the vectors

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ -1 \\ -1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 3 \\ -4 \\ 3 \\ 5 \\ -3 \end{bmatrix}, \quad \mathbf{x}_4 = \begin{bmatrix} -1 \\ 8 \\ -5 \\ -6 \\ 1 \end{bmatrix} \in \mathbb{R}^5, \quad (2.81)$$

we are interested in finding out which vectors $\mathbf{x}_1, \dots, \mathbf{x}_4$ are a basis for U . For this, we need to check whether $\mathbf{x}_1, \dots, \mathbf{x}_4$ are linearly independent. Therefore, we need to solve

$$\sum_{i=1}^4 \lambda_i \mathbf{x}_i = \mathbf{0}, \quad (2.82)$$

which leads to a homogeneous system of equations with matrix

$$[\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4] = \begin{bmatrix} 1 & 2 & 3 & -1 \\ 2 & -1 & -4 & 8 \\ -1 & 1 & 3 & -5 \\ -1 & 2 & 5 & -6 \\ -1 & -2 & -3 & 1 \end{bmatrix}. \quad (2.83)$$

With the basic transformation rules for systems of linear equations, we obtain the row-echelon form

$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ 2 & -1 & -4 & 8 \\ -1 & 1 & 3 & -5 \\ -1 & 2 & 5 & -6 \\ -1 & -2 & -3 & 1 \end{bmatrix} \rightsquigarrow \dots \rightsquigarrow \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Since the pivot columns indicate which set of vectors is linearly independent, we see from the row-echelon form that $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_4$ are linearly independent (because the system of linear equations $\lambda_1 \mathbf{x}_1 + \lambda_2 \mathbf{x}_2 + \lambda_4 \mathbf{x}_4 = \mathbf{0}$ can only be solved with $\lambda_1 = \lambda_2 = \lambda_4 = 0$). Therefore, $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_4\}$ is a basis of U .

2.6.2 Rank

The number of linearly independent columns of a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ equals the number of linearly independent rows and is called the *rank* of $\mathbf{A}$ and is denoted by $\text{rk}(\mathbf{A})$.

Remark. The rank of a matrix has some important properties:

- $\text{rk}(\mathbf{A}) = \text{rk}(\mathbf{A}^\top)$, i.e., the column rank equals the row rank.
- The columns of $\mathbf{A} \in \mathbb{R}^{m \times n}$ span a subspace $U \subseteq \mathbb{R}^m$ with $\dim(U) = \text{rk}(\mathbf{A})$. Later we will call this subspace the *image* or *range*. A basis of U can be found by applying Gaussian elimination to $\mathbf{A}$ to identify the pivot columns.
- The rows of $\mathbf{A} \in \mathbb{R}^{m \times n}$ span a subspace $W \subseteq \mathbb{R}^n$ with $\dim(W) = \text{rk}(\mathbf{A})$. A basis of W can be found by applying Gaussian elimination to $\mathbf{A}^\top$.
- For all $\mathbf{A} \in \mathbb{R}^{n \times n}$ it holds that $\mathbf{A}$ is regular (invertible) if and only if $\text{rk}(\mathbf{A}) = n$.
- For all $\mathbf{A} \in \mathbb{R}^{m \times n}$ and all $\mathbf{b} \in \mathbb{R}^m$ it holds that the linear equation system $\mathbf{A}\mathbf{x} = \mathbf{b}$ can be solved if and only if $\text{rk}(\mathbf{A}) = \text{rk}(\mathbf{A}|\mathbf{b})$, where $\mathbf{A}|\mathbf{b}$ denotes the augmented system.
- For $\mathbf{A} \in \mathbb{R}^{m \times n}$ the subspace of solutions for $\mathbf{A}\mathbf{x} = \mathbf{0}$ possesses dimension $n - \text{rk}(\mathbf{A})$. Later, we will call this subspace the *kernel* or the *null space*.
- A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ has *full rank* if its rank equals the largest possible rank for a matrix of the same dimensions. This means that the rank of a full-rank matrix is the lesser of the number of rows and columns, i.e., $\text{rk}(\mathbf{A}) = \min(m, n)$. A matrix is said to be *rank deficient* if it does not have full rank.



Example 2.18 (Rank)

▪ $\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$

$\mathbf{A}$ has two linearly independent rows/columns so that $\text{rk}(\mathbf{A}) = 2$.

$$\blacksquare \mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix}.$$

We use Gaussian elimination to determine the rank:

$$\begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix} \rightsquigarrow \dots \rightsquigarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}. \quad (2.84)$$

Here, we see that the number of linearly independent rows and columns is 2, such that $\text{rk}(\mathbf{A}) = 2$.

2.7 Linear Mappings

In the following, we will study mappings on vector spaces that preserve their structure, which will allow us to define the concept of a coordinate. In the beginning of the chapter, we said that vectors are objects that can be added together and multiplied by a scalar, and the resulting object is still a vector. We wish to preserve this property when applying the mapping: Consider two real vector spaces V, W . A mapping $\Phi : V \rightarrow W$ preserves the structure of the vector space if

$$\Phi(\mathbf{x} + \mathbf{y}) = \Phi(\mathbf{x}) + \Phi(\mathbf{y}) \quad (2.85)$$

$$\Phi(\lambda \mathbf{x}) = \lambda \Phi(\mathbf{x}) \quad (2.86)$$

for all $\mathbf{x}, \mathbf{y} \in V$ and $\lambda \in \mathbb{R}$. We can summarize this in the following definition:

Definition 2.15 (Linear Mapping). For vector spaces V, W , a mapping $\Phi : V \rightarrow W$ is called a *linear mapping* (or *vector space homomorphism/linear transformation*) if

$$\forall \mathbf{x}, \mathbf{y} \in V \forall \lambda, \psi \in \mathbb{R} : \Phi(\lambda \mathbf{x} + \psi \mathbf{y}) = \lambda \Phi(\mathbf{x}) + \psi \Phi(\mathbf{y}). \quad (2.87)$$

It turns out that we can represent linear mappings as matrices (Section 2.7.1). Recall that we can also collect a set of vectors as columns of a matrix. When working with matrices, we have to keep in mind what the matrix represents: a linear mapping or a collection of vectors. We will see more about linear mappings in Chapter 4. Before we continue, we will briefly introduce special mappings.

Definition 2.16 (Injective, Surjective, Bijective). Consider a mapping $\Phi : \mathcal{V} \rightarrow \mathcal{W}$, where $\mathcal{V}, \mathcal{W}$ can be arbitrary sets. Then Φ is called

- *Injective* if $\forall \mathbf{x}, \mathbf{y} \in \mathcal{V} : \Phi(\mathbf{x}) = \Phi(\mathbf{y}) \implies \mathbf{x} = \mathbf{y}$.
- *Surjective* if $\Phi(\mathcal{V}) = \mathcal{W}$.
- *Bijective* if it is injective and surjective.

linear mapping
vector space
homomorphism
linear
transformation

injective
surjective
bijective

If Φ is surjective, then every element in $\mathcal{W}$ can be “reached” from $\mathcal{V}$ using Φ . A bijective Φ can be “undone”, i.e., there exists a mapping $\Psi : \mathcal{W} \rightarrow \mathcal{V}$ so that $\Psi \circ \Phi(x) = x$. This mapping Ψ is then called the inverse of Φ and normally denoted by Φ^{-1} .

With these definitions, we introduce the following special cases of linear mappings between vector spaces V and W :

- *Isomorphism*: $\Phi : V \rightarrow W$ linear and bijective
- *Endomorphism*: $\Phi : V \rightarrow V$ linear
- *Automorphism*: $\Phi : V \rightarrow V$ linear and bijective
- We define $\text{id}_V : V \rightarrow V, x \mapsto x$ as the *identity mapping* or *identity automorphism* in V .

isomorphism

endomorphism

automorphism

identity mapping

identity

automorphism

Example 2.19 (Homomorphism)

The mapping $\Phi : \mathbb{R}^2 \rightarrow \mathbb{C}, \Phi(x) = x_1 + ix_2$, is a homomorphism:

$$\begin{aligned} \Phi \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right) &= (x_1 + y_1) + i(x_2 + y_2) = x_1 + ix_2 + y_1 + iy_2 \\ &= \Phi \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) + \Phi \left(\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \right) \\ \Phi \left(\lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) &= \lambda x_1 + \lambda i x_2 = \lambda(x_1 + ix_2) = \lambda \Phi \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right). \end{aligned} \quad (2.88)$$

This also justifies why complex numbers can be represented as tuples in $\mathbb{R}^2$: There is a bijective linear mapping that converts the elementwise addition of tuples in $\mathbb{R}^2$ into the set of complex numbers with the corresponding addition. Note that we only showed linearity, but not the bijection.

Theorem 2.17 (Theorem 3.59 in Axler (2015)). *Finite-dimensional vector spaces V and W are isomorphic if and only if $\dim(V) = \dim(W)$.*

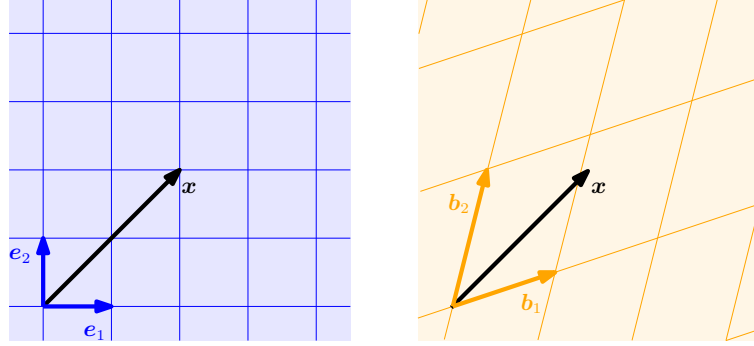
Theorem 2.17 states that there exists a linear, bijective mapping between two vector spaces of the same dimension. Intuitively, this means that vector spaces of the same dimension are kind of the same thing, as they can be transformed into each other without incurring any loss.

Theorem 2.17 also gives us the justification to treat $\mathbb{R}^{m \times n}$ (the vector space of $m \times n$ -matrices) and $\mathbb{R}^{mn}$ (the vector space of vectors of length mn) the same, as their dimensions are mn , and there exists a linear, bijective mapping that transforms one into the other.

Remark. Consider vector spaces V, W, X . Then:

- For linear mappings $\Phi : V \rightarrow W$ and $\Psi : W \rightarrow X$, the mapping $\Psi \circ \Phi : V \rightarrow X$ is also linear.
- If $\Phi : V \rightarrow W$ is an isomorphism, then $\Phi^{-1} : W \rightarrow V$ is an isomorphism, too.

Figure 2.8 Two different coordinate systems defined by two sets of basis vectors. A vector x has different coordinate representations depending on which coordinate system is chosen.



- If $\Phi : V \rightarrow W$, $\Psi : V \rightarrow W$ are linear, then $\Phi + \Psi$ and $\lambda\Phi$, $\lambda \in \mathbb{R}$, are linear, too.

◇

2.7.1 Matrix Representation of Linear Mappings

Any n -dimensional vector space is isomorphic to $\mathbb{R}^n$ (Theorem 2.17). We consider a basis $\{b_1, \dots, b_n\}$ of an n -dimensional vector space V . In the following, the order of the basis vectors will be important. Therefore, we write

$$B = (b_1, \dots, b_n) \quad (2.89)$$

ordered basis

and call this n -tuple an *ordered basis* of V .

Remark (Notation). We are at the point where notation gets a bit tricky. Therefore, we summarize some parts here. $B = (b_1, \dots, b_n)$ is an ordered basis, $\mathcal{B} = \{b_1, \dots, b_n\}$ is an (unordered) basis, and $B = [b_1, \dots, b_n]$ is a matrix whose columns are the vectors $b_1, \dots, b_n$. ◇

Definition 2.18 (Coordinates). Consider a vector space V and an ordered basis $B = (b_1, \dots, b_n)$ of V . For any $x \in V$ we obtain a unique representation (linear combination)

$$x = \alpha_1 b_1 + \dots + \alpha_n b_n \quad (2.90)$$

coordinate

of x with respect to B . Then $\alpha_1, \dots, \alpha_n$ are the *coordinates* of x with respect to B , and the vector

$$\alpha = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \in \mathbb{R}^n \quad (2.91)$$

coordinate vector
coordinate
representation

is the *coordinate vector/coordinate representation* of x with respect to the ordered basis B .

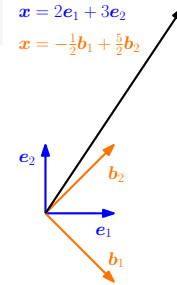
A basis effectively defines a coordinate system. We are familiar with the Cartesian coordinate system in two dimensions, which is spanned by the canonical basis vectors e_1, e_2 . In this coordinate system, a vector $x \in \mathbb{R}^2$ has a representation that tells us how to linearly combine e_1 and e_2 to obtain x . However, any basis of $\mathbb{R}^2$ defines a valid coordinate system, and the same vector x from before may have a different coordinate representation in the (b_1, b_2) basis. In Figure 2.8, the coordinates of x with respect to the standard basis (e_1, e_2) is $[2, 2]^\top$. However, with respect to the basis (b_1, b_2) the same vector x is represented as $[1.09, 0.72]^\top$, i.e., $x = 1.09b_1 + 0.72b_2$. In the following sections, we will discover how to obtain this representation.

Example 2.20

Let us have a look at a geometric vector $x \in \mathbb{R}^2$ with coordinates $[2, 3]^\top$ with respect to the standard basis (e_1, e_2) of $\mathbb{R}^2$. This means, we can write $x = 2e_1 + 3e_2$. However, we do not have to choose the standard basis to represent this vector. If we use the basis vectors $b_1 = [1, -1]^\top, b_2 = [1, 1]^\top$ we will obtain the coordinates $\frac{1}{2}[-1, 5]^\top$ to represent the same vector with respect to (b_1, b_2) (see Figure 2.9).

Remark. For an n -dimensional vector space V and an ordered basis B of V , the mapping $\Phi : \mathbb{R}^n \rightarrow V, \Phi(e_i) = b_i, i = 1, \dots, n$, is linear (and because of Theorem 2.17 an isomorphism), where $(e_1, \dots, e_n)$ is the standard basis of $\mathbb{R}^n$.

Figure 2.9
Different coordinate representations of a vector x , depending on the choice of basis.



Now we are ready to make an explicit connection between matrices and linear mappings between finite-dimensional vector spaces.

Definition 2.19 (Transformation Matrix). Consider vector spaces V, W with corresponding (ordered) bases $B = (b_1, \dots, b_n)$ and $C = (c_1, \dots, c_m)$. Moreover, we consider a linear mapping $\Phi : V \rightarrow W$. For $j \in \{1, \dots, n\}$,

$$\Phi(b_j) = \alpha_{1j}c_1 + \dots + \alpha_{mj}c_m = \sum_{i=1}^m \alpha_{ij}c_i \quad (2.92)$$

is the unique representation of $\Phi(b_j)$ with respect to C . Then, we call the $m \times n$ -matrix A_Φ , whose elements are given by

$$A_\Phi(i, j) = \alpha_{ij}, \quad (2.93)$$

the *transformation matrix* of Φ (with respect to the ordered bases B of V and C of W).

transformation
matrix

The coordinates of $\Phi(b_j)$ with respect to the ordered basis C of W are the j -th column of A_Φ . Consider (finite-dimensional) vector spaces V, W with ordered bases B, C and a linear mapping $\Phi : V \rightarrow W$ with

transformation matrix $\mathbf{A}_\Phi$. If $\hat{\mathbf{x}}$ is the coordinate vector of $\mathbf{x} \in V$ with respect to B and $\hat{\mathbf{y}}$ the coordinate vector of $\mathbf{y} = \Phi(\mathbf{x}) \in W$ with respect to C , then

$$\hat{\mathbf{y}} = \mathbf{A}_\Phi \hat{\mathbf{x}}. \quad (2.94)$$

This means that the transformation matrix can be used to map coordinates with respect to an ordered basis in V to coordinates with respect to an ordered basis in W .

Example 2.21 (Transformation Matrix)

Consider a homomorphism $\Phi : V \rightarrow W$ and ordered bases $B = (\mathbf{b}_1, \dots, \mathbf{b}_3)$ of V and $C = (\mathbf{c}_1, \dots, \mathbf{c}_4)$ of W . With

$$\begin{aligned} \Phi(\mathbf{b}_1) &= \mathbf{c}_1 - \mathbf{c}_2 + 3\mathbf{c}_3 - \mathbf{c}_4 \\ \Phi(\mathbf{b}_2) &= 2\mathbf{c}_1 + \mathbf{c}_2 + 7\mathbf{c}_3 + 2\mathbf{c}_4 \\ \Phi(\mathbf{b}_3) &= 3\mathbf{c}_2 + \mathbf{c}_3 + 4\mathbf{c}_4 \end{aligned} \quad (2.95)$$

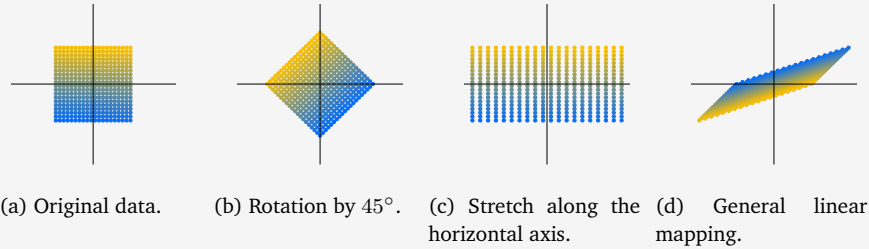
the transformation matrix $\mathbf{A}_\Phi$ with respect to B and C satisfies $\Phi(\mathbf{b}_k) = \sum_{i=1}^4 \alpha_{ik} \mathbf{c}_i$ for $k = 1, \dots, 3$ and is given as

$$\mathbf{A}_\Phi = [\alpha_1, \alpha_2, \alpha_3] = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \\ 3 & 7 & 1 \\ -1 & 2 & 4 \end{bmatrix}, \quad (2.96)$$

where the α_j , $j = 1, 2, 3$, are the coordinate vectors of $\Phi(\mathbf{b}_j)$ with respect to C .

Example 2.22 (Linear Transformations of Vectors)

Figure 2.10 Three examples of linear transformations of the vectors shown as dots in (a); (b) Rotation by 45° ; (c) Stretching of the horizontal coordinates by 2; (d) Combination of reflection, rotation and stretching.



We consider three linear transformations of a set of vectors in $\mathbb{R}^2$ with the transformation matrices

$$\mathbf{A}_1 = \begin{bmatrix} \cos(\frac{\pi}{4}) & -\sin(\frac{\pi}{4}) \\ \sin(\frac{\pi}{4}) & \cos(\frac{\pi}{4}) \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{A}_3 = \frac{1}{2} \begin{bmatrix} 3 & -1 \\ 1 & -1 \end{bmatrix}. \quad (2.97)$$

Figure 2.10 gives three examples of linear transformations of a set of vectors. Figure 2.10(a) shows 400 vectors in $\mathbb{R}^2$, each of which is represented by a dot at the corresponding (x_1, x_2) -coordinates. The vectors are arranged in a square. When we use matrix A_1 in (2.97) to linearly transform each of these vectors, we obtain the rotated square in Figure 2.10(b). If we apply the linear mapping represented by A_2 , we obtain the rectangle in Figure 2.10(c) where each x_1 -coordinate is stretched by 2. Figure 2.10(d) shows the original square from Figure 2.10(a) when linearly transformed using A_3 , which is a combination of a reflection, a rotation, and a stretch.

2.7.2 Basis Change

In the following, we will have a closer look at how transformation matrices of a linear mapping $\Phi : V \rightarrow W$ change if we change the bases in V and W . Consider two ordered bases

$$B = (\mathbf{b}_1, \dots, \mathbf{b}_n), \quad \tilde{B} = (\tilde{\mathbf{b}}_1, \dots, \tilde{\mathbf{b}}_n) \quad (2.98)$$

of V and two ordered bases

$$C = (\mathbf{c}_1, \dots, \mathbf{c}_m), \quad \tilde{C} = (\tilde{\mathbf{c}}_1, \dots, \tilde{\mathbf{c}}_m) \quad (2.99)$$

of W . Moreover, $A_\Phi \in \mathbb{R}^{m \times n}$ is the transformation matrix of the linear mapping $\Phi : V \rightarrow W$ with respect to the bases B and C , and $\tilde{A}_\Phi \in \mathbb{R}^{m \times n}$ is the corresponding transformation mapping with respect to $\tilde{B}$ and $\tilde{C}$. In the following, we will investigate how A and $\tilde{A}$ are related, i.e., how/whether we can transform A_Φ into $\tilde{A}_\Phi$ if we choose to perform a basis change from B, C to $\tilde{B}, \tilde{C}$.

Remark. We effectively get different coordinate representations of the identity mapping id_V . In the context of Figure 2.9, this would mean to map coordinates with respect to $(\mathbf{e}_1, \mathbf{e}_2)$ onto coordinates with respect to $(\mathbf{b}_1, \mathbf{b}_2)$ without changing the vector $\mathbf{x}$. By changing the basis and correspondingly the representation of vectors, the transformation matrix with respect to this new basis can have a particularly simple form that allows for straightforward computation. $\diamond$

Example 2.23 (Basis Change)

Consider a transformation matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (2.100)$$

with respect to the canonical basis in $\mathbb{R}^2$. If we define a new basis

$$B = \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) \quad (2.101)$$

we obtain a diagonal transformation matrix

$$\tilde{\mathbf{A}} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \quad (2.102)$$

with respect to B , which is easier to work with than $\mathbf{A}$.

In the following, we will look at mappings that transform coordinate vectors with respect to one basis into coordinate vectors with respect to a different basis. We will state our main result first and then provide an explanation.

Theorem 2.20 (Basis Change). *For a linear mapping $\Phi : V \rightarrow W$, ordered bases*

$$B = (\mathbf{b}_1, \dots, \mathbf{b}_n), \quad \tilde{B} = (\tilde{\mathbf{b}}_1, \dots, \tilde{\mathbf{b}}_n) \quad (2.103)$$

of V and

$$C = (\mathbf{c}_1, \dots, \mathbf{c}_m), \quad \tilde{C} = (\tilde{\mathbf{c}}_1, \dots, \tilde{\mathbf{c}}_m) \quad (2.104)$$

of W , and a transformation matrix $\mathbf{A}_\Phi$ of Φ with respect to B and C , the corresponding transformation matrix $\tilde{\mathbf{A}}_\Phi$ with respect to the bases $\tilde{B}$ and $\tilde{C}$ is given as

$$\tilde{\mathbf{A}}_\Phi = \mathbf{T}^{-1} \mathbf{A}_\Phi \mathbf{S}. \quad (2.105)$$

Here, $\mathbf{S} \in \mathbb{R}^{n \times n}$ is the transformation matrix of id_V that maps coordinates with respect to $\tilde{B}$ onto coordinates with respect to B , and $\mathbf{T} \in \mathbb{R}^{m \times m}$ is the transformation matrix of id_W that maps coordinates with respect to $\tilde{C}$ onto coordinates with respect to C .

Proof Following Drumm and Weil (2001), we can write the vectors of the new basis $\tilde{B}$ of V as a linear combination of the basis vectors of B , such that

$$\tilde{\mathbf{b}}_j = s_{1j}\mathbf{b}_1 + \dots + s_{nj}\mathbf{b}_n = \sum_{i=1}^n s_{ij}\mathbf{b}_i, \quad j = 1, \dots, n. \quad (2.106)$$

Similarly, we write the new basis vectors $\tilde{C}$ of W as a linear combination of the basis vectors of C , which yields

$$\tilde{\mathbf{c}}_k = t_{1k}\mathbf{c}_1 + \dots + t_{mk}\mathbf{c}_m = \sum_{l=1}^m t_{lk}\mathbf{c}_l, \quad k = 1, \dots, m. \quad (2.107)$$

We define $\mathbf{S} = ((s_{ij})) \in \mathbb{R}^{n \times n}$ as the transformation matrix that maps coordinates with respect to $\tilde{B}$ onto coordinates with respect to B and $\mathbf{T} = ((t_{lk})) \in \mathbb{R}^{m \times m}$ as the transformation matrix that maps coordinates with respect to $\tilde{C}$ onto coordinates with respect to C . In particular, the j th column of $\mathbf{S}$ is the coordinate representation of $\tilde{\mathbf{b}}_j$ with respect to B and

the k th column of $\mathbf{T}$ is the coordinate representation of $\tilde{\mathbf{c}}_k$ with respect to C . Note that both $\mathbf{S}$ and $\mathbf{T}$ are regular.

We are going to look at $\Phi(\tilde{\mathbf{b}}_j)$ from two perspectives. First, applying the mapping Φ , we get that for all $j = 1, \dots, n$

$$\Phi(\tilde{\mathbf{b}}_j) = \sum_{k=1}^m \underbrace{\tilde{a}_{kj} \tilde{\mathbf{c}}_k}_{\in W} \stackrel{(2.107)}{=} \sum_{k=1}^m \tilde{a}_{kj} \sum_{l=1}^m t_{lk} \mathbf{c}_l = \sum_{l=1}^m \left(\sum_{k=1}^m t_{lk} \tilde{a}_{kj} \right) \mathbf{c}_l, \quad (2.108)$$

where we first expressed the new basis vectors $\tilde{\mathbf{c}}_k \in W$ as linear combinations of the basis vectors $\mathbf{c}_l \in W$ and then swapped the order of summation.

Alternatively, when we express the $\tilde{\mathbf{b}}_j \in V$ as linear combinations of $\mathbf{b}_i \in V$, we arrive at

$$\Phi(\tilde{\mathbf{b}}_j) \stackrel{(2.106)}{=} \Phi \left(\sum_{i=1}^n s_{ij} \mathbf{b}_i \right) = \sum_{i=1}^n s_{ij} \Phi(\mathbf{b}_i) = \sum_{i=1}^n s_{ij} \sum_{l=1}^m a_{li} \mathbf{c}_l \quad (2.109a)$$

$$= \sum_{l=1}^m \left(\sum_{i=1}^n a_{li} s_{ij} \right) \mathbf{c}_l, \quad j = 1, \dots, n, \quad (2.109b)$$

where we exploited the linearity of Φ . Comparing (2.108) and (2.109b), it follows for all $j = 1, \dots, n$ and $l = 1, \dots, m$ that

$$\sum_{k=1}^m t_{lk} \tilde{a}_{kj} = \sum_{i=1}^n a_{li} s_{ij} \quad (2.110)$$

and, therefore,

$$\mathbf{T} \tilde{\mathbf{A}}_\Phi = \mathbf{A}_\Phi \mathbf{S} \in \mathbb{R}^{m \times n}, \quad (2.111)$$

such that

$$\tilde{\mathbf{A}}_\Phi = \mathbf{T}^{-1} \mathbf{A}_\Phi \mathbf{S}, \quad (2.112)$$

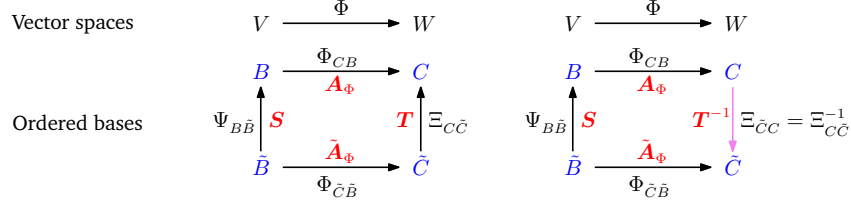
which proves Theorem 2.20. $\square$

Theorem 2.20 tells us that with a basis change in V (B is replaced with $\tilde{B}$) and W (C is replaced with $\tilde{C}$), the transformation matrix $\mathbf{A}_\Phi$ of a linear mapping $\Phi : V \rightarrow W$ is replaced by an equivalent matrix $\tilde{\mathbf{A}}_\Phi$ with

$$\tilde{\mathbf{A}}_\Phi = \mathbf{T}^{-1} \mathbf{A}_\Phi \mathbf{S}. \quad (2.113)$$

Figure 2.11 illustrates this relation: Consider a homomorphism $\Phi : V \rightarrow W$ and ordered bases $B, \tilde{B}$ of V and $C, \tilde{C}$ of W . The mapping Φ_{CB} is an instantiation of Φ and maps basis vectors of B onto linear combinations of basis vectors of C . Assume that we know the transformation matrix $\mathbf{A}_\Phi$ of Φ_{CB} with respect to the ordered bases B, C . When we perform a basis change from B to $\tilde{B}$ in V and from C to $\tilde{C}$ in W , we can determine the

Figure 2.11 For a homomorphism $\Phi : V \rightarrow W$ and ordered bases $B, \tilde{B}$ of V and $C, \tilde{C}$ of W (marked in blue), we can express the mapping $\Phi_{\tilde{C}\tilde{B}}$ with respect to the bases $\tilde{B}, \tilde{C}$ equivalently as a composition of the homomorphisms $\Phi_{\tilde{C}\tilde{B}} = \Xi_{\tilde{C}C} \circ \Phi_{CB} \circ \Psi_{B\tilde{B}}$ with respect to the bases in the subscripts. The corresponding transformation matrices are in red.



corresponding transformation matrix $\tilde{\mathbf{A}}_\Phi$ as follows: First, we find the matrix representation of the linear mapping $\Psi_{B\tilde{B}} : V \rightarrow V$ that maps coordinates with respect to the new basis $\tilde{B}$ onto the (unique) coordinates with respect to the “old” basis B (in V). Then, we use the transformation matrix $\mathbf{A}_\Phi$ of $\Phi_{CB} : V \rightarrow W$ to map these coordinates onto the coordinates with respect to C in W . Finally, we use a linear mapping $\Xi_{C\tilde{C}} : W \rightarrow W$ to map the coordinates with respect to C onto coordinates with respect to $\tilde{C}$. Therefore, we can express the linear mapping $\Phi_{\tilde{C}\tilde{B}}$ as a composition of linear mappings that involve the “old” basis:

$$\Phi_{\tilde{C}\tilde{B}} = \Xi_{C\tilde{C}} \circ \Phi_{CB} \circ \Psi_{B\tilde{B}} = \Xi_{C\tilde{C}}^{-1} \circ \Phi_{CB} \circ \Psi_{B\tilde{B}}. \quad (2.114)$$

Concretely, we use $\Psi_{B\tilde{B}} = \text{id}_V$ and $\Xi_{C\tilde{C}} = \text{id}_W$, i.e., the identity mappings that map vectors onto themselves, but with respect to a different basis.

equivalent

Definition 2.21 (Equivalence). Two matrices $\mathbf{A}, \tilde{\mathbf{A}} \in \mathbb{R}^{m \times n}$ are *equivalent* if there exist regular matrices $\mathbf{S} \in \mathbb{R}^{m \times n}$ and $\mathbf{T} \in \mathbb{R}^{n \times m}$, such that $\tilde{\mathbf{A}} = \mathbf{T}^{-1} \mathbf{A} \mathbf{S}$.

similar

Definition 2.22 (Similarity). Two matrices $\mathbf{A}, \tilde{\mathbf{A}} \in \mathbb{R}^{n \times n}$ are *similar* if there exists a regular matrix $\mathbf{S} \in \mathbb{R}^{n \times n}$ with $\tilde{\mathbf{A}} = \mathbf{S}^{-1} \mathbf{A} \mathbf{S}$.

Remark. Similar matrices are always equivalent. However, equivalent matrices are not necessarily similar. $\diamond$

Remark. Consider vector spaces V, W, X . From the remark that follows Theorem 2.17, we already know that for linear mappings $\Phi : V \rightarrow W$ and $\Psi : W \rightarrow X$ the mapping $\Psi \circ \Phi : V \rightarrow X$ is also linear. With transformation matrices $\mathbf{A}_\Phi$ and $\mathbf{A}_\Psi$ of the corresponding mappings, the overall transformation matrix is $\mathbf{A}_{\Psi \circ \Phi} = \mathbf{A}_\Psi \mathbf{A}_\Phi$. $\diamond$

In light of this remark, we can look at basis changes from the perspective of composing linear mappings:

- $\mathbf{A}_\Phi$ is the transformation matrix of a linear mapping $\Phi_{CB} : V \rightarrow W$ with respect to the bases B, C .
- $\tilde{\mathbf{A}}_\Phi$ is the transformation matrix of the linear mapping $\Phi_{\tilde{C}\tilde{B}} : V \rightarrow W$ with respect to the bases $\tilde{B}, \tilde{C}$.
- $\mathbf{S}$ is the transformation matrix of a linear mapping $\Psi_{B\tilde{B}} : V \rightarrow V$ (automorphism) that represents $\tilde{B}$ in terms of B . Normally, $\Psi = \text{id}_V$ is the identity mapping in V .

- T is the transformation matrix of a linear mapping $\Xi_{C\tilde{C}} : W \rightarrow W$ (automorphism) that represents $\tilde{C}$ in terms of C . Normally, $\Xi = \text{id}_W$ is the identity mapping in W .

If we (informally) write down the transformations just in terms of bases, then $A_\Phi : B \rightarrow C$, $\tilde{A}_\Phi : \tilde{B} \rightarrow \tilde{C}$, $S : \tilde{B} \rightarrow B$, $T : \tilde{C} \rightarrow C$ and $T^{-1} : C \rightarrow \tilde{C}$, and

$$\tilde{B} \rightarrow \tilde{C} = \tilde{B} \rightarrow B \rightarrow C \rightarrow \tilde{C} \quad (2.115)$$

$$\tilde{A}_\Phi = T^{-1} A_\Phi S. \quad (2.116)$$

Note that the execution order in (2.116) is from right to left because vectors are multiplied at the right-hand side so that $x \mapsto Sx \mapsto A_\Phi(Sx) \mapsto T^{-1}(A_\Phi(Sx)) = \tilde{A}_\Phi x$.

Example 2.24 (Basis Change)

Consider a linear mapping $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ whose transformation matrix is

$$A_\Phi = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \\ 3 & 7 & 1 \\ -1 & 2 & 4 \end{bmatrix} \quad (2.117)$$

with respect to the standard bases

$$B = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right), \quad C = \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right). \quad (2.118)$$

We seek the transformation matrix $\tilde{A}_\Phi$ of Φ with respect to the new bases

$$\tilde{B} = \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right) \in \mathbb{R}^3, \quad \tilde{C} = \left(\begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right). \quad (2.119)$$

Then,

$$S = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.120)$$

where the i th column of S is the coordinate representation of $\tilde{b}_i$ in terms of the basis vectors of B . Since B is the standard basis, the coordinate representation is straightforward to find. For a general basis B , we would need to solve a linear equation system to find the λ_i such that

$\sum_{i=1}^3 \lambda_i \mathbf{b}_i = \tilde{\mathbf{b}}_j, j = 1, \dots, 3$. Similarly, the j th column of $\mathbf{T}$ is the coordinate representation of $\tilde{\mathbf{c}}_j$ in terms of the basis vectors of C .

Therefore, we obtain

$$\tilde{\mathbf{A}}_\Phi = \mathbf{T}^{-1} \mathbf{A}_\Phi \mathbf{S} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 0 & 4 & 2 \\ 10 & 8 & 4 \\ 1 & 6 & 3 \end{bmatrix} \quad (2.121a)$$

$$= \begin{bmatrix} -4 & -4 & -2 \\ 6 & 0 & 0 \\ 4 & 8 & 4 \\ 1 & 6 & 3 \end{bmatrix}. \quad (2.121b)$$

In Chapter 4, we will be able to exploit the concept of a basis change to find a basis with respect to which the transformation matrix of an endomorphism has a particularly simple (diagonal) form. In Chapter 10, we will look at a data compression problem and find a convenient basis onto which we can project the data while minimizing the compression loss.

2.7.3 Image and Kernel

The image and kernel of a linear mapping are vector subspaces with certain important properties. In the following, we will characterize them more carefully.

Definition 2.23 (Image and Kernel).

For $\Phi : V \rightarrow W$, we define the *kernel/null space*

$$\ker(\Phi) := \Phi^{-1}(\mathbf{0}_W) = \{\mathbf{v} \in V : \Phi(\mathbf{v}) = \mathbf{0}_W\} \quad (2.122)$$

and the *image/range*

$$\text{Im}(\Phi) := \Phi(V) = \{\mathbf{w} \in W | \exists \mathbf{v} \in V : \Phi(\mathbf{v}) = \mathbf{w}\}. \quad (2.123)$$

We also call V and W also the *domain* and *codomain* of Φ , respectively.

Intuitively, the kernel is the set of vectors $\mathbf{v} \in V$ that Φ maps onto the neutral element $\mathbf{0}_W \in W$. The image is the set of vectors $\mathbf{w} \in W$ that can be “reached” by Φ from any vector in V . An illustration is given in Figure 2.12.

Remark. Consider a linear mapping $\Phi : V \rightarrow W$, where V, W are vector spaces.

- It always holds that $\Phi(\mathbf{0}_V) = \mathbf{0}_W$ and, therefore, $\mathbf{0}_V \in \ker(\Phi)$. In particular, the null space is never empty.
- $\text{Im}(\Phi) \subseteq W$ is a subspace of W , and $\ker(\Phi) \subseteq V$ is a subspace of V .

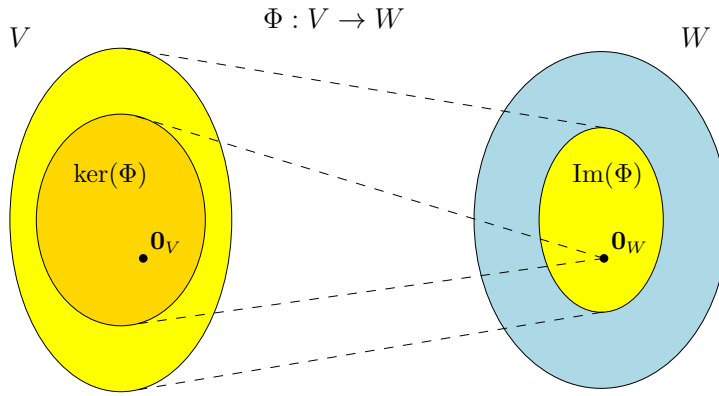


Figure 2.12 Kernel and image of a linear mapping $\Phi : V \rightarrow W$.

- Φ is injective (one-to-one) if and only if $\ker(\Phi) = \{\mathbf{0}\}$.

◇

Remark (Null Space and Column Space). Let us consider $\mathbf{A} \in \mathbb{R}^{m \times n}$ and a linear mapping $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\mathbf{x} \mapsto \mathbf{Ax}$.

- For $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$, where $\mathbf{a}_i$ are the columns of $\mathbf{A}$, we obtain

$$\text{Im}(\Phi) = \{\mathbf{Ax} : \mathbf{x} \in \mathbb{R}^n\} = \left\{ \sum_{i=1}^n x_i \mathbf{a}_i : x_1, \dots, x_n \in \mathbb{R} \right\} \quad (2.124a)$$

$$= \text{span}[\mathbf{a}_1, \dots, \mathbf{a}_n] \subseteq \mathbb{R}^m, \quad (2.124b)$$

i.e., the image is the span of the columns of $\mathbf{A}$, also called the *column space*. Therefore, the column space (image) is a subspace of $\mathbb{R}^m$, where m is the “height” of the matrix.

- $\text{rk}(\mathbf{A}) = \dim(\text{Im}(\Phi))$.
- The kernel/null space $\ker(\Phi)$ is the general solution to the homogeneous system of linear equations $\mathbf{Ax} = \mathbf{0}$ and captures all possible linear combinations of the elements in $\mathbb{R}^n$ that produce $\mathbf{0} \in \mathbb{R}^m$.
- The kernel is a subspace of $\mathbb{R}^n$, where n is the “width” of the matrix.
- The kernel focuses on the relationship among the columns, and we can use it to determine whether/how we can express a column as a linear combination of other columns.

◇

Example 2.25 (Image and Kernel of a Linear Mapping)

The mapping

$$\Phi : \mathbb{R}^4 \rightarrow \mathbb{R}^2, \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 2 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 - x_3 \\ x_1 + x_4 \end{bmatrix} \quad (2.125a)$$

$$= x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (2.125b)$$

is linear. To determine $\text{Im}(\Phi)$, we can take the span of the columns of the transformation matrix and obtain

$$\text{Im}(\Phi) = \text{span} \left[\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right]. \quad (2.126)$$

To compute the kernel (null space) of Φ , we need to solve $\mathbf{A}\mathbf{x} = \mathbf{0}$, i.e., we need to solve a homogeneous equation system. To do this, we use Gaussian elimination to transform $\mathbf{A}$ into reduced row-echelon form:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \rightsquigarrow \cdots \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}. \quad (2.127)$$

This matrix is in reduced row-echelon form, and we can use the Minus-1 Trick to compute a basis of the kernel (see Section 2.3.3). Alternatively, we can express the non-pivot columns (columns 3 and 4) as linear combinations of the pivot columns (columns 1 and 2). The third column $\mathbf{a}_3$ is equivalent to $-\frac{1}{2}$ times the second column $\mathbf{a}_2$. Therefore, $\mathbf{0} = \mathbf{a}_3 + \frac{1}{2}\mathbf{a}_2$. In the same way, we see that $\mathbf{a}_4 = \mathbf{a}_1 - \frac{1}{2}\mathbf{a}_2$ and, therefore, $\mathbf{0} = \mathbf{a}_1 - \frac{1}{2}\mathbf{a}_2 - \mathbf{a}_4$. Overall, this gives us the kernel (null space) as

$$\ker(\Phi) = \text{span} \left[\begin{bmatrix} 0 \\ \frac{1}{2} \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \\ 1 \end{bmatrix} \right]. \quad (2.128)$$

rank-nullity
theorem

Theorem 2.24 (Rank-Nullity Theorem). *For vector spaces V, W and a linear mapping $\Phi : V \rightarrow W$ it holds that*

$$\dim(\ker(\Phi)) + \dim(\text{Im}(\Phi)) = \dim(V). \quad (2.129)$$

fundamental
theorem of linear
mappings

The rank-nullity theorem is also referred to as the *fundamental theorem of linear mappings* (Axler, 2015, theorem 3.22). The following are direct consequences of Theorem 2.24:

- If $\dim(\text{Im}(\Phi)) < \dim(V)$, then $\ker(\Phi)$ is non-trivial, i.e., the kernel contains more than $\mathbf{0}_V$ and $\dim(\ker(\Phi)) \geq 1$.
- If $\mathbf{A}_\Phi$ is the transformation matrix of Φ with respect to an ordered basis and $\dim(\text{Im}(\Phi)) < \dim(V)$, then the system of linear equations $\mathbf{A}_\Phi \mathbf{x} = \mathbf{0}$ has infinitely many solutions.
- If $\dim(V) = \dim(W)$, then the following three-way equivalence holds:
 - Φ is injective
 - Φ is surjective
 - Φ is bijective
 since $\text{Im}(\Phi) \subseteq W$.

2.8 Affine Spaces

In the following, we will have a closer look at spaces that are offset from the origin, i.e., spaces that are no longer vector subspaces. Moreover, we will briefly discuss properties of mappings between these affine spaces, which resemble linear mappings.

Remark. In the machine learning literature, the distinction between linear and affine is sometimes not clear so that we can find references to affine spaces/mappings as linear spaces/mappings. $\diamond$

2.8.1 Affine Subspaces

Definition 2.25 (Affine Subspace). Let V be a vector space, $\mathbf{x}_0 \in V$ and $U \subseteq V$ a subspace. Then the subset

$$L = \mathbf{x}_0 + U := \{\mathbf{x}_0 + \mathbf{u} : \mathbf{u} \in U\} \quad (2.130a)$$

$$= \{\mathbf{v} \in V \mid \exists \mathbf{u} \in U : \mathbf{v} = \mathbf{x}_0 + \mathbf{u}\} \subseteq V \quad (2.130b)$$

is called *affine subspace* or *linear manifold* of V . U is called *direction* or *direction space*, and $\mathbf{x}_0$ is called *support point*. In Chapter 12, we refer to such a subspace as a *hyperplane*.

affine subspace
linear manifold
direction
direction space
support point
hyperplane

Note that the definition of an affine subspace excludes $\mathbf{0}$ if $\mathbf{x}_0 \notin U$. Therefore, an affine subspace is not a (linear) subspace (vector subspace) of V for $\mathbf{x}_0 \notin U$.

Examples of affine subspaces are points, lines, and planes in $\mathbb{R}^3$, which do not (necessarily) go through the origin.

Remark. Consider two affine subspaces $L = \mathbf{x}_0 + U$ and $\tilde{L} = \tilde{\mathbf{x}}_0 + \tilde{U}$ of a vector space V . Then, $L \subseteq \tilde{L}$ if and only if $U \subseteq \tilde{U}$ and $\mathbf{x}_0 - \tilde{\mathbf{x}}_0 \in \tilde{U}$.

Affine subspaces are often described by *parameters*: Consider a k -dimensional affine space $L = \mathbf{x}_0 + U$ of V . If $(\mathbf{b}_1, \dots, \mathbf{b}_k)$ is an ordered basis of U , then every element $\mathbf{x} \in L$ can be uniquely described as

$$\mathbf{x} = \mathbf{x}_0 + \lambda_1 \mathbf{b}_1 + \dots + \lambda_k \mathbf{b}_k, \quad (2.131)$$

where $\lambda_1, \dots, \lambda_k \in \mathbb{R}$. This representation is called *parametric equation* of L with directional vectors $\mathbf{b}_1, \dots, \mathbf{b}_k$ and *parameters* $\lambda_1, \dots, \lambda_k$. $\diamond$

parametric equation
parameters

Example 2.26 (Affine Subspaces)

- One-dimensional affine subspaces are called *lines* and can be written as $\mathbf{y} = \mathbf{x}_0 + \lambda \mathbf{b}_1$, where $\lambda \in \mathbb{R}$ and $U = \text{span}[\mathbf{b}_1] \subseteq \mathbb{R}^n$ is a one-dimensional subspace of $\mathbb{R}^n$. This means that a line is defined by a support point $\mathbf{x}_0$ and a vector $\mathbf{b}_1$ that defines the direction. See Figure 2.13 for an illustration.

line

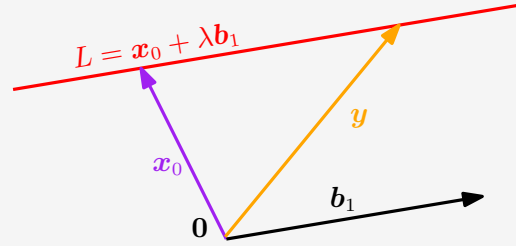
plane

- Two-dimensional affine subspaces of $\mathbb{R}^n$ are called *planes*. The parametric equation for planes is $\mathbf{y} = \mathbf{x}_0 + \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2$, where $\lambda_1, \lambda_2 \in \mathbb{R}$ and $U = \text{span}[\mathbf{b}_1, \mathbf{b}_2] \subseteq \mathbb{R}^n$. This means that a plane is defined by a support point $\mathbf{x}_0$ and two linearly independent vectors $\mathbf{b}_1, \mathbf{b}_2$ that span the direction space.

hyperplane

- In $\mathbb{R}^n$, the $(n - 1)$ -dimensional affine subspaces are called *hyperplanes*, and the corresponding parametric equation is $\mathbf{y} = \mathbf{x}_0 + \sum_{i=1}^{n-1} \lambda_i \mathbf{b}_i$, where $\mathbf{b}_1, \dots, \mathbf{b}_{n-1}$ form a basis of an $(n - 1)$ -dimensional subspace U of $\mathbb{R}^n$. This means that a hyperplane is defined by a support point $\mathbf{x}_0$ and $(n - 1)$ linearly independent vectors $\mathbf{b}_1, \dots, \mathbf{b}_{n-1}$ that span the direction space. In $\mathbb{R}^2$, a line is also a hyperplane. In $\mathbb{R}^3$, a plane is also a hyperplane.

Figure 2.13 Lines are affine subspaces. Vectors $\mathbf{y}$ on a line $\mathbf{x}_0 + \lambda \mathbf{b}_1$ lie in an affine subspace L with support point $\mathbf{x}_0$ and direction $\mathbf{b}_1$.



Remark (Inhomogeneous systems of linear equations and affine subspaces). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{x} \in \mathbb{R}^m$, the solution of the system of linear equations $\mathbf{A}\boldsymbol{\lambda} = \mathbf{x}$ is either the empty set or an affine subspace of $\mathbb{R}^n$ of dimension $n - \text{rk}(\mathbf{A})$. In particular, the solution of the linear equation $\lambda_1 \mathbf{b}_1 + \dots + \lambda_n \mathbf{b}_n = \mathbf{x}$, where $(\lambda_1, \dots, \lambda_n) \neq (0, \dots, 0)$, is a hyperplane in $\mathbb{R}^n$.

In $\mathbb{R}^n$, every k -dimensional affine subspace is the solution of an inhomogeneous system of linear equations $\mathbf{A}\mathbf{x} = \mathbf{b}$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$ and $\text{rk}(\mathbf{A}) = n - k$. Recall that for homogeneous equation systems $\mathbf{A}\mathbf{x} = \mathbf{0}$ the solution was a vector subspace, which we can also think of as a special affine space with support point $\mathbf{x}_0 = \mathbf{0}$. $\diamond$

2.8.2 Affine Mappings

Similar to linear mappings between vector spaces, which we discussed in Section 2.7, we can define affine mappings between two affine spaces. Linear and affine mappings are closely related. Therefore, many properties that we already know from linear mappings, e.g., that the composition of linear mappings is a linear mapping, also hold for affine mappings.

Definition 2.26 (Affine Mapping). For two vector spaces V, W , a linear

mapping $\Phi : V \rightarrow W$, and $\mathbf{a} \in W$, the mapping

$$\phi : V \rightarrow W \quad (2.132)$$

$$\mathbf{x} \mapsto \mathbf{a} + \Phi(\mathbf{x}) \quad (2.133)$$

is an *affine mapping* from V to W . The vector $\mathbf{a}$ is called the *translation vector* of ϕ .

affine mapping
translation vector

- Every affine mapping $\phi : V \rightarrow W$ is also the composition of a linear mapping $\Phi : V \rightarrow W$ and a translation $\tau : W \rightarrow W$ in W , such that $\phi = \tau \circ \Phi$. The mappings Φ and τ are uniquely determined.
- The composition $\phi' \circ \phi$ of affine mappings $\phi : V \rightarrow W$, $\phi' : W \rightarrow X$ is affine.
- Affine mappings keep the geometric structure invariant. They also preserve the dimension and parallelism.

2.9 Further Reading

There are many resources for learning linear algebra, including the textbooks by Strang (2003), Golan (2007), Axler (2015), and Liesen and Mehrmann (2015). There are also several online resources that we mentioned in the introduction to this chapter. We only covered Gaussian elimination here, but there are many other approaches for solving systems of linear equations, and we refer to numerical linear algebra textbooks by Stoer and Burlirsch (2002), Golub and Van Loan (2012), and Horn and Johnson (2013) for an in-depth discussion.

In this book, we distinguish between the topics of linear algebra (e.g., vectors, matrices, linear independence, basis) and topics related to the geometry of a vector space. In Chapter 3, we will introduce the inner product, which induces a norm. These concepts allow us to define angles, lengths and distances, which we will use for orthogonal projections. Projections turn out to be key in many machine learning algorithms, such as linear regression and principal component analysis, both of which we will cover in Chapters 9 and 10, respectively.

Exercises

2.1 We consider $(\mathbb{R} \setminus \{-1\}, \star)$, where

$$a \star b := ab + a + b, \quad a, b \in \mathbb{R} \setminus \{-1\} \quad (2.134)$$

- Show that $(\mathbb{R} \setminus \{-1\}, \star)$ is an Abelian group.
- Solve

$$3 \star x \star x = 15$$

in the Abelian group $(\mathbb{R} \setminus \{-1\}, \star)$, where $\star$ is defined in (2.134).

2.2 Let n be in $\mathbb{N} \setminus \{0\}$. Let k, x be in $\mathbb{Z}$. We define the congruence class $\bar{k}$ of the integer k as the set

$$\begin{aligned} \bar{k} &= \{x \in \mathbb{Z} \mid x - k = 0 \pmod{n}\} \\ &= \{x \in \mathbb{Z} \mid \exists a \in \mathbb{Z}: (x - k = n \cdot a)\}. \end{aligned}$$

We now define $\mathbb{Z}/n\mathbb{Z}$ (sometimes written $\mathbb{Z}_n$) as the set of all congruence classes modulo n . Euclidean division implies that this set is a finite set containing n elements:

$$\mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$$

For all $\bar{a}, \bar{b} \in \mathbb{Z}_n$, we define

$$\bar{a} \oplus \bar{b} := \overline{a + b}$$

- Show that $(\mathbb{Z}_n, \oplus)$ is a group. Is it Abelian?
- We now define another operation $\otimes$ for all $\bar{a}$ and $\bar{b}$ in $\mathbb{Z}_n$ as

$$\bar{a} \otimes \bar{b} = \overline{a \times b}, \quad (2.135)$$

where $a \times b$ represents the usual multiplication in $\mathbb{Z}$.

Let $n = 5$. Draw the times table of the elements of $\mathbb{Z}_5 \setminus \{\bar{0}\}$ under $\otimes$, i.e., calculate the products $\bar{a} \otimes \bar{b}$ for all $\bar{a}$ and $\bar{b}$ in $\mathbb{Z}_5 \setminus \{\bar{0}\}$.

Hence, show that $\mathbb{Z}_5 \setminus \{\bar{0}\}$ is closed under $\otimes$ and possesses a neutral element for $\otimes$. Display the inverse of all elements in $\mathbb{Z}_5 \setminus \{\bar{0}\}$ under $\otimes$. Conclude that $(\mathbb{Z}_5 \setminus \{\bar{0}\}, \otimes)$ is an Abelian group.

- Show that $(\mathbb{Z}_8 \setminus \{\bar{0}\}, \otimes)$ is not a group.
- We recall that the Bézout theorem states that two integers a and b are relatively prime (i.e., $\gcd(a, b) = 1$) if and only if there exist two integers u and v such that $au + bv = 1$. Show that $(\mathbb{Z}_n \setminus \{\bar{0}\}, \otimes)$ is a group if and only if $n \in \mathbb{N} \setminus \{0\}$ is prime.

2.3 Consider the set $\mathcal{G}$ of 3×3 matrices defined as follows:

$$\mathcal{G} = \left\{ \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 3} \mid x, y, z \in \mathbb{R} \right\}$$

We define $\cdot$ as the standard matrix multiplication.

Is $(\mathcal{G}, \cdot)$ a group? If yes, is it Abelian? Justify your answer.

2.4 Compute the following matrix products, if possible:

a.

$$\begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

b.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

c.

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

d.

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 4 & 1 & -1 & -4 \end{bmatrix} \begin{bmatrix} 0 & 3 \\ 1 & -1 \\ 2 & 1 \\ 5 & 2 \end{bmatrix}$$

e.

$$\begin{bmatrix} 0 & 3 \\ 1 & -1 \\ 2 & 1 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 & 2 \\ 4 & 1 & -1 & -4 \end{bmatrix}$$

2.5 Find the set S of all solutions in $\mathbf{x}$ of the following inhomogeneous linear systems $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ and $\mathbf{b}$ are defined as follows:

a.

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 2 & 5 & -7 & -5 \\ 2 & -1 & 1 & 3 \\ 5 & 2 & -4 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ -2 \\ 4 \\ 6 \end{bmatrix}$$

b.

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 & 0 & 1 \\ 1 & 1 & 0 & -3 & 0 \\ 2 & -1 & 0 & 1 & -1 \\ -1 & 2 & 0 & -2 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 6 \\ 5 \\ -1 \end{bmatrix}$$

2.6 Using Gaussian elimination, find all solutions of the inhomogeneous equation system $\mathbf{Ax} = \mathbf{b}$ with

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

- 2.7 Find all solutions in $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3$ of the equation system $\mathbf{A}\mathbf{x} = 12\mathbf{x}$, where

$$\mathbf{A} = \begin{bmatrix} 6 & 4 & 3 \\ 6 & 0 & 9 \\ 0 & 8 & 0 \end{bmatrix}$$

and $\sum_{i=1}^3 x_i = 1$.

- 2.8 Determine the inverses of the following matrices if possible:

a.

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix}$$

b.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

- 2.9 Which of the following sets are subspaces of $\mathbb{R}^3$?

a. $A = \{(\lambda, \lambda + \mu^3, \lambda - \mu^3) \mid \lambda, \mu \in \mathbb{R}\}$

b. $B = \{(\lambda^2, -\lambda^2, 0) \mid \lambda \in \mathbb{R}\}$

c. Let γ be in $\mathbb{R}$.

$$C = \{(\xi_1, \xi_2, \xi_3) \in \mathbb{R}^3 \mid \xi_1 - 2\xi_2 + 3\xi_3 = \gamma\}$$

d. $D = \{(\xi_1, \xi_2, \xi_3) \in \mathbb{R}^3 \mid \xi_2 \in \mathbb{Z}\}$

- 2.10 Are the following sets of vectors linearly independent?

a.

$$\mathbf{x}_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 3 \\ -3 \\ 8 \end{bmatrix}$$

b.

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

- 2.11 Write

$$\mathbf{y} = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}$$

as linear combination of

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

2.12 Consider two subspaces of $\mathbb{R}^4$:

$$U_1 = \text{span}\left[\begin{bmatrix} 1 \\ 1 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}\right], \quad U_2 = \text{span}\left[\begin{bmatrix} -1 \\ -2 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 6 \\ -2 \\ -1 \end{bmatrix}\right].$$

Determine a basis of $U_1 \cap U_2$.

2.13 Consider two subspaces U_1 and U_2 , where U_1 is the solution space of the homogeneous equation system $A_1x = 0$ and U_2 is the solution space of the homogeneous equation system $A_2x = 0$ with

$$A_1 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -2 & -1 \\ 2 & 1 & 3 \\ 1 & 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3 & -3 & 0 \\ 1 & 2 & 3 \\ 7 & -5 & 2 \\ 3 & -1 & 2 \end{bmatrix}.$$

- Determine the dimension of U_1, U_2 .
 - Determine bases of U_1 and U_2 .
 - Determine a basis of $U_1 \cap U_2$.
- 2.14 Consider two subspaces U_1 and U_2 , where U_1 is spanned by the columns of A_1 and U_2 is spanned by the columns of A_2 with

$$A_1 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -2 & -1 \\ 2 & 1 & 3 \\ 1 & 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3 & -3 & 0 \\ 1 & 2 & 3 \\ 7 & -5 & 2 \\ 3 & -1 & 2 \end{bmatrix}.$$

- Determine the dimension of U_1, U_2 .
 - Determine bases of U_1 and U_2 .
 - Determine a basis of $U_1 \cap U_2$.
- 2.15 Let $F = \{(x, y, z) \in \mathbb{R}^3 \mid x+y-z=0\}$ and $G = \{(a-b, a+b, a-3b) \mid a, b \in \mathbb{R}\}$.
- Show that F and G are subspaces of $\mathbb{R}^3$.
 - Calculate $F \cap G$ without resorting to any basis vector.
 - Find one basis for F and one for G , calculate $F \cap G$ using the basis vectors previously found and check your result with the previous question.
- 2.16 Are the following mappings linear?

- Let $a, b \in \mathbb{R}$.

$$\Phi : L^1([a, b]) \rightarrow \mathbb{R}$$

$$f \mapsto \Phi(f) = \int_a^b f(x)dx,$$

where $L^1([a, b])$ denotes the set of integrable functions on $[a, b]$.

-

$$\Phi : C^1 \rightarrow C^0$$

$$f \mapsto \Phi(f) = f',$$

where for $k \geq 1$, C^k denotes the set of k times continuously differentiable functions, and C^0 denotes the set of continuous functions.

c.

$$\begin{aligned}\Phi : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto \Phi(x) = \cos(x)\end{aligned}$$

d.

$$\begin{aligned}\Phi : \mathbb{R}^3 &\rightarrow \mathbb{R}^2 \\ \mathbf{x} &\mapsto \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 3 \end{bmatrix} \mathbf{x}\end{aligned}$$

e. Let θ be in $[0, 2\pi[$ and

$$\begin{aligned}\Phi : \mathbb{R}^2 &\rightarrow \mathbb{R}^2 \\ \mathbf{x} &\mapsto \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \mathbf{x}\end{aligned}$$

2.17 Consider the linear mapping

$$\begin{aligned}\Phi : \mathbb{R}^3 &\rightarrow \mathbb{R}^4 \\ \Phi \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) &= \begin{bmatrix} 3x_1 + 2x_2 + x_3 \\ x_1 + x_2 + x_3 \\ x_1 - 3x_2 \\ 2x_1 + 3x_2 + x_3 \end{bmatrix}\end{aligned}$$

- Find the transformation matrix $\mathbf{A}_\Phi$.
- Determine $\text{rk}(\mathbf{A}_\Phi)$.
- Compute the kernel and image of Φ . What are $\dim(\ker(\Phi))$ and $\dim(\text{Im}(\Phi))$?

2.18 Let E be a vector space. Let f and g be two automorphisms on E such that $f \circ g = \text{id}_E$ (i.e., $f \circ g$ is the identity mapping id_E). Show that $\ker(f) = \ker(g \circ f)$, $\text{Im}(g) = \text{Im}(g \circ f)$ and that $\ker(f) \cap \text{Im}(g) = \{\mathbf{0}_E\}$.

2.19 Consider an endomorphism $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ whose transformation matrix (with respect to the standard basis in $\mathbb{R}^3$) is

$$\mathbf{A}_\Phi = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

- a. Determine $\ker(\Phi)$ and $\text{Im}(\Phi)$.
- b. Determine the transformation matrix $\tilde{\mathbf{A}}_\Phi$ with respect to the basis

$$B = \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right),$$

i.e., perform a basis change toward the new basis B .

2.20 Let us consider $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}'_1, \mathbf{b}'_2$, 4 vectors of $\mathbb{R}^2$ expressed in the standard basis of $\mathbb{R}^2$ as

$$\mathbf{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad \mathbf{b}'_1 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}, \quad \mathbf{b}'_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and let us define two ordered bases $B = (\mathbf{b}_1, \mathbf{b}_2)$ and $B' = (\mathbf{b}'_1, \mathbf{b}'_2)$ of $\mathbb{R}^2$.

- a. Show that B and B' are two bases of $\mathbb{R}^2$ and draw those basis vectors.
- b. Compute the matrix P_1 that performs a basis change from B' to B .
- c. We consider c_1, c_2, c_3 , three vectors of $\mathbb{R}^3$ defined in the standard basis of $\mathbb{R}^3$ as

$$c_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, \quad c_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

and we define $C = (c_1, c_2, c_3)$.

- (i) Show that C is a basis of $\mathbb{R}^3$, e.g., by using determinants (see Section 4.1).
 - (ii) Let us call $C' = (c'_1, c'_2, c'_3)$ the standard basis of $\mathbb{R}^3$. Determine the matrix P_2 that performs the basis change from C to C' .
- d. We consider a homomorphism $\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, such that

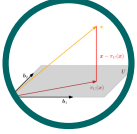
$$\begin{aligned} \Phi(b_1 + b_2) &= c_2 + c_3 \\ \Phi(b_1 - b_2) &= 2c_1 - c_2 + 3c_3 \end{aligned}$$

where $B = (b_1, b_2)$ and $C = (c_1, c_2, c_3)$ are ordered bases of $\mathbb{R}^2$ and $\mathbb{R}^3$, respectively.

Determine the transformation matrix A_Φ of Φ with respect to the ordered bases B and C .

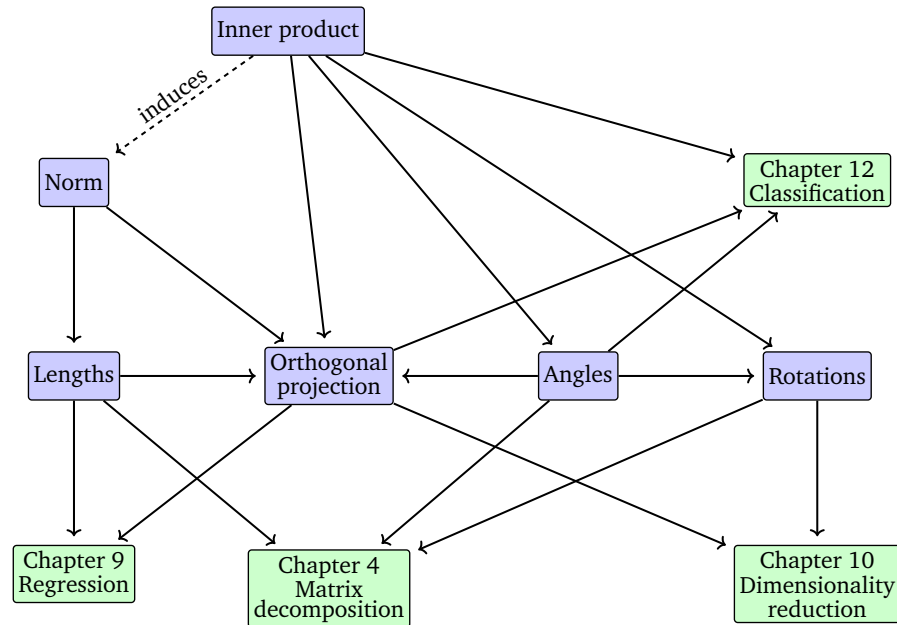
- e. Determine A' , the transformation matrix of Φ with respect to the bases B' and C' .
- f. Let us consider the vector $x \in \mathbb{R}^2$ whose coordinates in B' are $[2, 3]^\top$. In other words, $x = 2b'_1 + 3b'_2$.
 - (i) Calculate the coordinates of x in B .
 - (ii) Based on that, compute the coordinates of $\Phi(x)$ expressed in C .
 - (iii) Then, write $\Phi(x)$ in terms of c'_1, c'_2, c'_3 .
 - (iv) Use the representation of x in B' and the matrix A' to find this result directly.

Analytic Geometry



In Chapter 2, we studied vectors, vector spaces, and linear mappings at a general but abstract level. In this chapter, we will add some geometric interpretation and intuition to all of these concepts. In particular, we will look at geometric vectors and compute their lengths and distances or angles between two vectors. To be able to do this, we equip the vector space with an inner product that induces the geometry of the vector space. Inner products and their corresponding norms and metrics capture the intuitive notions of similarity and distances, which we use to develop the support vector machine in Chapter 12. We will then use the concepts of lengths and angles between vectors to discuss orthogonal projections, which will play a central role when we discuss principal component analysis in Chapter 10 and regression via maximum likelihood estimation in Chapter 9. Figure 3.1 gives an overview of how concepts in this chapter are related and how they are connected to other chapters of the book.

Figure 3.1 A mind map of the concepts introduced in this chapter, along with when they are used in other parts of the book.



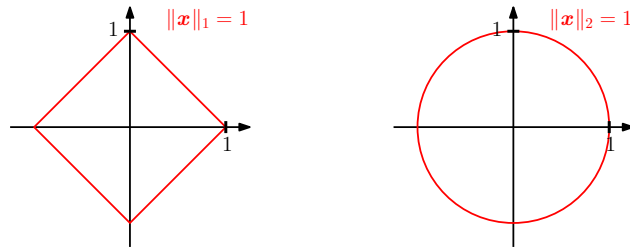


Figure 3.3 For different norms, the red lines indicate the set of vectors with norm 1. Left: Manhattan norm; Right: Euclidean distance.

3.1 Norms

When we think of geometric vectors, i.e., directed line segments that start at the origin, then intuitively the length of a vector is the distance of the “end” of this directed line segment from the origin. In the following, we will discuss the notion of the length of vectors using the concept of a norm.

Definition 3.1 (Norm). A *norm* on a vector space V is a function

$$\|\cdot\| : V \rightarrow \mathbb{R}, \quad (3.1)$$

$$x \mapsto \|x\|, \quad (3.2)$$

which assigns each vector x its *length* $\|x\| \in \mathbb{R}$, such that for all $\lambda \in \mathbb{R}$ and $x, y \in V$ the following hold:

- *Absolutely homogeneous:* $\|\lambda x\| = |\lambda| \|x\|$
- *Triangle inequality:* $\|x + y\| \leq \|x\| + \|y\|$
- *Positive definite:* $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$

In geometric terms, the triangle inequality states that for any triangle, the sum of the lengths of any two sides must be greater than or equal to the length of the remaining side; see Figure 3.2 for an illustration. Definition 3.1 is in terms of a general vector space V (Section 2.4), but in this book we will only consider a finite-dimensional vector space $\mathbb{R}^n$. Recall that for a vector $x \in \mathbb{R}^n$ we denote the elements of the vector using a subscript, that is, x_i is the i^{th} element of the vector x .

norm

length

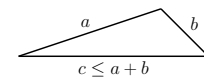
absolutely

homogeneous

triangle inequality

positive definite

Figure 3.2 Triangle inequality.



Example 3.1 (Manhattan Norm)

The *Manhattan norm* on $\mathbb{R}^n$ is defined for $x \in \mathbb{R}^n$ as

$$\|x\|_1 := \sum_{i=1}^n |x_i|, \quad (3.3)$$

where $|\cdot|$ is the absolute value. The left panel of Figure 3.3 shows all vectors $x \in \mathbb{R}^2$ with $\|x\|_1 = 1$. The Manhattan norm is also called ℓ_1 norm.

Manhattan norm

 ℓ_1 norm

Euclidean norm

Example 3.2 (Euclidean Norm)

The *Euclidean norm* of $\mathbf{x} \in \mathbb{R}^n$ is defined as

$$\|\mathbf{x}\|_2 := \sqrt{\sum_{i=1}^n x_i^2} = \sqrt{\mathbf{x}^\top \mathbf{x}} \quad (3.4)$$

Euclidean distance

and computes the *Euclidean distance* of $\mathbf{x}$ from the origin. The right panel of Figure 3.3 shows all vectors $\mathbf{x} \in \mathbb{R}^2$ with $\|\mathbf{x}\|_2 = 1$. The Euclidean norm is also called *ℓ_2 norm*.

 ℓ_2 norm

Remark. Throughout this book, we will use the Euclidean norm (3.4) by default if not stated otherwise. $\diamond$

3.2 Inner Products

Inner products allow for the introduction of intuitive geometrical concepts, such as the length of a vector and the angle or distance between two vectors. A major purpose of inner products is to determine whether vectors are orthogonal to each other.

3.2.1 Dot Product

scalar product
dot product

We may already be familiar with a particular type of inner product, the *scalar product/dot product* in $\mathbb{R}^n$, which is given by

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i. \quad (3.5)$$

We will refer to this particular inner product as the dot product in this book. However, inner products are more general concepts with specific properties, which we will now introduce.

3.2.2 General Inner Products

bilinear mapping

Recall the linear mapping from Section 2.7, where we can rearrange the mapping with respect to addition and multiplication with a scalar. A *bilinear mapping* Ω is a mapping with two arguments, and it is linear in each argument, i.e., when we look at a vector space V then it holds that for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$, $\lambda, \psi \in \mathbb{R}$ that

$$\Omega(\lambda \mathbf{x} + \psi \mathbf{y}, \mathbf{z}) = \lambda \Omega(\mathbf{x}, \mathbf{z}) + \psi \Omega(\mathbf{y}, \mathbf{z}) \quad (3.6)$$

$$\Omega(\mathbf{x}, \lambda \mathbf{y} + \psi \mathbf{z}) = \lambda \Omega(\mathbf{x}, \mathbf{y}) + \psi \Omega(\mathbf{x}, \mathbf{z}). \quad (3.7)$$

Here, (3.6) asserts that Ω is linear in the first argument, and (3.7) asserts that Ω is linear in the second argument (see also (2.87)).

Definition 3.2. Let V be a vector space and $\Omega : V \times V \rightarrow \mathbb{R}$ be a bilinear mapping that takes two vectors and maps them onto a real number. Then

- Ω is called *symmetric* if $\Omega(\mathbf{x}, \mathbf{y}) = \Omega(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in V$, i.e., the order of the arguments does not matter. symmetric
- Ω is called *positive definite* if positive definite

$$\forall \mathbf{x} \in V \setminus \{\mathbf{0}\} : \Omega(\mathbf{x}, \mathbf{x}) > 0, \quad \Omega(\mathbf{0}, \mathbf{0}) = 0. \quad (3.8)$$

Definition 3.3. Let V be a vector space and $\Omega : V \times V \rightarrow \mathbb{R}$ be a bilinear mapping that takes two vectors and maps them onto a real number. Then

- A positive definite, symmetric bilinear mapping $\Omega : V \times V \rightarrow \mathbb{R}$ is called an *inner product* on V . We typically write $\langle \mathbf{x}, \mathbf{y} \rangle$ instead of $\Omega(\mathbf{x}, \mathbf{y})$. inner product
- The pair $(V, \langle \cdot, \cdot \rangle)$ is called an *inner product space* or (real) *vector space with inner product*. If we use the dot product defined in (3.5), we call $(V, \langle \cdot, \cdot \rangle)$ a *Euclidean vector space*. inner product space
vector space with
inner product
Euclidean vector
space

We will refer to these spaces as inner product spaces in this book.

Example 3.3 (Inner Product That Is Not the Dot Product)

Consider $V = \mathbb{R}^2$. If we define

$$\langle \mathbf{x}, \mathbf{y} \rangle := x_1 y_1 - (x_1 y_2 + x_2 y_1) + 2x_2 y_2 \quad (3.9)$$

then $\langle \cdot, \cdot \rangle$ is an inner product but different from the dot product. The proof will be an exercise.

3.2.3 Symmetric, Positive Definite Matrices

Symmetric, positive definite matrices play an important role in machine learning, and they are defined via the inner product. In Section 4.3, we will return to symmetric, positive definite matrices in the context of matrix decompositions. The idea of symmetric positive semidefinite matrices is key in the definition of kernels (Section 12.4).

Consider an n -dimensional vector space V with an inner product $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ (see Definition 3.3) and an ordered basis $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ of V . Recall from Section 2.6.1 that any vectors $\mathbf{x}, \mathbf{y} \in V$ can be written as linear combinations of the basis vectors so that $\mathbf{x} = \sum_{i=1}^n \psi_i \mathbf{b}_i \in V$ and $\mathbf{y} = \sum_{j=1}^n \lambda_j \mathbf{b}_j \in V$ for suitable $\psi_i, \lambda_j \in \mathbb{R}$. Due to the bilinearity of the inner product, it holds for all $\mathbf{x}, \mathbf{y} \in V$ that

$$\langle \mathbf{x}, \mathbf{y} \rangle = \left\langle \sum_{i=1}^n \psi_i \mathbf{b}_i, \sum_{j=1}^n \lambda_j \mathbf{b}_j \right\rangle = \sum_{i=1}^n \sum_{j=1}^n \psi_i \langle \mathbf{b}_i, \mathbf{b}_j \rangle \lambda_j = \hat{\mathbf{x}}^\top \mathbf{A} \hat{\mathbf{y}}, \quad (3.10)$$

where $A_{ij} := \langle \mathbf{b}_i, \mathbf{b}_j \rangle$ and $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ are the coordinates of $\mathbf{x}$ and $\mathbf{y}$ with respect to the basis B . This implies that the inner product $\langle \cdot, \cdot \rangle$ is uniquely determined through $\mathbf{A}$. The symmetry of the inner product also means that $\mathbf{A}$

is symmetric. Furthermore, the positive definiteness of the inner product implies that

$$\forall \mathbf{x} \in V \setminus \{\mathbf{0}\} : \mathbf{x}^\top \mathbf{A} \mathbf{x} > 0. \quad (3.11)$$

symmetric, positive
definite
positive definite
symmetric, positive
semidefinite

Definition 3.4 (Symmetric, Positive Definite Matrix). A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ that satisfies (3.11) is called *symmetric, positive definite*, or just *positive definite*. If only $\geq$ holds in (3.11), then $\mathbf{A}$ is called *symmetric, positive semidefinite*.

Example 3.4 (Symmetric, Positive Definite Matrices)

Consider the matrices

$$\mathbf{A}_1 = \begin{bmatrix} 9 & 6 \\ 6 & 5 \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} 9 & 6 \\ 6 & 3 \end{bmatrix}. \quad (3.12)$$

$\mathbf{A}_1$ is positive definite because it is symmetric and

$$\mathbf{x}^\top \mathbf{A}_1 \mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 9 & 6 \\ 6 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (3.13a)$$

$$= 9x_1^2 + 12x_1x_2 + 5x_2^2 = (3x_1 + 2x_2)^2 + x_2^2 > 0 \quad (3.13b)$$

for all $\mathbf{x} \in V \setminus \{\mathbf{0}\}$. In contrast, $\mathbf{A}_2$ is symmetric but not positive definite because $\mathbf{x}^\top \mathbf{A}_2 \mathbf{x} = 9x_1^2 + 12x_1x_2 + 3x_2^2 = (3x_1 + 2x_2)^2 - x_2^2$ can be less than 0, e.g., for $\mathbf{x} = [2, -3]^\top$.

If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric, positive definite, then

$$\langle \mathbf{x}, \mathbf{y} \rangle = \hat{\mathbf{x}}^\top \mathbf{A} \hat{\mathbf{y}} \quad (3.14)$$

defines an inner product with respect to an ordered basis B , where $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are the coordinate representations of $\mathbf{x}, \mathbf{y} \in V$ with respect to B .

Theorem 3.5. For a real-valued, finite-dimensional vector space V and an ordered basis B of V , it holds that $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is an inner product if and only if there exists a symmetric, positive definite matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ with

$$\langle \mathbf{x}, \mathbf{y} \rangle = \hat{\mathbf{x}}^\top \mathbf{A} \hat{\mathbf{y}}. \quad (3.15)$$

The following properties hold if $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric and positive definite:

- The null space (kernel) of $\mathbf{A}$ consists only of $\mathbf{0}$ because $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$. This implies that $\mathbf{A} \mathbf{x} \neq \mathbf{0}$ if $\mathbf{x} \neq \mathbf{0}$.
- The diagonal elements a_{ii} of $\mathbf{A}$ are positive because $a_{ii} = \mathbf{e}_i^\top \mathbf{A} \mathbf{e}_i > 0$, where $\mathbf{e}_i$ is the i th vector of the standard basis in $\mathbb{R}^n$.

3.3 Lengths and Distances

In Section 3.1, we already discussed norms that we can use to compute the length of a vector. Inner products and norms are closely related in the sense that any inner product induces a norm

Inner products
induce norms.

$$\|x\| := \sqrt{\langle x, x \rangle} \quad (3.16)$$

in a natural way, such that we can compute lengths of vectors using the inner product. However, not every norm is induced by an inner product. The Manhattan norm (3.3) is an example of a norm without a corresponding inner product. In the following, we will focus on norms that are induced by inner products and introduce geometric concepts, such as lengths, distances, and angles.

Remark (Cauchy-Schwarz Inequality). For an inner product vector space $(V, \langle \cdot, \cdot \rangle)$ the induced norm $\|\cdot\|$ satisfies the *Cauchy-Schwarz inequality*

Cauchy-Schwarz
inequality

$$|\langle x, y \rangle| \leq \|x\| \|y\|. \quad (3.17)$$

◇

Example 3.5 (Lengths of Vectors Using Inner Products)

In geometry, we are often interested in lengths of vectors. We can now use an inner product to compute them using (3.16). Let us take $x = [1, 1]^\top \in \mathbb{R}^2$. If we use the dot product as the inner product, with (3.16) we obtain

$$\|x\| = \sqrt{x^\top x} = \sqrt{1^2 + 1^2} = \sqrt{2} \quad (3.18)$$

as the length of x . Let us now choose a different inner product:

$$\langle x, y \rangle := x^\top \begin{bmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix} y = x_1 y_1 - \frac{1}{2}(x_1 y_2 + x_2 y_1) + x_2 y_2. \quad (3.19)$$

If we compute the norm of a vector, then this inner product returns smaller values than the dot product if x_1 and x_2 have the same sign (and $x_1 x_2 > 0$); otherwise, it returns greater values than the dot product. With this inner product, we obtain

$$\langle x, x \rangle = x_1^2 - x_1 x_2 + x_2^2 = 1 - 1 + 1 = 1 \implies \|x\| = \sqrt{1} = 1, \quad (3.20)$$

such that x is “shorter” with this inner product than with the dot product.

Definition 3.6 (Distance and Metric). Consider an inner product space $(V, \langle \cdot, \cdot \rangle)$. Then

$$d(x, y) := \|x - y\| = \sqrt{\langle x - y, x - y \rangle} \quad (3.21)$$

is called the *distance* between x and y for $x, y \in V$. If we use the dot product as the inner product, then the distance is called *Euclidean distance*.

distance
Euclidean distance

The mapping

$$d : V \times V \rightarrow \mathbb{R} \quad (3.22)$$

$$(\mathbf{x}, \mathbf{y}) \mapsto d(\mathbf{x}, \mathbf{y}) \quad (3.23)$$

metric

is called a *metric*.

Remark. Similar to the length of a vector, the distance between vectors does not require an inner product: a norm is sufficient. If we have a norm induced by an inner product, the distance may vary depending on the choice of the inner product. $\diamond$

A metric d satisfies the following:

positive definite

1. d is *positive definite*, i.e., $d(\mathbf{x}, \mathbf{y}) \geq 0$ for all $\mathbf{x}, \mathbf{y} \in V$ and $d(\mathbf{x}, \mathbf{y}) = 0 \iff \mathbf{x} = \mathbf{y}$.

symmetric

2. d is *symmetric*, i.e., $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in V$.

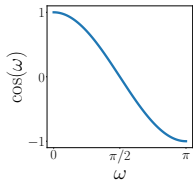
triangle inequality

3. *Triangle inequality*: $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$ for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$.

Remark. At first glance, the lists of properties of inner products and metrics look very similar. However, by comparing Definition 3.3 with Definition 3.6 we observe that $\langle \mathbf{x}, \mathbf{y} \rangle$ and $d(\mathbf{x}, \mathbf{y})$ behave in opposite directions. Very similar $\mathbf{x}$ and $\mathbf{y}$ will result in a large value for the inner product and a small value for the metric. $\diamond$

3.4 Angles and Orthogonality

Figure 3.4 When restricted to $[0, \pi]$ then $f(\omega) = \cos(\omega)$ returns a unique number in the interval $[-1, 1]$.



In addition to enabling the definition of lengths of vectors, as well as the distance between two vectors, inner products also capture the geometry of a vector space by defining the angle ω between two vectors. We use the Cauchy-Schwarz inequality (3.17) to define angles ω in inner product spaces between two vectors $\mathbf{x}, \mathbf{y}$, and this notion coincides with our intuition in $\mathbb{R}^2$ and $\mathbb{R}^3$. Assume that $\mathbf{x} \neq \mathbf{0}, \mathbf{y} \neq \mathbf{0}$. Then

$$-1 \leq \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} \leq 1. \quad (3.24)$$

Therefore, there exists a unique $\omega \in [0, \pi]$, illustrated in Figure 3.4, with

$$\cos \omega = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}. \quad (3.25)$$

angle

The number ω is the *angle* between the vectors $\mathbf{x}$ and $\mathbf{y}$. Intuitively, the angle between two vectors tells us how similar their orientations are. For example, using the dot product, the angle between $\mathbf{x}$ and $\mathbf{y} = 4\mathbf{x}$, i.e., $\mathbf{y}$ is a scaled version of $\mathbf{x}$, is 0: Their orientation is the same.

Example 3.6 (Angle between Vectors)

Let us compute the angle between $\mathbf{x} = [1, 1]^\top \in \mathbb{R}^2$ and $\mathbf{y} = [1, 2]^\top \in \mathbb{R}^2$; see Figure 3.5, where we use the dot product as the inner product. Then we get

$$\cos \omega = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\sqrt{\langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle}} = \frac{\mathbf{x}^\top \mathbf{y}}{\sqrt{\mathbf{x}^\top \mathbf{x} \mathbf{y}^\top \mathbf{y}}} = \frac{3}{\sqrt{10}}, \quad (3.26)$$

and the angle between the two vectors is $\arccos(\frac{3}{\sqrt{10}}) \approx 0.32$ rad, which corresponds to about 18° .

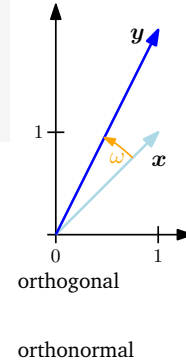
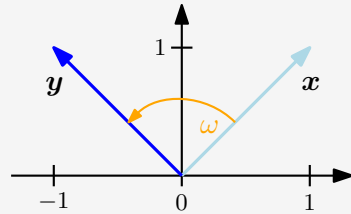
A key feature of the inner product is that it also allows us to characterize vectors that are orthogonal.

Definition 3.7 (Orthogonality). Two vectors $\mathbf{x}$ and $\mathbf{y}$ are *orthogonal* if and only if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$, and we write $\mathbf{x} \perp \mathbf{y}$. If additionally $\|\mathbf{x}\| = 1 = \|\mathbf{y}\|$, i.e., the vectors are unit vectors, then $\mathbf{x}$ and $\mathbf{y}$ are *orthonormal*.

An implication of this definition is that the $\mathbf{0}$ -vector is orthogonal to every vector in the vector space.

Remark. Orthogonality is the generalization of the concept of perpendicularity to bilinear forms that do not have to be the dot product. In our context, geometrically, we can think of orthogonal vectors as having a right angle with respect to a specific inner product. $\diamond$

Figure 3.5 The angle ω between two vectors $\mathbf{x}, \mathbf{y}$ is computed using the inner product.

**Example 3.7 (Orthogonal Vectors)**

Consider two vectors $\mathbf{x} = [1, 1]^\top, \mathbf{y} = [-1, 1]^\top \in \mathbb{R}^2$; see Figure 3.6. We are interested in determining the angle ω between them using two different inner products. Using the dot product as the inner product yields an angle ω between $\mathbf{x}$ and $\mathbf{y}$ of 90° , such that $\mathbf{x} \perp \mathbf{y}$. However, if we choose the inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{y}, \quad (3.27)$$

Figure 3.6 The angle ω between two vectors $\mathbf{x}, \mathbf{y}$ can change depending on the inner product.

we get that the angle ω between $\mathbf{x}$ and $\mathbf{y}$ is given by

$$\cos \omega = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|} = -\frac{1}{3} \implies \omega \approx 1.91 \text{ rad} \approx 109.5^\circ, \quad (3.28)$$

and $\mathbf{x}$ and $\mathbf{y}$ are not orthogonal. Therefore, vectors that are orthogonal with respect to one inner product do not have to be orthogonal with respect to a different inner product.

orthogonal matrix

Definition 3.8 (Orthogonal Matrix). A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is an *orthogonal matrix* if and only if its columns are orthonormal so that

$$\mathbf{A}\mathbf{A}^\top = \mathbf{I} = \mathbf{A}^\top \mathbf{A}, \quad (3.29)$$

which implies that

$$\mathbf{A}^{-1} = \mathbf{A}^\top, \quad (3.30)$$

i.e., the inverse is obtained by simply transposing the matrix.

It is convention to call these matrices “orthogonal” but a more precise description would be “orthonormal”. Transformations with orthogonal matrices preserve distances and angles.

Transformations by orthogonal matrices are special because the length of a vector $\mathbf{x}$ is not changed when transforming it using an orthogonal matrix $\mathbf{A}$. For the dot product, we obtain

$$\|\mathbf{Ax}\|^2 = (\mathbf{Ax})^\top (\mathbf{Ax}) = \mathbf{x}^\top \mathbf{A}^\top \mathbf{Ax} = \mathbf{x}^\top \mathbf{Ix} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|^2. \quad (3.31)$$

Moreover, the angle between any two vectors $\mathbf{x}, \mathbf{y}$, as measured by their inner product, is also unchanged when transforming both of them using an orthogonal matrix $\mathbf{A}$. Assuming the dot product as the inner product, the angle of the images $\mathbf{Ax}$ and $\mathbf{Ay}$ is given as

$$\cos \omega = \frac{(\mathbf{Ax})^\top (\mathbf{Ay})}{\|\mathbf{Ax}\| \|\mathbf{Ay}\|} = \frac{\mathbf{x}^\top \mathbf{A}^\top \mathbf{Ay}}{\sqrt{\mathbf{x}^\top \mathbf{A}^\top \mathbf{Ax} \mathbf{y}^\top \mathbf{A}^\top \mathbf{Ay}}} = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}, \quad (3.32)$$

which gives exactly the angle between $\mathbf{x}$ and $\mathbf{y}$. This means that orthogonal matrices $\mathbf{A}$ with $\mathbf{A}^\top = \mathbf{A}^{-1}$ preserve both angles and distances. It turns out that orthogonal matrices define transformations that are rotations (with the possibility of flips). In Section 3.9, we will discuss more details about rotations.

3.5 Orthonormal Basis

In Section 2.6.1, we characterized properties of basis vectors and found that in an n -dimensional vector space, we need n basis vectors, i.e., n vectors that are linearly independent. In Sections 3.3 and 3.4, we used inner products to compute the length of vectors and the angle between vectors. In the following, we will discuss the special case where the basis vectors are orthogonal to each other and where the length of each basis vector is 1. We will call this basis then an orthonormal basis.

Let us introduce this more formally.

Definition 3.9 (Orthonormal Basis). Consider an n -dimensional vector space V and a basis $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ of V . If

$$\langle \mathbf{b}_i, \mathbf{b}_j \rangle = 0 \quad \text{for } i \neq j \quad (3.33)$$

$$\langle \mathbf{b}_i, \mathbf{b}_i \rangle = 1 \quad (3.34)$$

for all $i, j = 1, \dots, n$ then the basis is called an *orthonormal basis* (ONB). If only (3.33) is satisfied, then the basis is called an *orthogonal basis*. Note that (3.34) implies that every basis vector has length/norm 1.

orthonormal basis
ONB
orthogonal basis

Recall from Section 2.6.1 that we can use Gaussian elimination to find a basis for a vector space spanned by a set of vectors. Assume we are given a set $\{\tilde{\mathbf{b}}_1, \dots, \tilde{\mathbf{b}}_n\}$ of non-orthogonal and unnormalized basis vectors. We concatenate them into a matrix $\tilde{\mathbf{B}} = [\tilde{\mathbf{b}}_1, \dots, \tilde{\mathbf{b}}_n]$ and apply Gaussian elimination to the augmented matrix (Section 2.3.2) $[\tilde{\mathbf{B}}\tilde{\mathbf{B}}^\top | \tilde{\mathbf{B}}]$ to obtain an orthonormal basis. This constructive way to iteratively build an orthonormal basis $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is called the *Gram-Schmidt process* (Strang, 2003).

Example 3.8 (Orthonormal Basis)

The canonical/standard basis for a Euclidean vector space $\mathbb{R}^n$ is an orthonormal basis, where the inner product is the dot product of vectors.

In $\mathbb{R}^2$, the vectors

$$\mathbf{b}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (3.35)$$

form an orthonormal basis since $\mathbf{b}_1^\top \mathbf{b}_2 = 0$ and $\|\mathbf{b}_1\| = 1 = \|\mathbf{b}_2\|$.

We will exploit the concept of an orthonormal basis in Chapter 12 and Chapter 10 when we discuss support vector machines and principal component analysis.

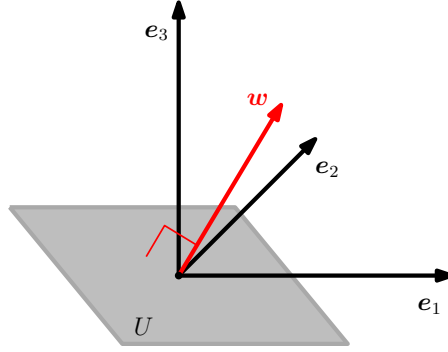
3.6 Orthogonal Complement

Having defined orthogonality, we will now look at vector spaces that are orthogonal to each other. This will play an important role in Chapter 10, when we discuss linear dimensionality reduction from a geometric perspective.

Consider a D -dimensional vector space V and an M -dimensional subspace $U \subseteq V$. Then its *orthogonal complement* $U^\perp$ is a $(D-M)$ -dimensional subspace of V and contains all vectors in V that are orthogonal to every vector in U . Furthermore, $U \cap U^\perp = \{\mathbf{0}\}$ so that any vector $\mathbf{x} \in V$ can be

orthogonal
complement

Figure 3.7 A plane U in a three-dimensional vector space can be described by its normal vector, which spans its orthogonal complement $U^\perp$.



uniquely decomposed into

$$\mathbf{x} = \sum_{m=1}^M \lambda_m \mathbf{b}_m + \sum_{j=1}^{D-M} \psi_j \mathbf{b}_j^\perp, \quad \lambda_m, \psi_j \in \mathbb{R}, \quad (3.36)$$

where $(\mathbf{b}_1, \dots, \mathbf{b}_M)$ is a basis of U and $(\mathbf{b}_1^\perp, \dots, \mathbf{b}_{D-M}^\perp)$ is a basis of $U^\perp$.

Therefore, the orthogonal complement can also be used to describe a plane U (two-dimensional subspace) in a three-dimensional vector space. More specifically, the vector $\mathbf{w}$ with $\|\mathbf{w}\| = 1$, which is orthogonal to the plane U , is the basis vector of $U^\perp$. Figure 3.7 illustrates this setting. All vectors that are orthogonal to $\mathbf{w}$ must (by construction) lie in the plane U . The vector $\mathbf{w}$ is called the *normal vector* of U .

normal vector

Generally, orthogonal complements can be used to describe hyperplanes in n -dimensional vector and affine spaces.

3.7 Inner Product of Functions

Thus far, we looked at properties of inner products to compute lengths, angles and distances. We focused on inner products of finite-dimensional vectors. In the following, we will look at an example of inner products of a different type of vectors: inner products of functions.

The inner products we discussed so far were defined for vectors with a finite number of entries. We can think of a vector $\mathbf{x} \in \mathbb{R}^n$ as a function with n function values. The concept of an inner product can be generalized to vectors with an infinite number of entries (countably infinite) and also continuous-valued functions (uncountably infinite). Then the sum over individual components of vectors (see Equation (3.5) for example) turns into an integral.

An inner product of two functions $u : \mathbb{R} \rightarrow \mathbb{R}$ and $v : \mathbb{R} \rightarrow \mathbb{R}$ can be defined as the definite integral

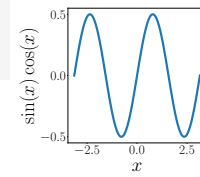
$$\langle u, v \rangle := \int_a^b u(x)v(x)dx \quad (3.37)$$

for lower and upper limits $a, b < \infty$, respectively. As with our usual inner product, we can define norms and orthogonality by looking at the inner product. If (3.37) evaluates to 0, the functions u and v are orthogonal. To make the preceding inner product mathematically precise, we need to take care of measures and the definition of integrals, leading to the definition of a Hilbert space. Furthermore, unlike inner products on finite-dimensional vectors, inner products on functions may diverge (have infinite value). All this requires diving into some more intricate details of real and functional analysis, which we do not cover in this book.

Example 3.9 (Inner Product of Functions)

If we choose $u = \sin(x)$ and $v = \cos(x)$, the integrand $f(x) = u(x)v(x)$ of (3.37), is shown in Figure 3.8. We see that this function is odd, i.e., $f(-x) = -f(x)$. Therefore, the integral with limits $a = -\pi, b = \pi$ of this product evaluates to 0. Therefore, $\sin$ and $\cos$ are orthogonal functions.

Figure 3.8 $f(x) = \sin(x)\cos(x)$.



Remark. It also holds that the collection of functions

$$\{1, \cos(x), \cos(2x), \cos(3x), \dots\} \quad (3.38)$$

is orthogonal if we integrate from $-\pi$ to π , i.e., any pair of functions are orthogonal to each other. The collection of functions in (3.38) spans a large subspace of the functions that are even and periodic on $[-\pi, \pi)$, and projecting functions onto this subspace is the fundamental idea behind Fourier series. $\diamond$

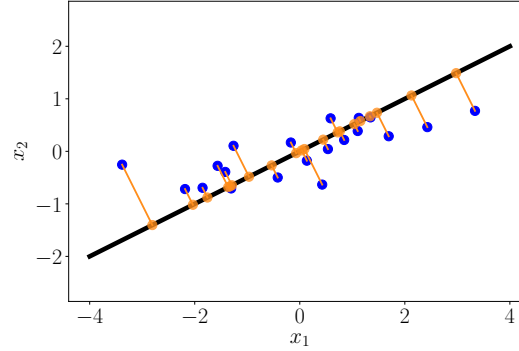
In Section 6.4.6, we will have a look at a second type of unconventional inner products: the inner product of random variables.

3.8 Orthogonal Projections

Projections are an important class of linear transformations (besides rotations and reflections) and play an important role in graphics, coding theory, statistics and machine learning. In machine learning, we often deal with data that is high-dimensional. High-dimensional data is often hard to analyze or visualize. However, high-dimensional data quite often possesses the property that only a few dimensions contain most information, and most other dimensions are not essential to describe key properties of the data. When we compress or visualize high-dimensional data, we will lose information. To minimize this compression loss, we ideally find the most informative dimensions in the data. As discussed in Chapter 1, data can be represented as vectors, and in this chapter, we will discuss some of the fundamental tools for data compression. More specifically, we can project the original high-dimensional data onto a lower-dimensional feature space and work in this lower-dimensional space to learn more about the dataset and extract relevant patterns. For example, machine

“Feature” is a common expression for data representation.

Figure 3.9
Orthogonal
projection (orange
dots) of a
two-dimensional
dataset (blue dots)
onto a
one-dimensional
subspace (straight
line).



learning algorithms, such as principal component analysis (PCA) by Pearson (1901) and Hotelling (1933) and deep neural networks (e.g., deep auto-encoders (Deng et al., 2010)), heavily exploit the idea of dimensionality reduction. In the following, we will focus on orthogonal projections, which we will use in Chapter 10 for linear dimensionality reduction and in Chapter 12 for classification. Even linear regression, which we discuss in Chapter 9, can be interpreted using orthogonal projections. For a given lower-dimensional subspace, orthogonal projections of high-dimensional data retain as much information as possible and minimize the difference/error between the original data and the corresponding projection. An illustration of such an orthogonal projection is given in Figure 3.9. Before we detail how to obtain these projections, let us define what a projection actually is.

projection

Definition 3.10 (Projection). Let V be a vector space and $U \subseteq V$ a subspace of V . A linear mapping $\pi : V \rightarrow U$ is called a *projection* if $\pi^2 = \pi \circ \pi = \pi$.

projection matrix

Since linear mappings can be expressed by transformation matrices (see Section 2.7), the preceding definition applies equally to a special kind of transformation matrices, the *projection matrices* P_π , which exhibit the property that $P_\pi^2 = P_\pi$.

line

In the following, we will derive orthogonal projections of vectors in the inner product space $(\mathbb{R}^n, \langle \cdot, \cdot \rangle)$ onto subspaces. We will start with one-dimensional subspaces, which are also called *lines*. If not mentioned otherwise, we assume the dot product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y}$ as the inner product.

3.8.1 Projection onto One-Dimensional Subspaces (Lines)

Assume we are given a line (one-dimensional subspace) through the origin with basis vector $\mathbf{b} \in \mathbb{R}^n$. The line is a one-dimensional subspace $U \subseteq \mathbb{R}^n$ spanned by $\mathbf{b}$. When we project $\mathbf{x} \in \mathbb{R}^n$ onto U , we seek the vector $\pi_U(\mathbf{x}) \in U$ that is closest to $\mathbf{x}$. Using geometric arguments, let

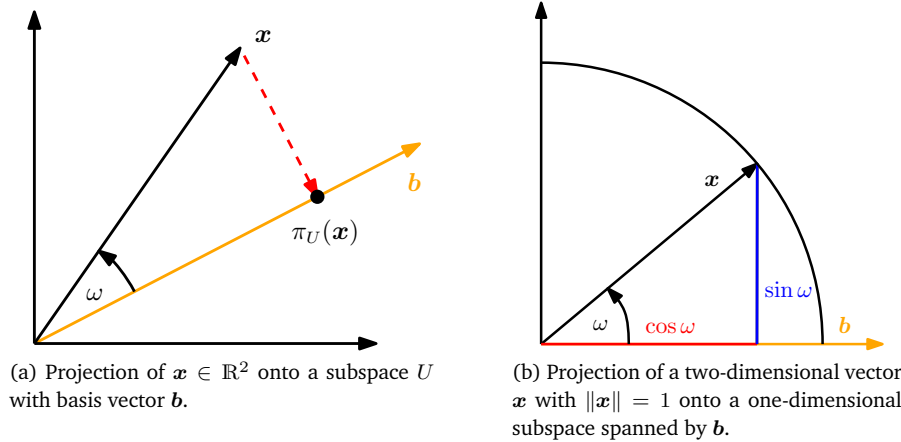


Figure 3.10
Examples of
projections onto
one-dimensional
subspaces.

us characterize some properties of the projection $\pi_U(\mathbf{x})$ (Figure 3.10(a) serves as an illustration):

- The projection $\pi_U(\mathbf{x})$ is closest to $\mathbf{x}$, where “closest” implies that the distance $\|\mathbf{x} - \pi_U(\mathbf{x})\|$ is minimal. It follows that the segment $\pi_U(\mathbf{x}) - \mathbf{x}$ from $\pi_U(\mathbf{x})$ to $\mathbf{x}$ is orthogonal to U , and therefore the basis vector $\mathbf{b}$ of U . The orthogonality condition yields $\langle \pi_U(\mathbf{x}) - \mathbf{x}, \mathbf{b} \rangle = 0$ since angles between vectors are defined via the inner product.
- The projection $\pi_U(\mathbf{x})$ of $\mathbf{x}$ onto U must be an element of U and, therefore, a multiple of the basis vector $\mathbf{b}$ that spans U . Hence, $\pi_U(\mathbf{x}) = \lambda \mathbf{b}$, for some $\lambda \in \mathbb{R}$.

λ is then the
coordinate of $\pi_U(\mathbf{x})$
with respect to $\mathbf{b}$.

In the following three steps, we determine the coordinate λ , the projection $\pi_U(\mathbf{x}) \in U$, and the projection matrix $\mathbf{P}_\pi$ that maps any $\mathbf{x} \in \mathbb{R}^n$ onto U :

1. Finding the coordinate λ . The orthogonality condition yields

$$\langle \mathbf{x} - \pi_U(\mathbf{x}), \mathbf{b} \rangle = 0 \stackrel{\pi_U(\mathbf{x}) = \lambda \mathbf{b}}{\iff} \langle \mathbf{x} - \lambda \mathbf{b}, \mathbf{b} \rangle = 0. \quad (3.39)$$

We can now exploit the bilinearity of the inner product and arrive at

$$\langle \mathbf{x}, \mathbf{b} \rangle - \lambda \langle \mathbf{b}, \mathbf{b} \rangle = 0 \iff \lambda = \frac{\langle \mathbf{x}, \mathbf{b} \rangle}{\langle \mathbf{b}, \mathbf{b} \rangle} = \frac{\langle \mathbf{b}, \mathbf{x} \rangle}{\|\mathbf{b}\|^2}. \quad (3.40)$$

With a general inner
product, we get
 $\lambda = \langle \mathbf{x}, \mathbf{b} \rangle$ if
 $\|\mathbf{b}\| = 1$.

In the last step, we exploited the fact that inner products are symmetric. If we choose $\langle \cdot, \cdot \rangle$ to be the dot product, we obtain

$$\lambda = \frac{\mathbf{b}^\top \mathbf{x}}{\mathbf{b}^\top \mathbf{b}} = \frac{\mathbf{b}^\top \mathbf{x}}{\|\mathbf{b}\|^2}. \quad (3.41)$$

If $\|\mathbf{b}\| = 1$, then the coordinate λ of the projection is given by $\mathbf{b}^\top \mathbf{x}$.

2. Finding the projection point $\pi_U(\mathbf{x}) \in U$. Since $\pi_U(\mathbf{x}) = \lambda \mathbf{b}$, we immediately obtain with (3.40) that

$$\pi_U(\mathbf{x}) = \lambda \mathbf{b} = \frac{\langle \mathbf{x}, \mathbf{b} \rangle}{\|\mathbf{b}\|^2} \mathbf{b} = \frac{\mathbf{b}^\top \mathbf{x}}{\|\mathbf{b}\|^2} \mathbf{b}, \quad (3.42)$$

where the last equality holds for the dot product only. We can also compute the length of $\pi_U(\mathbf{x})$ by means of Definition 3.1 as

$$\|\pi_U(\mathbf{x})\| = \|\lambda \mathbf{b}\| = |\lambda| \|\mathbf{b}\|. \quad (3.43)$$

Hence, our projection is of length $|\lambda|$ times the length of $\mathbf{b}$. This also adds the intuition that λ is the coordinate of $\pi_U(\mathbf{x})$ with respect to the basis vector $\mathbf{b}$ that spans our one-dimensional subspace U .

If we use the dot product as an inner product, we get

$$\|\pi_U(\mathbf{x})\| \stackrel{(3.42)}{=} \frac{|\mathbf{b}^\top \mathbf{x}|}{\|\mathbf{b}\|^2} \|\mathbf{b}\| \stackrel{(3.25)}{=} |\cos \omega| \|\mathbf{x}\| \|\mathbf{b}\| \frac{\|\mathbf{b}\|}{\|\mathbf{b}\|^2} = |\cos \omega| \|\mathbf{x}\|. \quad (3.44)$$

Here, ω is the angle between $\mathbf{x}$ and $\mathbf{b}$. This equation should be familiar from trigonometry: If $\|\mathbf{x}\| = 1$, then $\mathbf{x}$ lies on the unit circle. It follows that the projection onto the horizontal axis spanned by $\mathbf{b}$ is exactly $\cos \omega$, and the length of the corresponding vector $\pi_U(\mathbf{x}) = |\cos \omega|$. An illustration is given in Figure 3.10(b).

3. Finding the projection matrix $\mathbf{P}_\pi$. We know that a projection is a linear mapping (see Definition 3.10). Therefore, there exists a projection matrix $\mathbf{P}_\pi$, such that $\pi_U(\mathbf{x}) = \mathbf{P}_\pi \mathbf{x}$. With the dot product as inner product and

$$\pi_U(\mathbf{x}) = \lambda \mathbf{b} = \mathbf{b} \lambda = \mathbf{b} \frac{\mathbf{b}^\top \mathbf{x}}{\|\mathbf{b}\|^2} = \frac{\mathbf{b} \mathbf{b}^\top}{\|\mathbf{b}\|^2} \mathbf{x}, \quad (3.45)$$

we immediately see that

$$\mathbf{P}_\pi = \frac{\mathbf{b} \mathbf{b}^\top}{\|\mathbf{b}\|^2}. \quad (3.46)$$

Note that $\mathbf{b} \mathbf{b}^\top$ (and, consequently, $\mathbf{P}_\pi$) is a symmetric matrix (of rank 1), and $\|\mathbf{b}\|^2 = \langle \mathbf{b}, \mathbf{b} \rangle$ is a scalar.

The projection matrix $\mathbf{P}_\pi$ projects any vector $\mathbf{x} \in \mathbb{R}^n$ onto the line through the origin with direction $\mathbf{b}$ (equivalently, the subspace U spanned by $\mathbf{b}$).

Remark. The projection $\pi_U(\mathbf{x}) \in \mathbb{R}^n$ is still an n -dimensional vector and not a scalar. However, we no longer require n coordinates to represent the projection, but only a single one if we want to express it with respect to the basis vector $\mathbf{b}$ that spans the subspace U : λ . $\diamond$

The horizontal axis is a one-dimensional subspace.

Projection matrices are always symmetric.

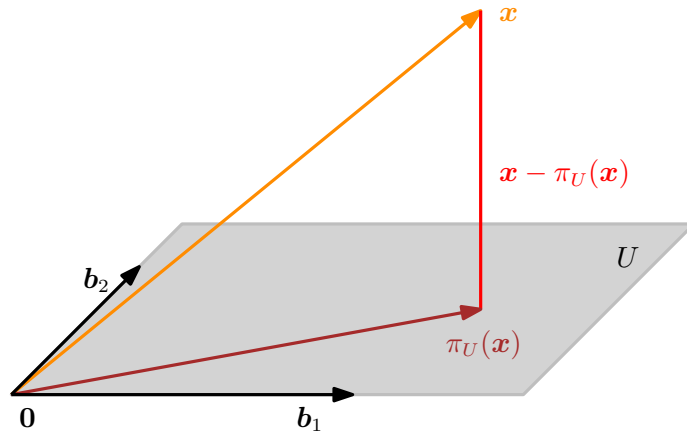


Figure 3.11
Projection onto a two-dimensional subspace U with basis b_1, b_2 . The projection $\pi_U(\mathbf{x})$ of $\mathbf{x} \in \mathbb{R}^3$ onto U can be expressed as a linear combination of b_1, b_2 and the displacement vector $\mathbf{x} - \pi_U(\mathbf{x})$ is orthogonal to both b_1 and b_2 .

Example 3.10 (Projection onto a Line)

Find the projection matrix P_π onto the line through the origin spanned by $\mathbf{b} = [1 \ 2 \ 2]^\top$. $\mathbf{b}$ is a direction and a basis of the one-dimensional subspace (line through origin).

With (3.46), we obtain

$$P_\pi = \frac{\mathbf{b}\mathbf{b}^\top}{\mathbf{b}^\top\mathbf{b}} = \frac{1}{9} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} [1 \ 2 \ 2] = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}. \quad (3.47)$$

Let us now choose a particular $\mathbf{x}$ and see whether it lies in the subspace spanned by $\mathbf{b}$. For $\mathbf{x} = [1 \ 1 \ 1]^\top$, the projection is

$$\pi_U(\mathbf{x}) = P_\pi \mathbf{x} = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 5 \\ 10 \\ 10 \end{bmatrix} \in \text{span} \left[\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \right]. \quad (3.48)$$

Note that the application of P_π to $\pi_U(\mathbf{x})$ does not change anything, i.e., $P_\pi \pi_U(\mathbf{x}) = \pi_U(\mathbf{x})$. This is expected because according to Definition 3.10, we know that a projection matrix P_π satisfies $P_\pi^2 \mathbf{x} = P_\pi \mathbf{x}$ for all $\mathbf{x}$.

Remark. With the results from Chapter 4, we can show that $\pi_U(\mathbf{x})$ is an eigenvector of P_π , and the corresponding eigenvalue is 1. $\diamond$

3.8.2 Projection onto General Subspaces

In the following, we look at orthogonal projections of vectors $\mathbf{x} \in \mathbb{R}^n$ onto lower-dimensional subspaces $U \subseteq \mathbb{R}^n$ with $\dim(U) = m \geq 1$. An illustration is given in Figure 3.11.

Assume that $(b_1, \dots, b_m)$ is an ordered basis of U . Any projection $\pi_U(\mathbf{x})$ onto U is necessarily an element of U . Therefore, they can be represented

If U is given by a set of spanning vectors, which are not a basis, make sure you determine a basis $b_1, \dots, b_m$ before proceeding.

The basis vectors form the columns of $B \in \mathbb{R}^{n \times m}$, where $B = [\mathbf{b}_1, \dots, \mathbf{b}_m]$.

as linear combinations of the basis vectors $\mathbf{b}_1, \dots, \mathbf{b}_m$ of U , such that $\pi_U(\mathbf{x}) = \sum_{i=1}^m \lambda_i \mathbf{b}_i$.

As in the 1D case, we follow a three-step procedure to find the projection $\pi_U(\mathbf{x})$ and the projection matrix P_π :

1. Find the coordinates $\lambda_1, \dots, \lambda_m$ of the projection (with respect to the basis of U), such that the linear combination

$$\pi_U(\mathbf{x}) = \sum_{i=1}^m \lambda_i \mathbf{b}_i = B\boldsymbol{\lambda}, \quad (3.49)$$

$$B = [\mathbf{b}_1, \dots, \mathbf{b}_m] \in \mathbb{R}^{n \times m}, \quad \boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_m]^\top \in \mathbb{R}^m, \quad (3.50)$$

is closest to $\mathbf{x} \in \mathbb{R}^n$. As in the 1D case, “closest” means “minimum distance”, which implies that the vector connecting $\pi_U(\mathbf{x}) \in U$ and $\mathbf{x} \in \mathbb{R}^n$ must be orthogonal to all basis vectors of U . Therefore, we obtain m simultaneous conditions (assuming the dot product as the inner product)

$$\langle \mathbf{b}_1, \mathbf{x} - \pi_U(\mathbf{x}) \rangle = \mathbf{b}_1^\top (\mathbf{x} - \pi_U(\mathbf{x})) = 0 \quad (3.51)$$

$$\vdots$$

$$\langle \mathbf{b}_m, \mathbf{x} - \pi_U(\mathbf{x}) \rangle = \mathbf{b}_m^\top (\mathbf{x} - \pi_U(\mathbf{x})) = 0 \quad (3.52)$$

which, with $\pi_U(\mathbf{x}) = B\boldsymbol{\lambda}$, can be written as

$$\mathbf{b}_1^\top (\mathbf{x} - B\boldsymbol{\lambda}) = 0 \quad (3.53)$$

$$\vdots$$

$$\mathbf{b}_m^\top (\mathbf{x} - B\boldsymbol{\lambda}) = 0 \quad (3.54)$$

such that we obtain a homogeneous linear equation system

$$\begin{bmatrix} \mathbf{b}_1^\top \\ \vdots \\ \mathbf{b}_m^\top \end{bmatrix} \begin{bmatrix} \mathbf{x} - B\boldsymbol{\lambda} \end{bmatrix} = \mathbf{0} \iff B^\top (\mathbf{x} - B\boldsymbol{\lambda}) = \mathbf{0} \quad (3.55)$$

$$\iff B^\top B\boldsymbol{\lambda} = B^\top \mathbf{x}. \quad (3.56)$$

normal equation

The last expression is called *normal equation*. Since $\mathbf{b}_1, \dots, \mathbf{b}_m$ are a basis of U and, therefore, linearly independent, $B^\top B \in \mathbb{R}^{m \times m}$ is regular and can be inverted. This allows us to solve for the coefficients/coordinates

$$\boldsymbol{\lambda} = (B^\top B)^{-1} B^\top \mathbf{x}. \quad (3.57)$$

pseudo-inverse

The matrix $(B^\top B)^{-1} B^\top$ is also called the *pseudo-inverse* of B , which can be computed for non-square matrices B . It only requires that $B^\top B$ is positive definite, which is the case if B is full rank. In practical applications (e.g., linear regression), we often add a “jitter term” ϵI to

$B^\top B$ to guarantee increased numerical stability and positive definiteness. This “ridge” can be rigorously derived using Bayesian inference. See Chapter 9 for details.

- Find the projection $\pi_U(\mathbf{x}) \in U$. We already established that $\pi_U(\mathbf{x}) = B\boldsymbol{\lambda}$. Therefore, with (3.57)

$$\pi_U(\mathbf{x}) = B(B^\top B)^{-1}B^\top \mathbf{x}. \quad (3.58)$$

- Find the projection matrix P_π . From (3.58), we can immediately see that the projection matrix that solves $P_\pi \mathbf{x} = \pi_U(\mathbf{x})$ must be

$$P_\pi = B(B^\top B)^{-1}B^\top. \quad (3.59)$$

Remark. The solution for projecting onto general subspaces includes the 1D case as a special case: If $\dim(U) = 1$, then $B^\top B \in \mathbb{R}$ is a scalar and we can rewrite the projection matrix in (3.59) $P_\pi = B(B^\top B)^{-1}B^\top$ as $P_\pi = \frac{BB^\top}{B^\top B}$, which is exactly the projection matrix in (3.46). $\diamond$

Example 3.11 (Projection onto a Two-dimensional Subspace)

For a subspace $U = \text{span}\left[\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}\right] \subseteq \mathbb{R}^3$ and $\mathbf{x} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^3$ find the coordinates $\boldsymbol{\lambda}$ of $\mathbf{x}$ in terms of the subspace U , the projection point $\pi_U(\mathbf{x})$ and the projection matrix P_π .

First, we see that the generating set of U is a basis (linear independence) and write the basis vectors of U into a matrix $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$.

Second, we compute the matrix $B^\top B$ and the vector $B^\top \mathbf{x}$ as

$$B^\top B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \quad B^\top \mathbf{x} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}. \quad (3.60)$$

Third, we solve the normal equation $B^\top B\boldsymbol{\lambda} = B^\top \mathbf{x}$ to find $\boldsymbol{\lambda}$:

$$\begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix} \iff \boldsymbol{\lambda} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}. \quad (3.61)$$

Fourth, the projection $\pi_U(\mathbf{x})$ of $\mathbf{x}$ onto U , i.e., into the column space of B , can be directly computed via

$$\pi_U(\mathbf{x}) = B\boldsymbol{\lambda} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}. \quad (3.62)$$

projection error

The projection error is also called the *reconstruction error*.

The corresponding *projection error* is the norm of the difference vector between the original vector and its projection onto U , i.e.,

$$\|\mathbf{x} - \pi_U(\mathbf{x})\| = \left\| \begin{bmatrix} 1 & -2 & 1 \end{bmatrix}^\top \right\| = \sqrt{6}. \quad (3.63)$$

Fifth, the projection matrix (for any $\mathbf{x} \in \mathbb{R}^3$) is given by

$$\mathbf{P}_\pi = \mathbf{B}(\mathbf{B}^\top \mathbf{B})^{-1} \mathbf{B}^\top = \frac{1}{6} \begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}. \quad (3.64)$$

To verify the results, we can (a) check whether the displacement vector $\pi_U(\mathbf{x}) - \mathbf{x}$ is orthogonal to all basis vectors of U , and (b) verify that $\mathbf{P}_\pi = \mathbf{P}_\pi^2$ (see Definition 3.10).

Remark. The projections $\pi_U(\mathbf{x})$ are still vectors in $\mathbb{R}^n$ although they lie in an m -dimensional subspace $U \subseteq \mathbb{R}^n$. However, to represent a projected vector we only need the m coordinates $\lambda_1, \dots, \lambda_m$ with respect to the basis vectors $\mathbf{b}_1, \dots, \mathbf{b}_m$ of U . $\diamond$

Remark. In vector spaces with general inner products, we have to pay attention when computing angles and distances, which are defined by means of the inner product. $\diamond$

We can find approximate solutions to unsolvable linear equation systems using projections.

Projections allow us to look at situations where we have a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ without a solution. Recall that this means that $\mathbf{b}$ does not lie in the span of $\mathbf{A}$, i.e., the vector $\mathbf{b}$ does not lie in the subspace spanned by the columns of $\mathbf{A}$. Given that the linear equation cannot be solved exactly, we can find an *approximate solution*. The idea is to find the vector in the subspace spanned by the columns of $\mathbf{A}$ that is closest to $\mathbf{b}$, i.e., we compute the orthogonal projection of $\mathbf{b}$ onto the subspace spanned by the columns of $\mathbf{A}$. This problem arises often in practice, and the solution is called the *least-squares solution* (assuming the dot product as the inner product) of an overdetermined system. This is discussed further in Section 9.4. Using reconstruction errors (3.63) is one possible approach to derive principal component analysis (Section 10.3).

least-squares solution

Remark. We just looked at projections of vectors $\mathbf{x}$ onto a subspace U with basis vectors $\{\mathbf{b}_1, \dots, \mathbf{b}_k\}$. If this basis is an ONB, i.e., (3.33) and (3.34) are satisfied, the projection equation (3.58) simplifies greatly to

$$\pi_U(\mathbf{x}) = \mathbf{B}\mathbf{B}^\top \mathbf{x} \quad (3.65)$$

since $\mathbf{B}^\top \mathbf{B} = \mathbf{I}$ with coordinates

$$\boldsymbol{\lambda} = \mathbf{B}^\top \mathbf{x}. \quad (3.66)$$

This means that we no longer have to compute the inverse from (3.58), which saves computation time. $\diamond$

3.8.3 Gram-Schmidt Orthogonalization

Projections are at the core of the Gram-Schmidt method that allows us to constructively transform any basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ of an n -dimensional vector space V into an orthogonal/orthonormal basis $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ of V . This basis always exists (Liesen and Mehrmann, 2015) and $\text{span}[\mathbf{b}_1, \dots, \mathbf{b}_n] = \text{span}[\mathbf{u}_1, \dots, \mathbf{u}_n]$. The *Gram-Schmidt orthogonalization* method iteratively constructs an orthogonal basis $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ from any basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ of V as follows:

Gram-Schmidt
orthogonalization

$$\mathbf{u}_1 := \mathbf{b}_1 \quad (3.67)$$

$$\mathbf{u}_k := \mathbf{b}_k - \pi_{\text{span}[\mathbf{u}_1, \dots, \mathbf{u}_{k-1}]}(\mathbf{b}_k), \quad k = 2, \dots, n. \quad (3.68)$$

In (3.68), the k th basis vector $\mathbf{b}_k$ is projected onto the subspace spanned by the first $k - 1$ constructed orthogonal vectors $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$; see Section 3.8.2. This projection is then subtracted from $\mathbf{b}_k$ and yields a vector $\mathbf{u}_k$ that is orthogonal to the $(k - 1)$ -dimensional subspace spanned by $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$. Repeating this procedure for all n basis vectors $\mathbf{b}_1, \dots, \mathbf{b}_n$ yields an orthogonal basis $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ of V . If we normalize the $\mathbf{u}_k$, we obtain an ONB where $\|\mathbf{u}_k\| = 1$ for $k = 1, \dots, n$.

Example 3.12 (Gram-Schmidt Orthogonalization)

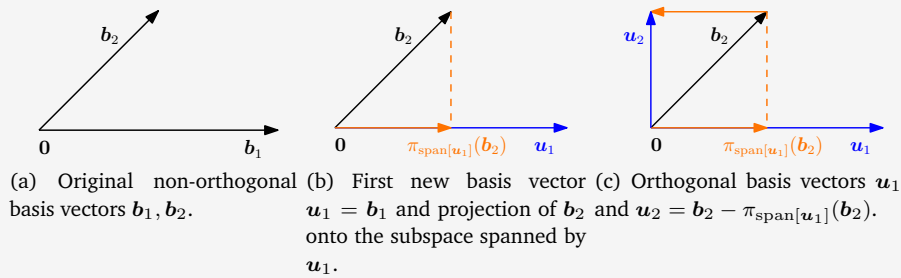


Figure 3.12
Gram-Schmidt
orthogonalization.
(a) non-orthogonal
basis $(\mathbf{b}_1, \mathbf{b}_2)$ of $\mathbb{R}^2$;
(b) first constructed
basis vector $\mathbf{u}_1$ and
orthogonal
projection of $\mathbf{b}_2$
onto $\text{span}[\mathbf{u}_1]$;
(c) orthogonal basis
 $(\mathbf{u}_1, \mathbf{u}_2)$ of $\mathbb{R}^2$.

Consider a basis $(\mathbf{b}_1, \mathbf{b}_2)$ of $\mathbb{R}^2$, where

$$\mathbf{b}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \quad (3.69)$$

see also Figure 3.12(a). Using the Gram-Schmidt method, we construct an orthogonal basis $(\mathbf{u}_1, \mathbf{u}_2)$ of $\mathbb{R}^2$ as follows (assuming the dot product as the inner product):

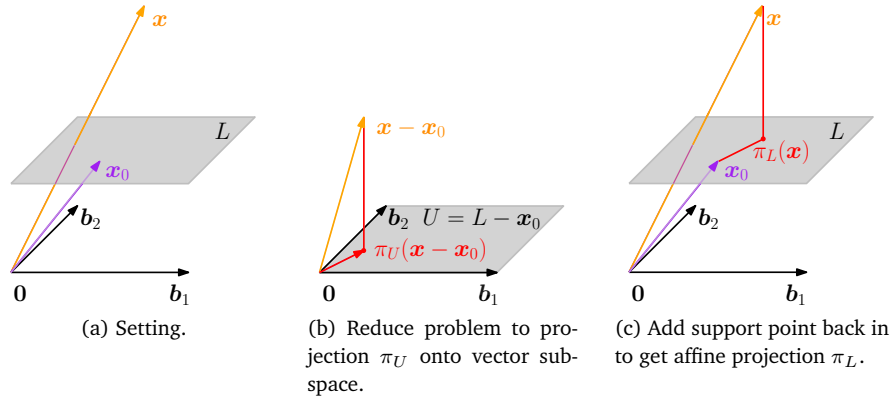
$$\mathbf{u}_1 := \mathbf{b}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad (3.70)$$

$$\mathbf{u}_2 := \mathbf{b}_2 - \pi_{\text{span}[\mathbf{u}_1]}(\mathbf{b}_2) \stackrel{(3.45)}{=} \mathbf{b}_2 - \frac{\mathbf{u}_1 \mathbf{u}_1^\top}{\|\mathbf{u}_1\|^2} \mathbf{b}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (3.71)$$

Figure 3.13

Projection onto an affine space.

(a) original setting;
 (b) setting shifted by $-\mathbf{x}_0$ so that $\mathbf{x} - \mathbf{x}_0$ can be projected onto the direction space U ;
 (c) projection is translated back to $\mathbf{x}_0 + \pi_U(\mathbf{x} - \mathbf{x}_0)$, which gives the final orthogonal projection $\pi_L(\mathbf{x})$.



These steps are illustrated in Figures 3.12(b) and (c). We immediately see that $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthogonal, i.e., $\mathbf{u}_1^\top \mathbf{u}_2 = 0$.

3.8.4 Projection onto Affine Subspaces

Thus far, we discussed how to project a vector onto a lower-dimensional subspace U . In the following, we provide a solution to projecting a vector onto an affine subspace.

Consider the setting in Figure 3.13(a). We are given an affine space $L = \mathbf{x}_0 + U$, where $\mathbf{b}_1, \mathbf{b}_2$ are basis vectors of U . To determine the orthogonal projection $\pi_L(\mathbf{x})$ of $\mathbf{x}$ onto L , we transform the problem into a problem that we know how to solve: the projection onto a vector subspace. In order to get there, we subtract the support point $\mathbf{x}_0$ from $\mathbf{x}$ and from L , so that $L - \mathbf{x}_0 = U$ is exactly the vector subspace U . We can now use the orthogonal projections onto a subspace we discussed in Section 3.8.2 and obtain the projection $\pi_U(\mathbf{x} - \mathbf{x}_0)$, which is illustrated in Figure 3.13(b). This projection can now be translated back into L by adding $\mathbf{x}_0$, such that we obtain the orthogonal projection onto an affine space L as

$$\pi_L(\mathbf{x}) = \mathbf{x}_0 + \pi_U(\mathbf{x} - \mathbf{x}_0), \quad (3.72)$$

where $\pi_U(\cdot)$ is the orthogonal projection onto the subspace U , i.e., the direction space of L ; see Figure 3.13(c).

From Figure 3.13, it is also evident that the distance of $\mathbf{x}$ from the affine space L is identical to the distance of $\mathbf{x} - \mathbf{x}_0$ from U , i.e.,

$$d(\mathbf{x}, L) = \|\mathbf{x} - \pi_L(\mathbf{x})\| = \|\mathbf{x} - (\mathbf{x}_0 + \pi_U(\mathbf{x} - \mathbf{x}_0))\| \quad (3.73a)$$

$$= d(\mathbf{x} - \mathbf{x}_0, \pi_U(\mathbf{x} - \mathbf{x}_0)) = d(\mathbf{x} - \mathbf{x}_0, U). \quad (3.73b)$$

We will use projections onto an affine subspace to derive the concept of a separating hyperplane in Section 12.1.

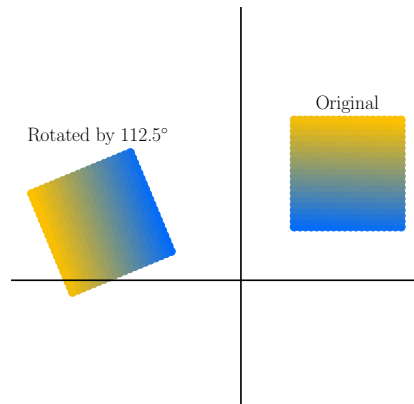


Figure 3.14 A rotation rotates objects in a plane about the origin. If the rotation angle is positive, we rotate counterclockwise.

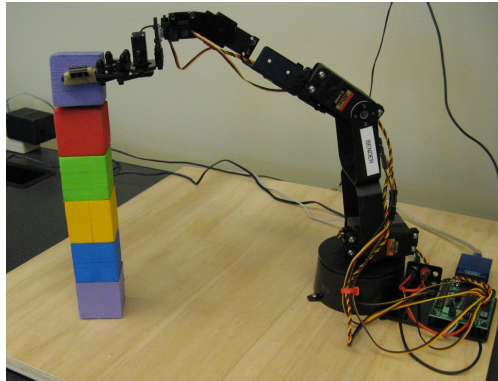


Figure 3.15 The robotic arm needs to rotate its joints in order to pick up objects or to place them correctly. Figure taken from (Deisenroth et al., 2015).

3.9 Rotations

Length and angle preservation, as discussed in Section 3.4, are the two characteristics of linear mappings with orthogonal transformation matrices. In the following, we will have a closer look at specific orthogonal transformation matrices, which describe rotations.

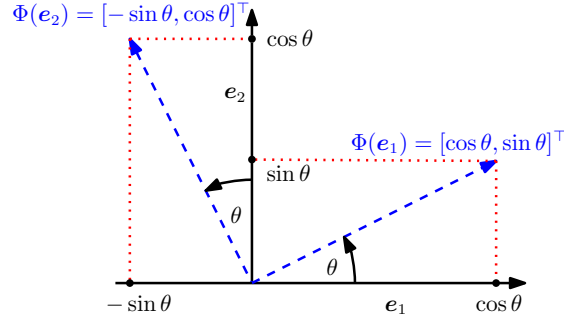
A *rotation* is a linear mapping (more specifically, an automorphism of a Euclidean vector space) that rotates a plane by an angle θ about the origin, i.e., the origin is a fixed point. For a positive angle $\theta > 0$, by common convention, we rotate in a counterclockwise direction. An example is shown in Figure 3.14, where the transformation matrix is

rotation

$$\mathbf{R} = \begin{bmatrix} -0.38 & -0.92 \\ 0.92 & -0.38 \end{bmatrix}. \quad (3.74)$$

Important application areas of rotations include computer graphics and robotics. For example, in robotics, it is often important to know how to rotate the joints of a robotic arm in order to pick up or place an object, see Figure 3.15.

Figure 3.16
Rotation of the
standard basis in $\mathbb{R}^2$
by an angle θ .



3.9.1 Rotations in $\mathbb{R}^2$

Consider the standard basis $\left\{ e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ of $\mathbb{R}^2$, which defines the standard coordinate system in $\mathbb{R}^2$. We aim to rotate this coordinate system by an angle θ as illustrated in Figure 3.16. Note that the rotated vectors are still linearly independent and, therefore, are a basis of $\mathbb{R}^2$. This means that the rotation performs a basis change.

Rotations Φ are linear mappings so that we can express them by a *rotation matrix* $\mathbf{R}(\theta)$. Trigonometry (see Figure 3.16) allows us to determine the coordinates of the rotated axes (the image of Φ) with respect to the standard basis in $\mathbb{R}^2$. We obtain

$$\Phi(e_1) = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \quad \Phi(e_2) = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}. \quad (3.75)$$

Therefore, the rotation matrix that performs the basis change into the rotated coordinates $\mathbf{R}(\theta)$ is given as

$$\mathbf{R}(\theta) = [\Phi(e_1) \quad \Phi(e_2)] = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (3.76)$$

3.9.2 Rotations in $\mathbb{R}^3$

In contrast to the $\mathbb{R}^2$ case, in $\mathbb{R}^3$ we can rotate any two-dimensional plane about a one-dimensional axis. The easiest way to specify the general rotation matrix is to specify how the images of the standard basis e_1, e_2, e_3 are supposed to be rotated, and making sure these images $\mathbf{R}e_1, \mathbf{R}e_2, \mathbf{R}e_3$ are orthonormal to each other. We can then obtain a general rotation matrix $\mathbf{R}$ by combining the images of the standard basis.

To have a meaningful rotation angle, we have to define what “counterclockwise” means when we operate in more than two dimensions. We use the convention that a “counterclockwise” (planar) rotation about an axis refers to a rotation about an axis when we look at the axis “head on, from the end toward the origin”. In $\mathbb{R}^3$, there are therefore three (planar) rotations about the three standard basis vectors (see Figure 3.17):

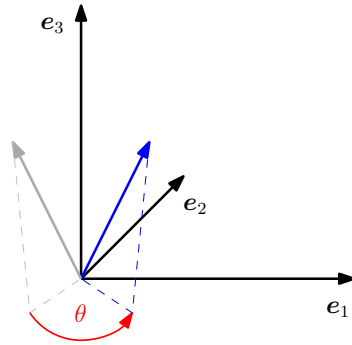


Figure 3.17
Rotation of a vector (gray) in $\mathbb{R}^3$ by an angle θ about the e_3 -axis. The rotated vector is shown in blue.

- Rotation about the e_1 -axis

$$R_1(\theta) = [\Phi(e_1) \quad \Phi(e_2) \quad \Phi(e_3)] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}. \quad (3.77)$$

Here, the e_1 coordinate is fixed, and the counterclockwise rotation is performed in the e_2e_3 plane.

- Rotation about the e_2 -axis

$$R_2(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}. \quad (3.78)$$

If we rotate the e_1e_3 plane about the e_2 axis, we need to look at the e_2 axis from its “tip” toward the origin.

- Rotation about the e_3 -axis

$$R_3(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (3.79)$$

Figure 3.17 illustrates this.

3.9.3 Rotations in n Dimensions

The generalization of rotations from 2D and 3D to n -dimensional Euclidean vector spaces can be intuitively described as fixing $n - 2$ dimensions and restrict the rotation to a two-dimensional plane in the n -dimensional space. As in the three-dimensional case, we can rotate any plane (two-dimensional subspace of $\mathbb{R}^n$).

Definition 3.11 (Givens Rotation). Let V be an n -dimensional Euclidean vector space and $\Phi : V \rightarrow V$ an automorphism with transformation ma-

trix

$$\mathbf{R}_{ij}(\theta) := \begin{bmatrix} \mathbf{I}_{i-1} & \mathbf{0} & \cdots & \cdots & \mathbf{0} \\ \mathbf{0} & \cos \theta & \mathbf{0} & -\sin \theta & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{j-i-1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \sin \theta & \mathbf{0} & \cos \theta & \mathbf{0} \\ \mathbf{0} & \cdots & \cdots & \mathbf{0} & \mathbf{I}_{n-j} \end{bmatrix} \in \mathbb{R}^{n \times n}, \quad (3.80)$$

Givens rotation

for $1 \leq i < j \leq n$ and $\theta \in \mathbb{R}$. Then $\mathbf{R}_{ij}(\theta)$ is called a *Givens rotation*. Essentially, $\mathbf{R}_{ij}(\theta)$ is the identity matrix $\mathbf{I}_n$ with

$$r_{ii} = \cos \theta, \quad r_{ij} = -\sin \theta, \quad r_{ji} = \sin \theta, \quad r_{jj} = \cos \theta. \quad (3.81)$$

In two dimensions (i.e., $n = 2$), we obtain (3.76) as a special case.

3.9.4 Properties of Rotations

Rotations exhibit a number of useful properties, which can be derived by considering them as orthogonal matrices (Definition 3.8):

- Rotations preserve distances, i.e., $\|\mathbf{x} - \mathbf{y}\| = \|\mathbf{R}_\theta(\mathbf{x}) - \mathbf{R}_\theta(\mathbf{y})\|$. In other words, rotations leave the distance between any two points unchanged after the transformation.
- Rotations preserve angles, i.e., the angle between $\mathbf{R}_\theta \mathbf{x}$ and $\mathbf{R}_\theta \mathbf{y}$ equals the angle between $\mathbf{x}$ and $\mathbf{y}$.
- Rotations in three (or more) dimensions are generally not commutative. Therefore, the order in which rotations are applied is important, even if they rotate about the same point. Only in two dimensions vector rotations are commutative, such that $\mathbf{R}(\phi)\mathbf{R}(\theta) = \mathbf{R}(\theta)\mathbf{R}(\phi)$ for all $\phi, \theta \in [0, 2\pi)$. They form an Abelian group (with multiplication) only if they rotate about the same point (e.g., the origin).

3.10 Further Reading

In this chapter, we gave a brief overview of some of the important concepts of analytic geometry, which we will use in later chapters of the book. For a broader and more in-depth overview of some of the concepts we presented, we refer to the following excellent books: Axler (2015) and Boyd and Vandenberghe (2018).

Inner products allow us to determine specific bases of vector (sub)spaces, where each vector is orthogonal to all others (orthogonal bases) using the Gram-Schmidt method. These bases are important in optimization and numerical algorithms for solving linear equation systems. For instance, Krylov subspace methods, such as conjugate gradients or the generalized minimal residual method (GMRES), minimize residual errors that are orthogonal to each other (Stoer and Burlirsch, 2002).

In machine learning, inner products are important in the context of

kernel methods (Schölkopf and Smola, 2002). Kernel methods exploit the fact that many linear algorithms can be expressed purely by inner product computations. Then, the “kernel trick” allows us to compute these inner products implicitly in a (potentially infinite-dimensional) feature space, without even knowing this feature space explicitly. This allowed the “non-linearization” of many algorithms used in machine learning, such as kernel-PCA (Schölkopf et al., 1997) for dimensionality reduction. Gaussian processes (Rasmussen and Williams, 2006) also fall into the category of kernel methods and are the current state of the art in probabilistic regression (fitting curves to data points). The idea of kernels is explored further in Chapter 12.

Projections are often used in computer graphics, e.g., to generate shadows. In optimization, orthogonal projections are often used to (iteratively) minimize residual errors. This also has applications in machine learning, e.g., in linear regression where we want to find a (linear) function that minimizes the residual errors, i.e., the lengths of the orthogonal projections of the data onto the linear function (Bishop, 2006). We will investigate this further in Chapter 9. PCA (Pearson, 1901; Hotelling, 1933) also uses projections to reduce the dimensionality of high-dimensional data. We will discuss this in more detail in Chapter 10.